

L.U.C.I.D.

Laser-Ultraviolet

Cyanobacteria-concentration

Identification and Detection

College of Optics and Photonics
Department of Electrical and Computer Engineering

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Chapter 1: Executive Summary

Cyanobacteria, also known as blue-green algae, is the leading cause for harmful algal blooms (HABs). HABs can devastate aquatic ecosystems by depleting the oxygen content in water and therefore kill other lifeforms in the water, like other plants and aquatic animals. The extent of impact for HABs also extends to land animals, including humans. Cyanobacteria produce cyanotoxins which are harmful when ingested, and the death of aquatic animals like fish depletes food sources for humans and other animals, consequently disturbing entire ecosystems. HABs can be prevented, but they must be detected on time because cyanobacteria can reproduce quickly, and by the time the algae are visible it is already too late.

The L.U.C.I.D. device proposes a way to analyze water samples with algae for early cyanobacteria presence detection. Cyanobacteria have unique spectral properties which make it differentiable from the standard green-algae – spectral properties that are dependent on the concentration of cyanobacteria in water, thus making it possible to correlate spectral intensity with concentration levels. Furthermore, fluorescence can be induced in cyanobacteria by using a UV-laser and emit light at a wavelength of 650 nm.¹ The same can be done with green algae, which fluoresces at a wavelength of 680 nm under the same excitation source.¹ Therefore, the L.U.C.I.D. device will be an optical device that consists of exciting the sample, and a spectrometer to analyze the wavelength of the same sample.

The electronics and software of the L.U.C.I.D. system make it a device that is user-friendly and safe to use with water samples. The device is able to deliver appropriate optical power to the sample while having robust electronics to obtain and interpret the very same data from the optical signals. Additionally, the device comes with its own software to neatly display results like optical power being delivered to the sample, the spectrum, presence of cyanobacteria or green algae, and at what concentrations of cyanobacteria are in the sample being tested. With these features, the L.U.C.I.D. device delivers precise and convenient early harmful algal bloom detection through using spectroscopy and being portable.

Chapter 2: Project Description

This chapter focuses on providing more specific background information on the L.U.C.I.D. device, such as the motivations, current existing products, past projects, L.U.C.I.D.'s goals and objectives, and requirements/specification of the technology in the device.

2.1 Project Motivation and Background

Modern day agriculture focuses on the frequent use of fertilizers, which are commonly composed of nitrogen, potassium, and phosphorous¹⁹ because they are the three essential nutrients necessary for any plant life. When used in excess amounts, these nutrients often leak into fresh bodies of water like lakes and ponds. Fresh bodies of water with an excess amount of nutrients create the perfect environment for various types of algae to rapidly grow, a phenomenon known as algal blooms. Algal blooms are typically harmful because they block out large amounts of sunlight, and when they die the bacteria that decompose the algae deplete the water's supply of oxygen, and the body of water becomes eutrophic. A lack of oxygen in the water creates dangerous drinking water and also kills other organisms like fish, thus devastating entire aquatic ecosystems.

Cyanobacteria specifically, also known as blue-green algae, is the leading cause for harmful algal blooms (HABs).²¹ It has unique properties, such as it is first and foremost a bacterium and not a plant or an alga. It is important to distinguish cyanobacteria as a bacterium because its fundamental nucleus and membrane are unlike plants. Essentially, the cyanobacteria use the excess nutrients in the water and photosynthesis to rapidly reproduce, which is easy for them since they do not need to germinate and take the time to grow like typical plants do. Furthermore, unlike the typical green algae, cyanobacteria produce harmful toxins²² which is why it is incredibly important to be able to detect cyanobacteria presence before eutrophication can occur (the process by which an aquatic ecosystem becomes eutrophic).

Green algae, cyanobacteria, and other plants in aquatic ecosystems contain chlorophyll. Cyanobacteria in addition to chlorophyll, contains a complementary blue pigment called phycocyanin.¹ Phycocyanin can be excited optically using a UV laser, which causes the pigment to emit light at a wavelength of approximately 650 nanometers.¹ The emitting light can then be detected optically, and the amount of cyanobacteria can be estimated using the amount of fluorescing light being detected. Therefore, it is entirely possible to detect even a small presence of cyanobacteria. The amount of cyanobacteria that is dangerous varies from country to country, state to state, county to county, the species of cyanobacteria, whether the cyanobacteria is consumed by a human, and even the amount of time that cyanobacteria has been present in a body of water. Therefore, pinning a specific concentration level as harmful depends on a variety of factors that make it difficult

to quantify a sort of threshold below which is considered safe. It can, however, be agreed upon that a concentration above 1 mg/mL²⁰ is very dangerous and is indicative of an already collapsing aquatic ecosystem; concentrations below 1 mg/mL can be labeled as concerning until the concentration reaches such a low level that it is now safe. As a result, the L.U.C.I.D. system places a threshold level at 1 mg/mL, where concentrations measured above 1 mg/mL are dangerous and concentrations measured below 1 mg/mL are either concerning or safe. This is also an important threshold for our project because 1 mg/mL is a high enough concentration that can be used for sensor calibration, so detecting concentrations below 1 mg/mL, to the microgram range is highly likely and more representative of concentrations found in natural bodies of water.

Aquatic ecosystems also play a key role in upholding non-aquatic lifeforms. For example, lakes and ponds are freshwater sources for animals to drink from, including us humans. Additionally, fresh bodies of water are home to amphibians like frogs and turtles, not just strictly aquatic lifeforms. The lifeforms that live in these aquatic ecosystems also function as a food source for life outside of the water. It is now evident that the devastation of an aquatic ecosystem is devastation to all ecosystems and forms of life, therefore making it a priority to prevent eutrophication as much as possible and detecting cyanobacteria presence holds the key to doing so.

2.2 Existing Works

This section will cover currently existing similar projects and works to our system, which include past senior design projects, current industry products, and our improvements to those systems

2.2.1 Past Projects and Improvements

There have been senior design projects in the past that also focus on assessing water quality parameters, primarily on detecting chlorophyll. Two projects that have measured such parameters are “Absorption Dependent Intensity Detector for Chlorophyll” and “Detection of Algae in Fish Tanks Using Optical Absorption.” Both projects focus on the absorption of light to detect chlorophyll and green-algae presence. Although these previous projects can be used to detect cyanobacteria, there are several limitations of these projects that prompt the use of a different detection method.

For one, these projects focus only on the detection of chlorophyll. In natural bodies of water, and more controlled environments like a fishtank, lots of light will be lost due to scattering which makes measuring absorption losses more difficult. Second, chlorophyll is a poor parameter to use in measuring cyanobacteria because it is not possible to distinguish the chlorophyll from cyanobacteria and chlorophyll from green algae. Chlorophyll is also contained in quite literally every plant, making it impossible to distinguish cyanobacteria, green algae, and other types of plant life

that found its way into a body of water. Additionally, cyanobacteria are a much stronger indicator of eutrophication than the chlorophyll in the typical green algae. Therefore, detecting the presence of cyanobacteria through its unique property given by phycocyanin is plausible and a clearer indicator of eutrophication, hence why we are focusing on it for this project.

To uniquely determine cyanobacteria presence, and distinguish it from green algae, we plan to use the fluorescence properties of both. Cyanobacteria fluoresces at 650 nanometers due to phycocyanin presence, whereas green algae fluoresce at 680 nanometers. The unique wavelengths at which they both fluoresce make it possible to distinguish between the two through fluorescence spectroscopy. Therefore, for our project we are focusing on fluorescence as opposed to absorption. Like absorption, the spectral peak at each wavelength is determined by the concentration levels of cyanobacteria and green algae, thus the concentrations can be determined via spectral peaks and their direct correlation with peak intensity. The general optical setup and spectrometer that will be used to make this possible is shown below in figure 1.

2.2.2 Existing Products

One currently existing product on the market is the In-Situ Aqua TROLL Blue Green Algae Sensor. This product is used to detect cyanobacteria by determining phycocyanin presence in water. Optically, the product is very simple. It primarily consists of an LED and two sensors: one reference sensor and one primary detector. The LED wavelength is not specified. This detector has a quick response time of less than one second and can detect about 1 ug/L of cyanobacteria. This product, however, is expensive and must be installed in water for several months at a time. This is limiting, and the optics will also not be able to differentiate between green algae or cyanobacteria. Furthermore, with prolonged exposure in water the optical surfaces may become dirty and distort or entirely prevent the sensors from giving accurate readings, which is a limitation that the L.U.C.I.D does not need to worry about.

Another existing product is YSI 6131 Blue-Green Algae Sensor. This device has great sensitivity and is able to provide early warnings for high blue-green algae content. It can also detect up to 280,000 cells/mL. It is not cost-effective, and similar to the In-Situ Aqua TROLL, must be installed in a body of water which means it must be mounted properly and retrieved every few months. The optical methods behind the device are not discussed, and thus making an optical comparison is not possible. Compared to this device, the LU.C.I.D. system is able to not only be portable but also will not be affected by outside environmental or aquatic factors.

A third existing product is the Aquaread Blue-Green Algae Sensor. This system has robust sapphire lenses which allow for good optical transmission – again, the specifics of the optical set up in this device are not listed. It has an upper

detection limit of 300,000 cells/mL. This device is also not able to distinguish between cyanobacteria and green algae while being very expensive.

The L.U.C.I.D console consists of robust optics that do not need to be in a real body of water for performing measurements. The use of a laser instead of an LED allows for better excitation of the sample which is crucial in detection, and the use of a diffraction grating allows for the differentiation of cyanobacteria (blue-green algae) as well as regular green algae, all while behind under \$1,000. The L.U.C.I.D console is a cost-effective and optically advanced device.

2.3 Goals and Objectives

The use of fertilizer in agriculture is not coming to an end soon, and with a constantly increasing population, the need for more food and agriculture increases thus amplifying the possibility of eutrophication occurring. As a result, there are a few goals that have guided the construction of the L.U.C.I.D. system.

2.3.1 Goals

Basic Goals

- Build fluorescence intensity vs concentration curve.
- Ability to detect at 1 mg/mL concentration of cyanobacteria
- Have an organized User Interface (UI) to easily navigate between device features.
- Develop an interface that will process the fluorescent data and translate it to graphs and charts from real-time data visualization.
- Determine what levels of concentration are safe, concerning, and dangerous.

Advanced Goals

- Differentiate between green algae and cyanobacteria.
- Determine concentration of green algae.

Stretch Goals

- Determine concentration of cyanobacteria and green algae in mixed sample.
- Reduce detection limit to something on the order of pg/mL.
- Create a compact and portable device for field use.
- Integrate a machine learning model to use the previously analyzed data from previous tests and implement predictive analytics.

2.3.2 Objectives

Basic Objectives:

- Design optics for laser excitation (i.e. provide enough laser power to sample)
- Design sensitive light collection system/spectrometer
- Induce fluorescence of cyanobacteria using a 405 nm laser
- Calibrate each sensors' response at varying concentration levels and store them in a database that can be referenced when doing real-time measurements.

Advanced Objectives:

- Calibrate wavelength to pixel on CMOS sensor to detect wavelength differences.
- Build concentration vs fluorescent intensity curve for green algae.

Stretch Objectives

- Derive a way to measure smaller concentrations.
- Create a more sensitive system to detect pg/mL concentrations.
- Reduce focal lengths of lenses and reduce optical path length for portability.

2.4 Requirements and Specifications

The following figures below (figs. 1-3) outline the hardware along with system requirements.

2.4.1 Optical Hardware Description

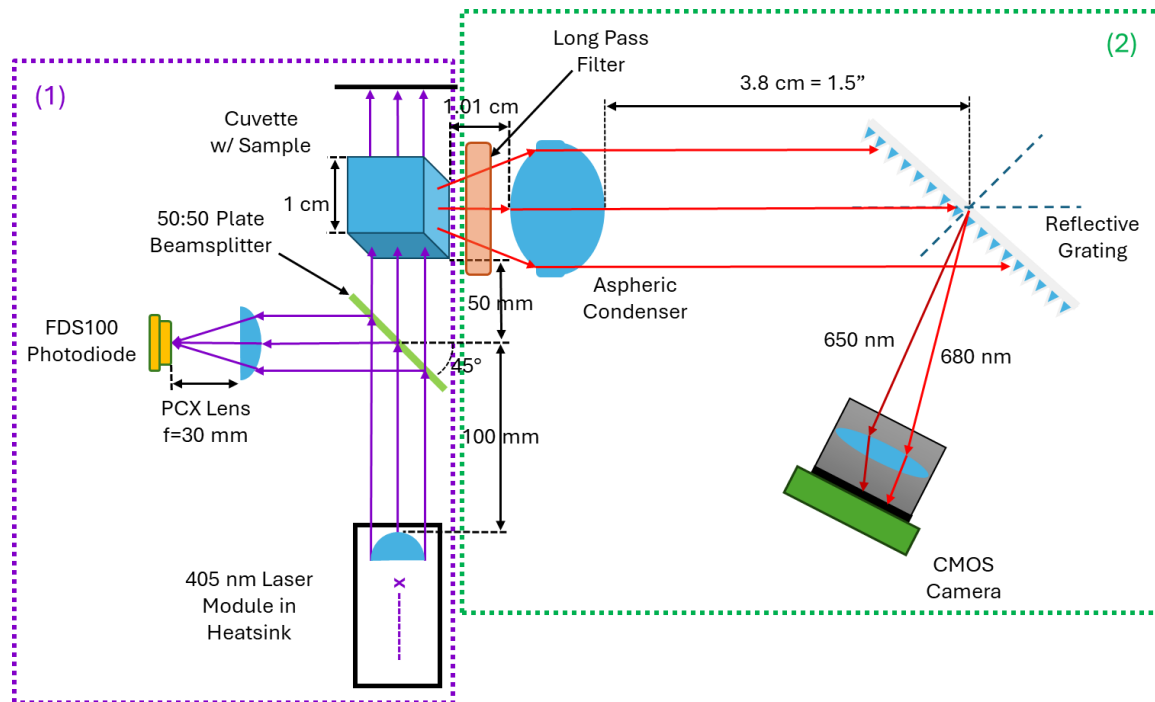


Figure 1. Optical Design Schematic

The optical system can be generalized in two components: (1) sample excitation (in purple) and (2) spectrum measurement (in green) as shown above in figure 1. In order to actively excite the phycocyanin in the cyanobacteria, sufficient enough laser power must be provided, hence our choice of the 405 nanometer laser. A 50/50 beamsplitter splits the beam in two directions – one reflects 50% of the beam to a photodiode to measure laser power stability, and the second transmits the other 50% of the beam towards the sample. The sample receives about 10 mW of power in order to induce fluorescence – 10 mW is a standard power and is not too low to detect but also not high enough to damage the sample.²³ There is a series of optics for collimating the laser beam due to its high divergence, and for focusing it onto the photodiode. The spectrum measurement optics focus on collecting the fluorescing light, which should be about 650-680nm¹. It consists of, a long pass filter so the excitation noise of the laser does not play a role in the photodetection of fluorescing light. A focusing optic with a high numerical aperture is used to collect as much light as possible and collimate the signal onto a reflective diffraction grating. At the diffracting angle, the grating will split apart the constituent wavelengths which are directed towards the CMOS sensor. From there, the electrical components will measure output voltage and the data will go through a series of processing in order to output the incoming spectrum.

2.4.2 Electrical Hardware Description

For our electrical component selection we have two major components that we will be making which are a MCU and a powersupply.

For the MCU we plan to base our project around using a esp32 as the brains of the operation. The esp32 will interface directly with the optical componets of our project and will be used to transfer the data from the optical mesurements into data we can use for our software to determine if there is cyanobacteria. The esp32 will be able to directly interface with a computer and can accurately give real time information about what the system is doing.

For the power supply we plan to make a board that will most likely stepdown 9V or 12V to a 5V and a 3.3V. We plan to use very little power for the system and to be able to run everything. The power supply will power the MCU and may possible power the photoics if needed.

2.4.3 System Hardware Blockdiagram and Description

The following table below is a rough block diagram of our project with included work distribution.

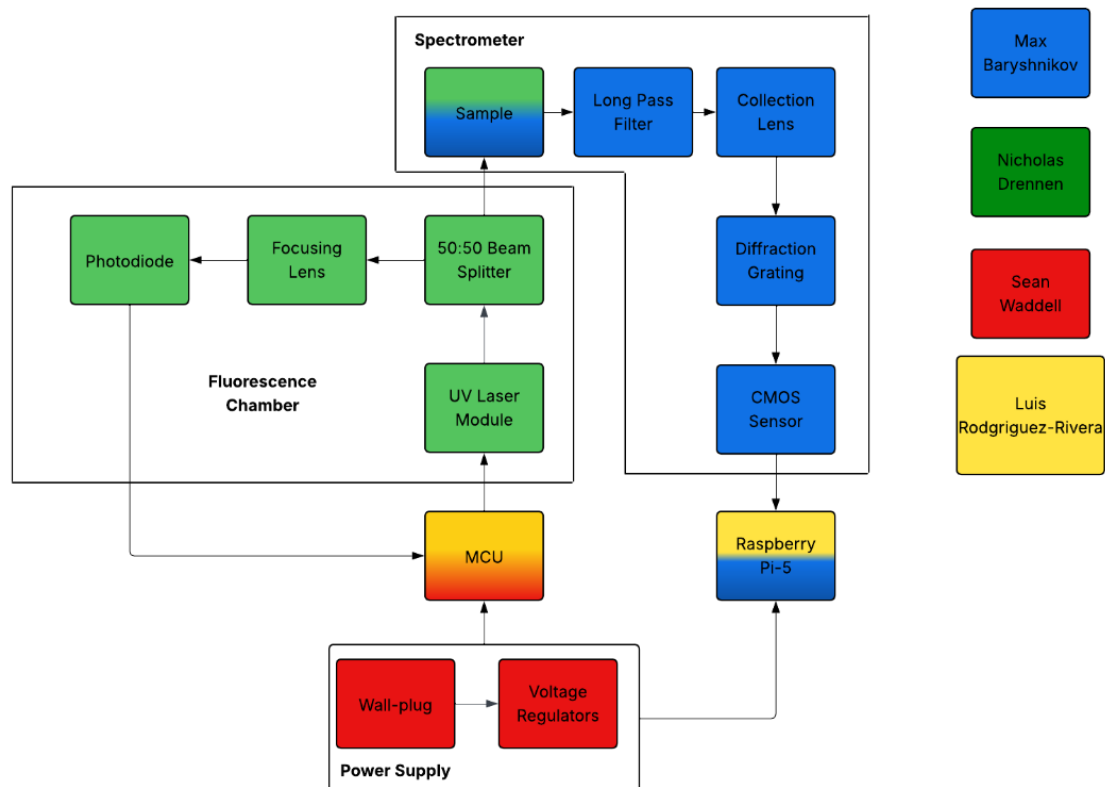


Figure 2. Hardware Block Diagram with allocated tasking

The roles of our electrical engineer and computer engineer will be to update and receive in real time from our optical design portion. We will be receiving a signal in a current response from the photodiode our job will be to boost and amplify the signal to make sure we are not having noise interference and confidence in the results. Then taking this current signal we will use a Transimpedance Amplifier (TIA) to convert this current signal to voltage signal that we can read directly from our Micro Controller Unit (MCU). We will also have the job of programming a motor that will alterate the filters to determine cyanobacteria.

2.4.4 Software Flowcharts

Below is the software flowchart indicating the connections between components and how they will be processed through coding and software.

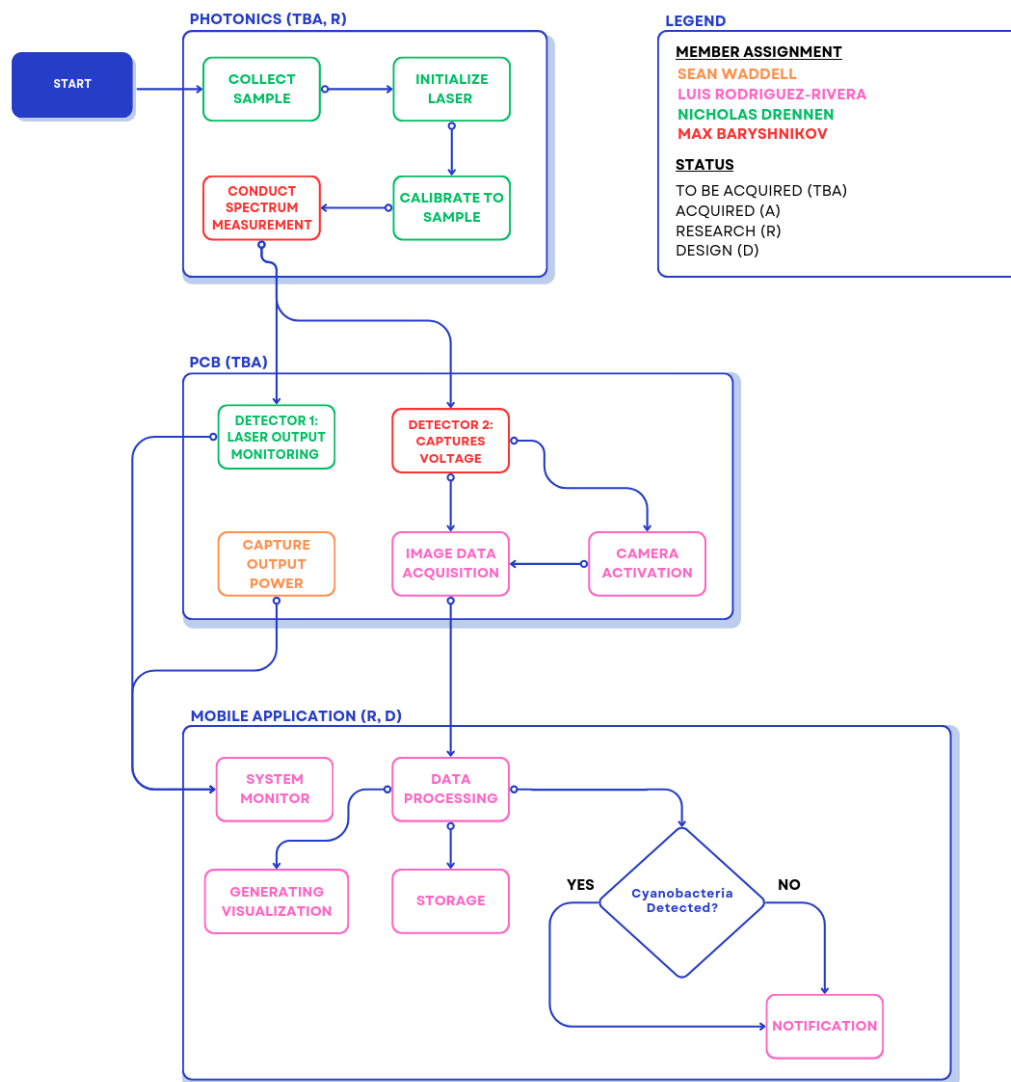


Figure 3. Software Flowcharts

The software for the LUCID system will be designed to handle real-time data collection, processing, and storage from the hardware components, through communication with the PCB. The PCB design will be using the ESP32 microcontroller to allow for communication between the physical sensors and the software. Through the implementation of ESP32's Analog-to-Digital Converter (ADC), we will read the voltage values coming from the sensors and process the data. This data will represent the fluorescence intensity of phycocyanin, our main indicator of the presence of cyanobacteria. Once collected, the ESP32 will transmit the processed data wirelessly to the mobile application via a communication protocol like HTTP through Wi-Fi or Bluetooth for real-time visualization and storage. The processed data will be transferred to a storage point on the mobile device. If necessary, an external hard drive may be needed as an alternative if the processing times are slow, and to ensure historical trend tracking.

2.4.5 Mobile Application Flowchart

Since most of the data collected by the L.U.C.I.D. system is processed through software for storage and display purposes, it is important to showcase that in figures 4 and 5 below. Figure 4 showcases how the mobile app works while figure 5 showcases what the actual app UI may look like.

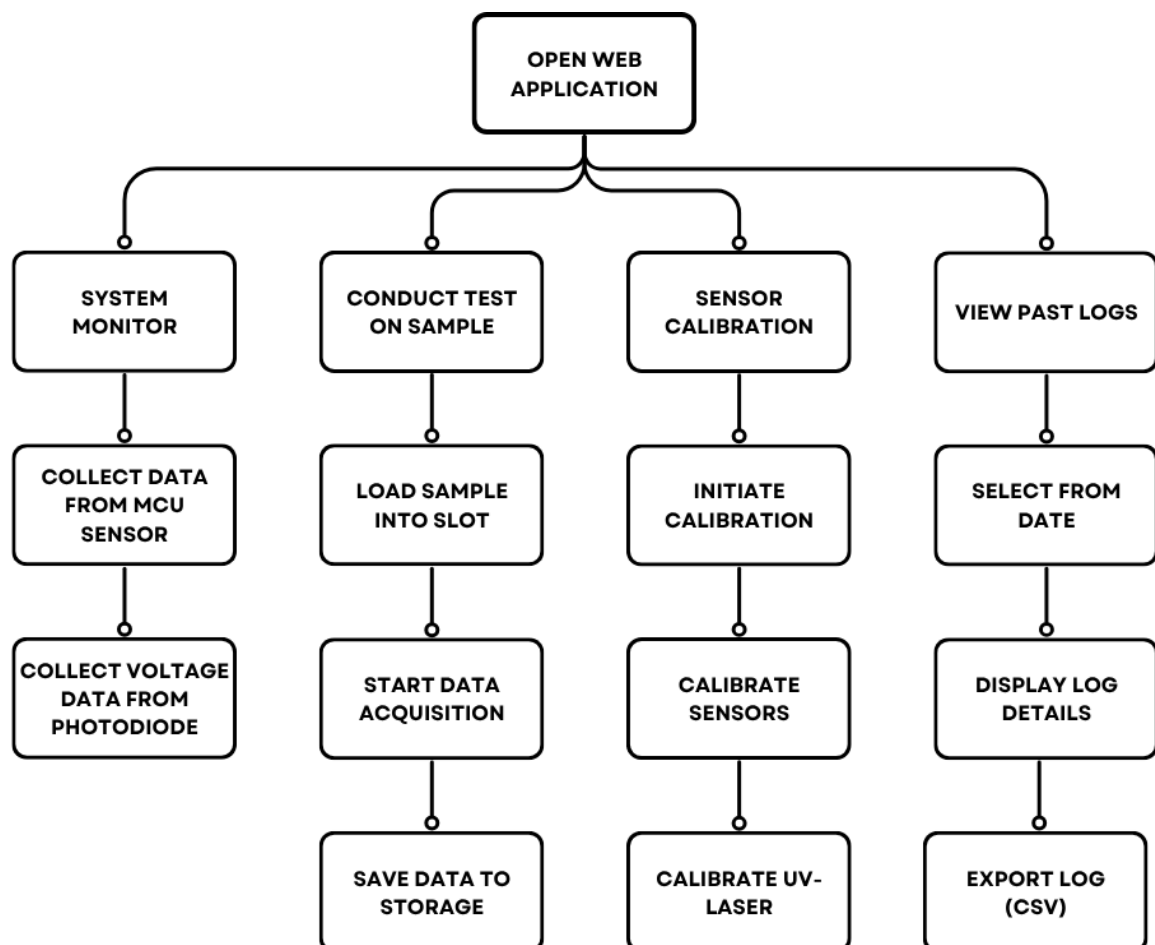


Figure 4. Mobile Application Flowchart

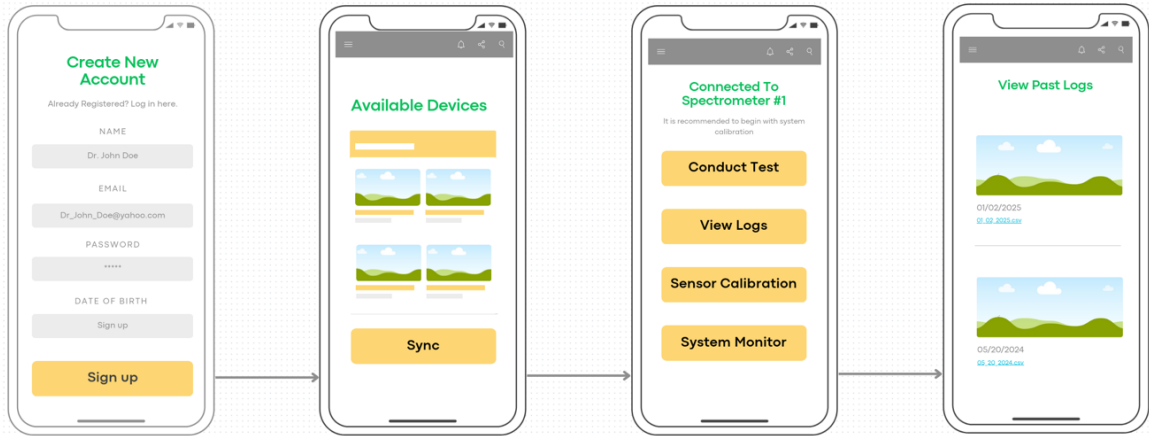


Figure 5. Mobile Application Mockup

On the mobile application, the software will be designed to provide an intuitive and responsive user interface. The mobile application will be the access point for the user to monitor the sensor data in real-time, visualize trends, and access historical data of previous tests. The app will need to store the collected sensor data locally in an SQLite database. This would enable offline access and export options for further analysis of the results. The app will receive live voltage readings, graphical representations of previous results, and calibration.

2.4.6 Engineering Requirements and Specifications

The table below summarizes the requirements and specifications of the system. Table 1 focuses on mainly the components of the system which include optical, electrical, and computer components while table 2 showcases the requirements and specs of the entire system, with 3 demonstratable items highlighted in yellow

Table 1. Engineering requirement specifications for the components that have been used in this project to build the envisioned system

Category	Component	Parameter	Specification	Unit
Optics	Beam Splitter	Power Split Ratio	50:50	R:T
	Cuvette	Dimensions	12 x 12 x 44	(L x W x H) mm
	Aspheric Condenser	a. Focal Length b. Diameter c. Coating d. Numerical Aperture	a. 17.5 b. 3 c. ARC 600-1050 d. 0.86	a. mm b. cm c. nm d. N/A

	Blazed Diffraction Grating	a. Groove Density b. Size c. Efficiency at 650 nm and 680 nm	a. 1200 b. 25x25 c. 70	a. grooves/mm b. mm x mm c. %
Optoelectronics	405 nm Laser Diode	Power	20	mW
	FDS100 Photodiode	a. Responsivity at 405 nm b. Active Area c. NEP	a. 0.2 b. 3.6x3.6 c. 1.2E-14	a. A/W b. mm x mm c. W/Hz ^{1/2}
	Arducam for Pi 5	a. Resolution b. Video c. Video FPS	a. 2592x1944 b. 1080 c. 30	a. pixels b. pixels c. fps
Electronics	ESP32-S3	Speed Flash storage Ram	240 16 512	a. MHz b. MB c. KB
	Raspberry Pi 5	a. Speed b. Storage c. Ram	a. 2.4 b. SD Card c. 16	a. GHz b. Micro c. GB

Table 2. Three demonstratables for the engineering requirements.

Engineering Requirement	Requirement Specification	Unit
The system shall provide the required amount of optical power to the cyanobacteria sample to induce fluorescence	>10	mW
The system shall be able to detect below the threshold concentration of cyanobacteria with 90% accuracy	<1	mg/mL
The system shall display concentration results within a reasonable time frame	<45	seconds
The system shall consume a reasonable amount of voltage	12-24	Volts
The device shall be portable	15 x 30 x 30	(L x W x H) inches

Chapter 3: Research & Technology Investigation

The following chapter goes into the detail of the research process behind what makes the L.U.C.I.D. device functional which includes optical, electronic, and software technologies. The process of selecting each component and comparisons to other components are also explained to reason why the L.U.C.I.D. device is constructed the way that it is. Preliminary research and design to figure out necessary component parameters is also discussed to provide a fundamental basis for our component selection.

3.1 Fluorescence Chamber Optics

In order to induce fluorescence of our cyanobacteria sample, we designed an optical system that properly excites the sample without damaging it as well as seen in the figure below. Major components we have included in the optical design are the light source used for the excitation of the cyanobacteria, free-space optical components (lenses and beamsplitters) to collimate and guide the excitation light to the sample, and a photodetector to monitor the stability of the laser output power. While each kind of component has certain unique aspects that will be examined, price was a major player in the selection process of every component.

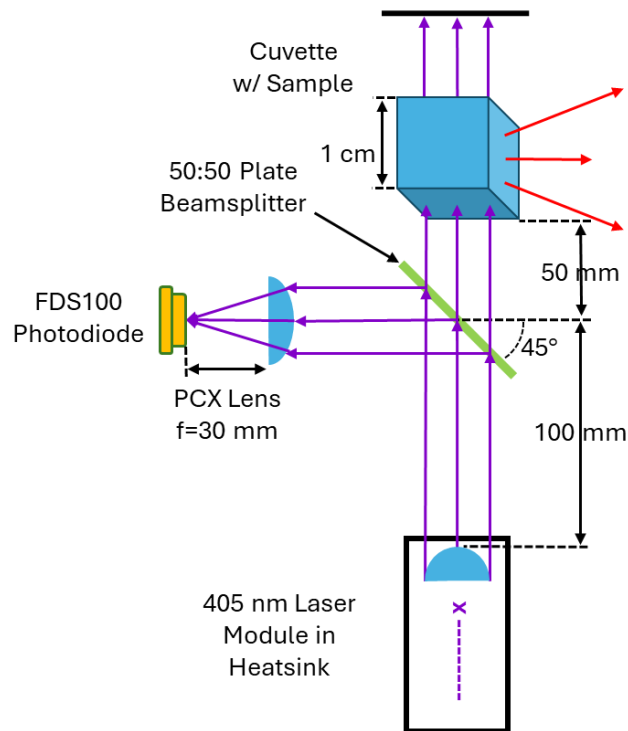


Figure 6. Fluorescence chamber optical design.

3.1.1 Light Sources

The determination of the light source needed to induce fluorescence in a sample of cyanobacteria is a crucial decision. The design of the fluorescence chamber light source has 3 important factors that need to be considered: excitation wavelength, optical power, and beam shape.

To determine what wavelength of light we want to use to excite the cyanobacteria, we must first take a look into its biological pigmentations in figure 6. Cyanobacteria commonly contain phycobiliproteins, like phycocyanin and allophycocyanin, which typically absorb in the UV to blue range of wavelengths (250-450 nm). A 405 nm light source is typically used for phycobiliprotein fluorescence.

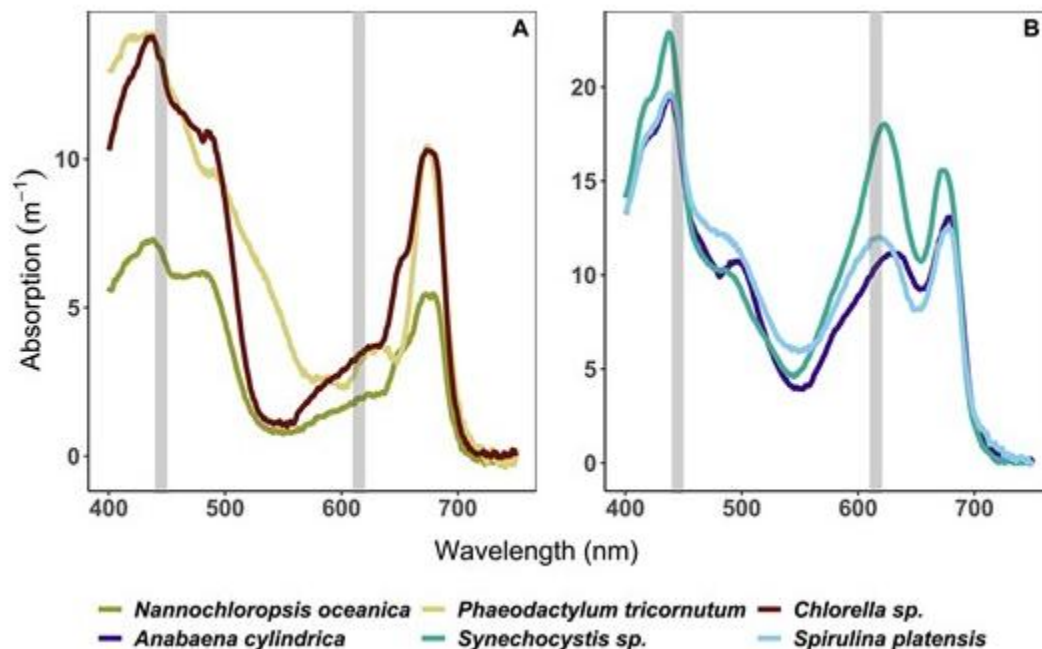


Figure 7. Absorption spectra of algae (A) and cyanobacteria (B) species. [19]

Once a wavelength is selected for proper fluorescence of the cyanobacteria, incident optical power must be considered. During the research phase, we found that the optical power output of the light sourced used in fluorescence experiments of bacterium ranged between 5-10 mW of optical power. This low power level is delicate enough for biological samples to fluoresce without causing photobleaching or photodamage.

3.1.1.1 Light Emitting Diodes

LEDs are an inexpensive and small option to use for the design of our system. LEDs are used in many of the already existing products on the market that are used for cyanobacteria detection and provide the necessary optical power that we require. While they are a cheap and effective option, they lack some of the necessary parameters for the optical design for the sample excitation.

The biggest difference between our design and pre-existing products is that we want to use the emitted fluorescence of the cyanobacteria to determine concentration levels in a sample of water. Encapsulated LEDs emit light in a narrow cone, and we want to keep the light from our source to be collimated so that we are able to use as much optical power as possible. To do this with an LED, another subsystem of lenses would need to be included in our design and purchased to achieve a collimated light source. Collimated LEDs do exist on the market; however, the exorbitant price increase removes them from our options.

3.1.1.2 Laser Diodes

Laser diodes are another commodity light source that are easily found on the market. While their average price can be higher than LEDs, there are options on the market that can suit both the needs of our design and our budget. Laser diodes come in a variety of wavelengths including the 405 nm light source we are looking for, and a range of optical powers for each wavelength. They also come in bare TO-can packages with different pin configurations depending on the size of the packaging.

Laser diodes are made up of semiconductors with a very tiny emitting aperture. This small aperture causes the emitted light to disperse quickly and therefore, the beam divergence in the far-field will be very large. The emitting aperture is a rectangular shape and will cause the emitted beam to be elliptical in shape. This option would require optics used to shape the beam.

3.1.1.3 Laser Pointers

Laser pointers are another light source option that was explored. Laser pointers provide an inexpensive alternative to LEDs and laser diodes, but they fall short of our specifications in many other aspects. Being relatively inexpensive, laser pointers are often lower quality light sources. They are generally designed in mind for a simple quick application, and not a continuous source of light. This lower quality is often seen in the deviation of the peak wavelength shift of 5 nm, positively and negatively, depending on environmental conditions like temperature. Laser pointers are manufactured in a wide range of wavelengths, so acquiring one that emits at the needed 405 nm wavelength was not a foreseeable issue.

What might be the biggest issue with a laser pointer is the power source needed. Some laser pointers operate on batteries where the ignition button needs to constantly be pressed down. Others laser pointers have the ability to be charged

or plugged into either a wall outlet or a USB port. Laser pointers provide an interesting option in that the more you try to pin down the perfect one, it will tend to have more sacrifices in order to emphasize one of its aspects.

3.1.1.4 Comparison

The following table examines the positives and negatives of the different light sources that were considered for the initial design. The highlighted option is the one we decided to move forward with.

Table 3. Final comparison of light sources

Light Source	405 nm	<\$100	Collimated	Needs Additional Components	± Wiring
LED	✓	✓	X	✓	✓
Laser Diode	✓	✓	✓	✓*	✓
Laser Pointer	✓	✓	✓	X	X

*Additional components required if using bare laser diode.

3.1.1.5 Laser Diode Comparison

It was determined that a laser diode would suit our cyanobacteria detection system the best. Next, the process to decide on a laser diode that best fits the specifications mentioned in table 1 began.

While researching the necessary components and laser diodes, we found another that option for laser diodes is a prepackaged module. Similar to a laser pointer, these modules feature a laser diode and driver circuit already installed and a tiny lens used to focus the laser and provide sufficient collimation to the emitted light. Unlike the laser pointer, the major differences are that the module is not battery powered, it has positive and negative wired connection and features a manually adjustable focusing lens. The most important aspect of the laser diode module is that it provides a source of collimated light that we can guide to our sample of cyanobacteria. With this small lens included in the laser module package already, the overall cost of our project design will decrease by as much as \$50.00 due to the purchase of a collimation lens now not being necessary. With this in mind, three different laser diode modules were examined and compared in choosing the best fit.

3.1.1.5.1 Civil Laser Blue Violet Laser Module

Apart from using a bare 3 pin TO-can laser diode this was by far the smallest option that was found. This option operates at our desired 405 nm wavelength and its max optical output power is 20 mW. The website states that this laser diode module boasts a huge >8000 hours of operating time. The package has a 12 mm diameter which will fit into the most commonly available aluminum heat sinks. It is also a relatively low-priced option with the per laser diode cost sitting at \$11.00. There is a minimum order quantity of 5, however, making it one of the most expensive of the three options.

Compared to the following two options, the biggest setbacks of this laser module are that the packaging does not come with a focusing lens included with the casing. Also, the module comes pre-wired with just positive and negative wiring, so we will have to be mindful the input voltage and current don't exceed what the laser diode can properly operate at. The manufacturer states that we will have to keep the max operating current under 80 mA and operating voltage under 5.6 V. Any input values higher than these can result in the component being damaged, which wouldn't be an issue with this option, since the minimum order size is 5 pieces. The lead time on this component from Civil Laser is only 3-5 business days, so ordering more also wouldn't be an issue.

3.1.1.5.2 Apinex LBM Violet Laser Module

As before, laser diode modules are definitely our best option when looking into a laser diode. The previous choice was very bare minimum providing no focus or collimation and with the exception of being pre-wired meaning there is an integrated laser driver circuit; this option would be a slight step up in comparison. The Apinex LBM laser module boasts the 405 nm emitted laser light that we desire and < 20 mW of optical power. The module isn't perfect in terms of power, but the added focusing lens included in the housing makes this the ideal choice for the design constraints. We will be taking a base power measurement with a photodiode to ensure that the laser is operating at a consistent power level. Again, we will need to be mindful of the input voltage and current so as not to damage the laser diode module. This laser diodes expected lifetime is >5000 hours which is more than suitable for our design constraints.

3.1.1.5.3 JOLOOYO 405 nm 20 mW Focusable Laser Diode Module

Lastly, we found an even cheaper version of the laser diode module we were looking for at well under a quarter of the price of the Apinex LBM (after shipping). This option from eBay fits our wavelength and power designs and also comes with free shipping; it is from China and could take minimum of 20 days, weekends included. The packaging comes preinstalled with a focus adjustable lens and the laser driver circuit with positive and negative wires. This laser diode has a max operating current and max operating voltage of <150 mA and 5 V that we will have to monitor during testing phases. This seller also gives the option of obtaining cross- or line- beam shapers for the laser diode module in place of the focusing

lens. This laser diode module is the total package for what we need: price, wavelength, optical power, and collimation all taken care of in one tiny module.

3.1.1.5.4 Final Selection of Laser Diode

Below is a table of the aforementioned laser diode options and our final selection. We opted for the JOLOOYO focusable laser module from eBay due to its unbeatable price and ability to mesh with our design.

Table 4. Comparison/Selection of Laser Diode Modules

Laser Diode	Wavelength	Price	Power (mW)	Size (mm x mm)	Needs Additional Components
Civil Laser Blue Violet Laser Module	405 nm $\pm$ 5 nm	\$11.00	20	12 x 15	✓
Apinex LBM Violet Laser Module	405 nm	\$29.50	<20	12 x 45	X
JOLOOYO Focusable Laser Module	405 nm	\$9.80	20	12 x 35	X

3.1.1.5.5 Heatsink

A heatsink is always required to operate a laser diode module for prolonged periods of time. Temperature has a profound effect on laser diodes as they are highly sensitive to temperature fluctuations. Increased temperature can lead to performance degradation by shifting the emitting wavelength, reducing efficiency, and causing mode hopping. The increased emission wavelength is also known as “redshift” and reduces the energy of emitted photons. Increased temperature can also affect the laser’s operational efficiency and output power by increasing the necessary current to induce lasing, also known as the threshold current as seen in figure 7 below. Once the threshold current, J_{th} , is met and lasing occurs at higher temperatures, the output optical power drops. Temperature increases lead to accelerated aging and potentially permanent damage to the laser diode; it can reduce the overall operational lifespan by potentially forming defects in the diode.

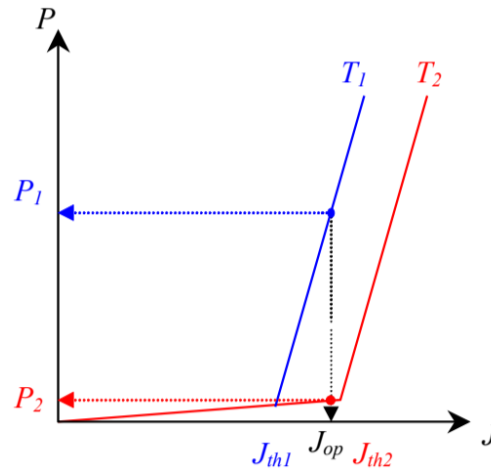


Figure 8. Increased temperature leads to lower output power and potentially no lasing

Commodity heatsinks exist on the market in the form of simple 12 mm diameter holes bored into chunks of aluminum. They can also be simple aluminum tubes with heat dissipation fins sticking out of the sides. These heat sinks are often aided by a thermal paste to help with heat diffusion. Some heatsinks can also be much more expensive and in depth than the cheap options on Amazon and eBay. The more expensive options can include a swivel mounted arm for ease of positioning or have an active cooling Thermoelectric Cooler (TEC) that electronically maintains the temperature of the active laser diode. We just need to keep our laser diode module under 35 degrees Celsius to ensure optimal operation and no prolonged and significant damage to the diode.

No comparison was made other than ensuring that the heat sink dimensions fit well with our chosen laser diode module. Any 12 mm diameter laser diode module will fit into any one of these options. We decided on the BEITESI due to its finned design and it being a simple cheap aluminum heatsink that is easily ordered from Amazon for only \$10.00 with free shipping and fast delivery.

Table 5. Comparison of Heatsinks

Brand	Circle Diameter	Price	Shipping
BEITESI	12 mm	\$9.98	Free 2-day
Q-BAIHE	12 mm	\$10.34	Free 2-day
DWS	12 mm	\$4.10	\$20.00 2-week

3.1.2 Beamsplitter

Beamsplitters are optical components that divide incoming light into two separate parts, Reflected and Transmitted. They are commonly used in laser systems, illumination setups, fluorescence applications, optical interferometry, and instrumentation in the life sciences and semiconductor fields. Light can be split based on overall intensity, wavelength, or polarization state.

Beamsplitters come in various types, including plate, cube, and prism configurations, and can feature a range of anti-reflection coatings suited for ultraviolet (UV), visible, or infrared (IR) wavelengths. Standard beamsplitters divide incident light by a fixed ratio, independent of wavelength or polarization, making them useful for illumination subassemblies or as one-way mirrors. Dichroic beamsplitters separate light by wavelength and are often used for combining laser beams or as hot/cold mirrors. Non-polarizing beamsplitters divide light by intensity and are well-suited for laser beam control. Polarizing beamsplitters, used in photonics instrumentation, split light based on polarization state.

Since we have chosen a 20 mW laser diode and don't want to damage the cyanobacteria by overexposing it with too much optical power, the system is designed with a beamsplitter to optically cut the power of the incident laser light from the diode. This allows us to take baseline power measurements and ensure the laser diode is operating at a stable level. During the selection of the beamsplitter, we had to decide what factors would be the most advantageous for our design: incident surface area, coating, and approximate power ratio split.

Selecting a laser diode module will keep the beam spot size small meaning the diameter of the surface that the light is incident on of the selected beamsplitter won't have as much constraint as initially thought. Ideally, the diameter would be a half inch, about 12.7 mm. Beamsplitter ratio, R:T, is defined by reflected optical power (R) to transmitted optical power (T). We want the split ratio of the selected beamsplitter to be 50:50 reflected power to transmitted power to split our 20 mW laser beam in half.

Keeping budget in mind first and foremost, the price of the selected beamsplitter would ideally be under \$100. Beamsplitters also come in many different shapes; from circular or rectangular plate beamsplitters, to cubes and triangular prisms of glass. However, this aspect really only affects the price of the component and not the overall optical design. Cube beamsplitters that fit our necessary parameters range anywhere from just under \$200 up to \$300, making it an easy option to rule out. Plate beamsplitters will undoubtedly be the type of beamsplitter that worked best for our project design and fit into our budget, starting at around \$50.

3.1.2.1 Edmund Optics #45-316

This beamsplitter from Edmund optics satisfied all of our design parameters and is priced at only \$50. This rectangular plate beamsplitter was by far the cheapest we found during the research phase, so if it is damaged in anyway, it will not be a

financial burden to have to order a second. It is a broadband beamsplitter (400 – 700 nm bandpass) with a 12.5 x 17.5 mm² surface area which will be plenty to interact with our laser light perfectly. The power split ratio right where we want it to be at 50% reflected to 50% transmitted. The 1 mm thickness of the material is a huge positive also. This small thickness will not decrease the laser beams optical power as it travels through the beamsplitter. The only downside to any of the plate beamsplitters is that alignment angle is crucial at must be mounted at exactly 45 degrees to the incident laser light. A small misalignment could result in improper performance of the beamsplitter and skew our results if we are not careful.

3.1.2.2 Thorlabs BSW04 UV Fused Silica Broadband Plate Beamsplitter

The Thorlabs BSW04 UVFS Plate Beamsplitter also boasts a lot of the desired parameters that we need to accomplish our basic goals. This circular plate beamsplitter has an interactive surface diameter of 0.5 inches which is exactly what we are looking for. It also supports the 50:50 reflected power to transmitted power that we need to split the optical power of our 20 mW laser diode. The beamsplitter will have to be aligned at exactly 45 degrees for the coating on the beamsplitter to interact as expected with our 405 nm laser light.

There are more cons than pros, however. The thickness of this particular beamsplitter is a whopping 3 mm. This large thickness could cause a slight loss of reflected or transmitted power from the coated surface as well as cause some diffraction of the laser beam. Lastly, the price is a huge concern at \$91.76 before shipping and handling. Any mistakes with this option could cause us to cross far over our budgeting threshold, making it an unnecessary risk.

3.1.2.3 Newport UVBS11-1

The Newport UV Beamsplitter is very similar to the Thorlabs option we just explored. It maintains an AR coating that allows the interaction with 250 - 450 nm wavelength range. Similarly, it has a 12.5 mm diameter and is optimized for performance at 45 degrees to the incident laser light to reflect 50% power and transmit 50% power. Thickness and price remain the limiting factor on this option as well costing \$120, this would be detrimental to damage or misuse in any way. The thickness is smaller than that of the Thorlabs option at just 2 mm. This will cause less error in power readings than the previous option, but it will not be as accurate as the Edmunds Optics option.

3.1.2.4 Comparison and Final Selection of Beamsplitters

Below is a final comparison and selection of the beamsplitter that is utilized in the design. The beamsplitter from Edmund Optics exceeded what little constraints we had with its selection.

Table 6. Comparison/Selection of Beamsplitters

Beamsplitter	Price	Surface Diameter	Shape	Glass Thickness	Power Split Ratio (R:T)
Edmund Optics #45-316	\$50.00	12.5 x 17.5 mm	Rectangular	1.00 mm	50:50
Thorlabs BSW04	\$91.76	0.5 inches	Circular	3.0 mm	50:50
Newport UVBS11-1	\$120	12.5 mm	Circular	2.0 mm	50:50

3.1.3 Focusing Lens

The purpose of the focusing lens is to focus the collimated beam from our UV laser onto our photodiode after it reflects off of the beamsplitter. Our device is designed for tabletop function, it is not handheld, so the focal length of this part of the system is not a major constraint. We designed the lens to have a focal length of less than 50 mm; this way we can keep the system decently compact. When choosing a lens shape, we want to choose a lens with a positive focal length, so that the light rays converge to a point on the optical axis and don't diverge away from the optical axis. This leaves us the choices of plano-convex lenses or biconvex lenses, both possessing positive focal lengths. The decision to use a plano-convex lens was made based on plano-convex lenses that are ideal for light collimation and focusing applications utilizing monochromatic illumination. Since we decided on a max focal length of 50 mm, we decided to start somewhere closer to the middle and settled on 30 mm. Observing the Lensmaker formula, and rearranging to solve for the radius of curvature,

$$R_1 = f(n - 1) \quad (1)$$

where f is focal length and n is the refractive index of N-BK7 glass (1.515). We solved and found a radius of curvature of 15.6 mm.

Using this radius of curvature, we then analyzed our designed lens using Zemax and observed the spot diagram of the gaussian beam to see how tight an image our lens would form on the photodiode active area, described in detail in section 6.2.1. We calculated an ideal spot size using the equation for spot size diameter, d ,

$$d = 1.22 \left(\frac{\lambda f}{D} \right) \quad (2)$$

Where f is the focal length, again using 30 mm, D is the diameter of the lens that we took from spec sheets from lens seller websites. We calculated this design would yield a spot diameter of $0.791\ \mu\text{m}$. Analyzing the spot diagram in Zemax, we determined that this design would indeed suit our needs as the spot size was only a fraction of the active area of the FDS100 silicon photodiode.

In table 7, we compare a couple of the lenses that were considered for the design of our fluorescence chamber. All lenses were made of N-BK7 material, with a 30 mm focal length, but they were all different diameters in size. We decided on the Thorlabs LA1289 due to its low price and compact size.

Table 7. Plano-convex focusing lens comparison and selection

PCX Lens	Material	Focal Length (mm)	Radius of Curvature (mm)	Diameter	Price
Thorlabs LA1289	N-BK7	30	15.5	½"	\$22.56
Thorlabs LA1805	N-BK7	30	15.5	1"	\$26.50
Edmund Optics 37-790	N-BK7	30	15.5	12.5 mm	30.21

3.2 Spectrometer Optics

After the samples of cyanobacteria, green algae, or mixed samples have been excited, the samples fluoresce at peak wavelengths of 650 nm and 680 nm. Although the emitting light is visible (depending on concentration), the light is incoherent, randomly polarized, and traveling in random directions. As a result, it was important to design a spectrometer that collects and collimates (i.e. give directionality) to as much of the fluorescing light as possible. This collected light then travels to a diffraction grating, where the fluorescing light is decomposed into its constituent wavelengths for detection.

3.2.1 Input Power Calculation

When dealing with fluorescence, it is generally known that the fluorescing light will be at a longer wavelength than the excitation source. In other words, the fluorescing light will have less energy than the 405 nm UV laser. To ensure that enough signal is provided into our spectrometer, a few preliminary calculations were made to ensure the feasibility of detecting the fluorescing light coming from the cyanobacteria.

The process of fluorescence can be broken down into three parts: excitation, absorption, and emission (i.e. fluorescence). To calculate how much optical power

comes from a concentration of 1 mg/mL of cyanobacteria, we first calculated how much of the excitation source is absorbed, and then how much of that absorbed light goes towards fluorescence. The amount of light absorbed by the cyanobacteria is dictated by its absorbance, which is given by the following formula

$$A = \epsilon c L \quad (3)$$

Where A is the absorbance, ϵ is the molar absorptivity, c is the concentration, and L is the optical path length the excitation light goes through. The values for each constant, are

$$\epsilon \approx \frac{10^6 L}{\text{mol} * \text{cm}}, c = 1 \frac{\text{mg}}{\text{mL}} = 0.002 \frac{\text{mol}}{L}, L = 1 \text{ cm}$$

Plugging those values into equation (1) gives

$$A = (10^6)(1)(0.002) = 2000$$

Absorbance can be thought of as an absorption coefficient and dictates how much power is absorbed per unit length of the sample. Then, it simply follows that the absorbed power is

$$P_{\text{absorbed}} = [P_{\text{in}}(1 - 10^{-A})]\eta_{\lambda} \quad (4)$$

The input power is $P_{\text{in}} \approx 10 \text{ mW}$, the absorbance coefficient A was calculated above, and η_{λ} is how efficiently the cyanobacteria absorbs the excitation light (at a wavelength λ) since there will be transmission and scattering losses, this efficiency is about 40% (0.4). Plugging these values into equation (2) above and the power absorbed is calculated to be

$$P_{\text{absorbed}} = 4 \text{ mW}$$

Lastly, the absorbed light will then go towards fluorescence. And because not all of the absorbed light will result in fluorescence, and because some fluorescing light will be reabsorbed into the cyanobacteria, the fluorescing power is

$$P_F = P_{\text{absorbed}} * \eta_F \quad (5)$$

Where η_f is the percentage of absorbed light that contributes to fluorescence and can be estimated to be about 30% (0.3). Using equation (3) and its known values, the fluorescing power can be calculated to be

$$P_F = 1.2 \text{ mW}$$

Hence, we expected an optical input power of approximately 1.2 mW into the spectrometer, which is a signal that is strong enough to be detected and even seen with the visible eye (see chapter 8 on feasibility and testing).

3.2.2 Light Collection Lenses

For the analysis and design of the necessary parameters for the collection lens, we began by analyzing standard formulas used in optics. Some of these formulas are not used with specific numerical values – this is because at first we prioritized understanding the mathematical relationship between different variables such as focal length and radius of curvature for lenses for example. All the formulas, however, are used in one way or another.

To collect as much of the fluorescing light as possible, there are a few parameters that this lens has met. One of the parameters is a high numerical aperture (NA). The formula for numerical aperture is (equation is used in section 3.2.6)

$$NA = n\sin(\theta) \quad (6)$$

Where n is the index of refraction of the medium ($n=1$ in our case because of air) and θ is the half-angle of the maximum acceptance angle. Since $NA \sim \theta$, the larger the NA the larger the acceptance angle, hence the light collection lens should ideally have a large numerical aperture. The device is enclosed; thus, background of ambient lighting is minimal, and was not heavily considered into the choice of numerical aperture.

In connection with the numerical aperture, the lens diameter also naturally factored into how much light passed through the lens and thus impacted the numerical aperture.

The collection lens must also be close to the sample to minimize any propagation losses. The closer to the sample the lens is, the more powerful it must be to collimate the fluorescing light. Therefore, the collection lens must also have a short focal length. One thing to consider, however, is that the focal length is different than the distance from the back surface of the lens to the sample. Most lenses also have what is called a “working distance”, more technically known as the “back focal length (BFL)”. Consequently, in tandem with choosing a collection lens with a short focal length, it must also have a short working distance. A lens with a short focal length also changes the numerical aperture, which is illustrated in the following equation

$$NA = \frac{D}{2f} \quad (7)$$

Where D is the lens diameter and f is the focal length, so choosing a short focal length lens will also further benefit us by increasing the size of the numerical aperture.

Another important parameter to consider is the losses due to reflection and aberrations. To minimize any reflection losses, the lens should be coated with an anti-reflection coating (ARC). Anti-reflection coatings can be optimized for specific wavelength ranges and given the parameters of our system in that we are dealing with visible light; the lens should have an ARC optimized for a range of 400-800 nm. Any losses due to aberration should also be considered, which can be accounted for via lens material (like using an achromat) or by lens shape (like aspherical lenses). Due to the fact that the collection lens will be collecting multiple wavelengths, the main aberration will be spherical aberration and chromatic aberrations and has to be factored into the optical design.

3.2.3 Spherical Lenses

The most standard shape of all lenses are spherical lenses. Spherical lenses include plano-convex lenses, biconvex lenses, plano-and-bi-concave lenses as well. We will not consider concave lenses in our design because we are trying to collimate the fluorescing light, which requires a lens with positive power (i.e. convex shape). For designing this spherical lens, we used the thin-lens lens maker equation where f is the focal length, R_1 and R_2 and the curvatures of the lens surfaces.¹⁰

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (8)$$

The first consideration for a spherical collection lens is the most basic – that is the planoconvex lens. Planoconvex lenses are cheap to manufacture but are known to have high spherical and chromatic aberrations. Since the collection lens will be collecting multiple wavelengths of light, a lens with high chromatic aberrations will be a limiting factor in the performance of the spectrometer. Additionally, planoconvex lenses are typically used for focusing monochromatic light, and not exactly to collect light because they do not have the highest numerical apertures. Plano-convex lenses also usually have lower power and longer focal lengths because only one surface is curved – meaning only one surface is bending the light. In other words, the thin lens formula simplifies down to

$$\frac{1}{f_{plano}} = (n - 1) \left(\frac{1}{R_1} \right) < \frac{1}{f_{biconvex}} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (9)$$

From the above equation, we mathematically demonstrated that plano-convex lenses have lower optical power due to one of the surfaces not contributing to the bending of the light as opposed to a lens that has two curved surfaces. Even if we designed and engineered the values specifically, the nature of the plano-convex

lens (as shown mathematically) limits the total amount of optical power that a plano-convex lens can achieve. They are, however, a cheap lens option which is beneficial for the project considering it is entirely self-funded.

The second consideration for a collection lens is a bi-convex lens. Contrary to the plano-convex lens, there are two curved surfaces which allow the lens to be designed with a higher power and therefore shorter focal lengths, also demonstrated in the above equation. Two curved surfaces also allow for a greater compensation of spherical aberration which is good but still exhibits chromatic aberration. In other words, the light collection efficiency is still impacted in a negative way. While the bi-convex lens minimizes some aberrations, reducing them further is desired. Ideally, eliminating them entirely would be ideal due to the sensitive nature of our project, but because that is not always possible there will have to be some compensation.

We can also choose to use a thicker lens; in which case thickness plays an important role. Mathematically, the thicker the lens the higher the optical power and the shorter the focal length. The thickness of the lens is another design consideration, and main problem with having a thicker lens is that losses will increase due to absorption (i.e. transmission losses). Due to the need for high-power lenses, we are choosing to settle with a thicker lens and are willing to spend some more money on high-efficiency (i.e. low transmission loss) lenses. The thickness of the lens will also negatively affect the number of aberrations, thus increasing it which is a bad trade-off. As a result, even using a bi-convex lens is not ideal for collecting as much of the fluorescing light as possible.

3.2.4 Aspheric Condenser Lenses

As discussed in section 3.2.3, spherical lenses can be limited (by their inherent nature) in focal length and therefore optical power, as well as spherical and chromatic aberrations. An alternative to spherical lenses are aspherical lenses. Spherical lenses have a constant radius of curvature no matter where on the surface the light is incident. On the contrary, the radius of curvature for aspherical lenses changes with position as shown in the figure below.

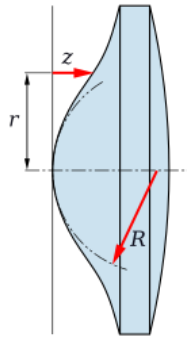


Figure 9. Aspheric condenser lens, with spherical lens outline inside to compare the asphere's nature with that of a spherical lens

With a constant radius of curvature, like in spherical lenses, if light comes in at steep angles it may miss the lens or not get focused properly. Aspherical lenses solve this problem by slightly protruding outwards and gives a greater chance for light rays from non-collimated sources (like a source of fluorescence) to hit the lens and therefore converge. As a result, aspherical lenses typically have a high numerical aperture, which corresponds to shorter focal lengths which is precisely what is needed for this project.

Spherical aberrations are eliminated through the use of aspherical lenses, which makes them an ideal option for collimating light because then we do not have to worry about as much light “missing” the lens. There are also fewer chromatic aberrations in aspherical lenses and considering that we are focusing on detecting two wavelengths (650 nm and 680 nm), the chromatic aberrations will be minimal, and the precision of the spectroscopic technique used in our project is enhanced.

3.2.5 Lens Shape Comparison and Choice

Below is a table quickly summarizing and comparing the different parameters for each lens choice discussed above (plan-convex, bi-convex, and aspherical).

Table 8. Different lens shape options comparing the most important parameters to the optical design of the project's spectrometer with our lens choice highlighted in yellow.
NOTE: what we consider as long and short is specific to our project.

Lens Type	Numerical Aperture	Focal Length (mm)	Working Distance	Aberrations
Plano-Convex	Low (<0.5)	Long ($>2\text{cm}$)	Long ($>1\text{ cm}$)	Spherical, coma, chromatic
Bi-Convex	Low-Med ($0.5\text{-}0.7$)	Long ($>2\text{cm}$)	Long ($>1\text{ cm}$)	Spherical, chromatic,

				better coma correction
Aspherical	High (>0.8)	Short-Long (few mm – 2 cm)	Very Short (order of a few mm)	No spherical aberration, low chromatic aberration, coma can be present

In comparing the different lens design types, their shapes, and parameters, we concluded that aspherical lenses were best suited to the needs of this project. They have a high numerical aperture, are already known for being good lenses at collimating light and have short focal lens or working distances all while keeping aberrations at a minimum.

3.2.6 Aspheric Condenser Design and Choice

Given our choice of using an aspheric lens to collect as much light as possible from the fluorescing cyanobacteria sample, these are the following necessary parameters that have been designed using Zemax and some basic math: lens diameter high NA, short focal length, and anti-reflection coating in the 400-700 nm wavelength range.

First and foremost, we designed our lens to be comparable in size to our cuvette, which is where the cyanobacteria sample rests. The cuvette stands vertically and has a height of 1.73 inches. Therefore, the diameter of the aspheric condenser was designed to be on the order of 1-2 inches as well.

Ideally, we wanted the lens to be as close to the cuvette as possible, or in other words, a short back focal length (BFL). Although, as mentioned earlier, the BFL should also be as short as possible we have to keep in mind design constraints. A lens with a BFL of a few millimeters is great in the sense that it is close to our sample but will be difficult to align with the cuvette because very small distances are challenging to work with. Consequently, a lens with a BFL on the order of 1-2 cm is more realistic and does not compromise too greatly on being too far from the cuvette.

The numerical aperture of our lens should also be high in order to encompass the full cuvette for as much light collection as possible. There are some calculations that can be performed to figure out the approximate value for numerical aperture that we are looking for.

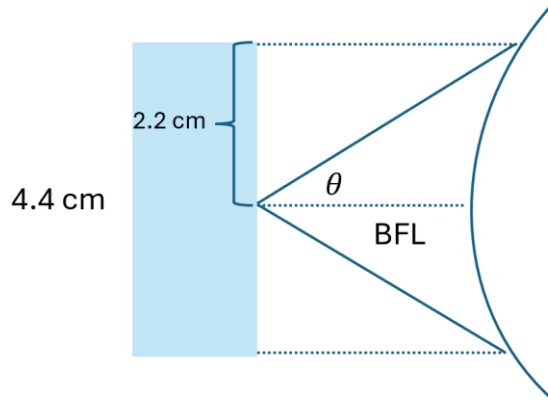


Figure 10. Diagram showing the cuvette (blue box to the left) and asphere lens surface, with back focal length and angle used to calculate numerical aperture

The diagram above considers the size of the cuvette, as well as the acceptance angle and BFL of the aspherical lens. The acceptance cone should roughly cover the height of the cuvette to consider for different amount of cyanobacteria. For initial analysis, suppose the BFL is about 1 cm. If the BFL is 1 cm, then it follows that

$$\theta = \tan^{-1}\left(\frac{2.2 \text{ cm}}{BFL}\right) = \tan^{-1}\left(\frac{2.2 \text{ cm}}{1 \text{ cm}}\right) = 65.56^\circ$$

$$\Rightarrow NA = \sin(\theta) = \sin(65.56^\circ) \approx 0.91$$

This is a relatively high numerical aperture and was difficult to find but still is the relative value that our actual choice of lens, which is discussed in a later section, should be around.

Lastly, some major design constraints were the coating and cost of the aspheric lens. Given that the aspheric lens will be collecting light in the red wavelength range (650-680 nm) the lens was designed to have an anti-reflection coating in this approximate range to allow as much of the fluorescing light through the lens as possible. Opting in for no-coating is not an option given the incredibly sensitive nature of fluorescence spectroscopy. The cost of the project will also depend on several factors such as coating, size, and availability. It is known that lenses may be expensive, thus our budget for this aspheric condenser is on the order of \$40-\$80. The cheaper the better, but compromises will have to be made.

The two main suppliers of aspherical condenser lenses are Edmund Optics and Thor Labs. These were four main options because other vendors on websites like eBay, although they offered significantly cheaper alternatives than Edmund Optics and Thorlabs, were limited in options and the quality of the lens was questionable.

Lots of sellers on eBay also did not provide a lot of technical information, such as BFL which are crucial parameters in the alignment of the L.U.C.I.D system. Below are a few tables listing options of aspheric condenser lenses from each vendor with parameters that closely match what was designed or calculated from above. One main compensation that had to be made was with the numerical aperture. A lens with a numerical aperture of 0.91 is difficult to find, and is unusual, so we had to compromise with mostly lower values for NA.

From Edmund optics, there were three options that were considered. The first aspheric condenser lens that we considered was from Edmund Optics (part #35-044). This lens option had the shortest focal length, shortest working distance, and was also the cheapest while also having a relatively high NA = 0.85 which was why it was an option that we had considered. The main drawback for this lens is its diameter of about 1.3 cm. As aforementioned, a small diameter is not ideal for light collection and in the case for this project is fairly limiting. The short working distance would also make this lens difficult to work with. The second option that we considered was lens #15-889. This lens has a numerical aperture of 0.89, which is closest to our calculated value of 0.91 of NA from figure 6. Its focal length is 11.1 mm with a similar working distance of ~5 mm, and has a bigger diameter of 2 cm. Although this lens has the largest NA that we were able to find on Edmund Optics' website, it was still small relative to the cuvette and had a working distance that is, unfortunately, difficult to work with. The lens being about half the size of the cuvette increased the chances of the lens not collecting enough light despite it having a large numerical aperture. The third option considered from Edmund Optics is lens #15-895. This aspheric condenser lens has a diameter that was much more acceptable than the other two previous options, it has a diameter of 3 cm which closely matches the 4.4 cm height of the cuvette. Additionally, the numerical aperture is still relatively high with a value of 0.86, which is not too far below our wanted value of 0.91. Although its focal length is longer at 17.5 mm, its working distance is 10.4 mm or about 1 cm, which is a distance that is easier to work with and align in the actual device. All of these comparisons and lens choices are summarized in table 9 below.

Table 9. Comparison and Selection of collection lenses from Edmund Optics

Edmund Optics					
Diameter (cm)	Numerical Aperture	Focal Length (mm)	Working Distance – BFL (mm)	Part #	Cost
1.3	0.85	7.7	4	35-044	\$43.00
2	0.89	11.1	4.9	15-889	\$65.00
3	0.86	17.5	10.4	15-895	\$71.00

There are several more lenses that we looked through from Thorlabs. The first lens that we considered from Thorlabs was #ACL12708U. This lens is budget friendly, but comes at the cost of being small with a diameter of 1.27 cm. Its NA is relatively large with a value of 0.78 and also has a short working distance, which is beneficial in one sense but like previously mentioned, is difficult to work with. It is also still budget friendly being under \$50.00. The second option from Thorlabs is a larger lens, and also has the largest numerical aperture value that is listed on Thorlabs' aspheric condenser page. Its NA is 0.79, and also has a more reasonable working distance of 7.3 mm – although this is still quite small it is less than 1 cm still. The third option from Thorlabs is the largest lens out of all the options from Thorlabs and Edmund Optics, lens #ACL50823U and has a lens diameter of 5.08 cm. This is larger than our cuvette by ~0.6 cm, and still maintains a decent value for numerical aperture having an NA of 0.75. Its working distance is also 1.7 cm which is even easier to work with in regards to optical alignment.

Table 10. Comparison of collection lenses from Thorlabs

Thorlabs					
Diameter (cm)	Numerical Aperture	Focal Length (mm)	Working Distance – BFL (mm)	Part #	Cost
1.27	0.78	8	3.7	ACL12708U	\$31.77
2.53	0.79	16	7.3	ACL25416U	\$20.16
5.08	0.75	32	17	ACL50823U	\$61.48

Between the options from Thorlabs and Edmund Optics, one of the more important parameters that we were hesitant to compromise on is numerical aperture. Thorlabs does not offer aspheric condensers with numerical aperture values greater than 0.79. The benefit of Thorlabs, as seen in lens #ACL50823U is that they can make relatively large lenses and at bit of a cheaper price than the Edmund Optics alternative by about \$10.00. That extra ten dollars is money that we were willing to spend in order to have an increased value for numerical aperture. As a result, we chose Edmund Optics aspheric condenser part #15-895 as highlighted in yellow in table 4. This lens has a high numerical aperture, large diameter relative to the size of the cuvette, and has a working-distance that is easy to work with. The compromise for this component was in the price, which is a bit on the pricey side at \$71.00, and was around ~\$85.00 after taxes and shipping costs.

3.2.7 Diffraction Gratings

Once the light collection lens collimates the fluorescing light emitting in all directions, the different constituent wavelengths must be split up in order to be detected individually. This section will go through the different types of gratings, as

well as the parameters that we considered while making as high an efficiency spectrometer as possible.

3.2.7.1 Grating Parameters and Design

Due to the fact that we were dealing with a relatively weak signal coming from the aspheric condenser collimating lens, the diffraction grating must have a few parameters to ensure low light losses: minimal losses/high efficiency at the desired wavelengths we want to detect, that is 650 *nm* and 680 *nm*. A second parameter is the groove density – the groove density is the number of grooves per millimeter and determines our ability to resolve two wavelengths (also known as chromatic resolving power). The third parameter is grating size. The amount of light being diffracted, and its efficiency, also depends on how many grooves the light is incident upon. The bigger the grating, and the higher the groove density the diffraction effects will be better as well as being more efficient. The position of the grating relative to the incident light coming from the aspheric condenser lens will also impact the ability to resolve between the 2 wavelengths.

In order to determine the number of slits that needed to be illuminated to resolve the wavelengths of 650 nm and 680 nm the resolving power of the grating must be obtained, which is⁹

$$R = \frac{\lambda}{\Delta\lambda} = mN \quad (11)$$

Where λ is the average wavelength

$$\lambda = \frac{650 + 680}{2} = 665 \text{ nm}$$

And $\Delta\lambda$ is the difference in wavelengths, or the spectral resolution in our case, which is 30 nm. N is the number of illuminated slits and m is the order of diffraction. If we want to use the first order diffracted mode, $m=1$, second order is $m=2$, etc. Now, using $m=1$ and solving for N,

$$N = \frac{\lambda}{\Delta\lambda} = \frac{665 \text{ nm}}{30 \text{ nm}} = 22.17 \text{ slits}$$

22.17 slits must be illuminated in order to spectrally resolve the wavelengths of 650 and 680 nm. If we were to use the second order diffracted mode, the number of slits would decrease by a factor of 2 and only about 11 slits would need to be illuminated. To ensure as many slits are illuminated as possible, the grating will be positioned at an angle in front of the collection lens as shown in the figure below

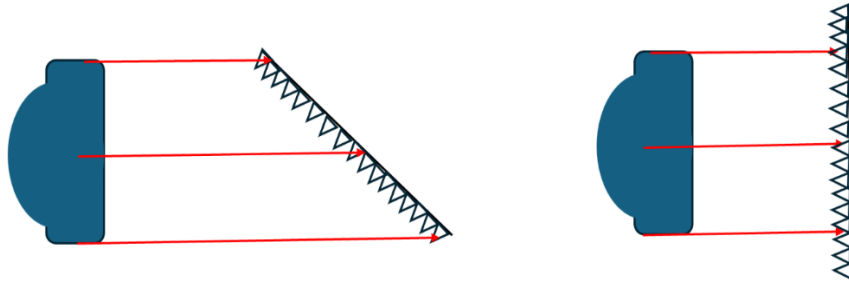


Figure 11. Grating positioned at an angle (left) covers more slits or grooves compared to a grating at normal incidence (right) relative to the incident light

Another important parameter we need to know is the angle at which the two wavelengths (650 nm and 680 nm) will diffract at. The angles at which the light diffracts determines the sensor placement. These angles are measured from the normal incidence of the grating and can be calculated from the grating equation for reflective gratings

$$m\lambda = d(\sin\theta_i + \sin\theta_m) \quad (12)$$

Solving for the angle at which a certain wavelength will diffract within a specific diffracted mode, m ,

$$\theta_m = \sin^{-1} \left(\frac{m\lambda}{d} - \sin\theta_i \right) \quad (13)$$

This parameter was calculated for each grating option covered below in section 3.1.2.3. Ideally, the larger the angle away from the normal line of the grating, the easier it will be to position the sensor and also to avoid overlapping wavelengths or modes. Another thing to note is that the large the angle of incidence is, the smaller the angle of diffraction will be. Smaller angles of diffraction, especially using the first-order diffraction mode are difficult to work with. However, if the angle of incidence is too close to normal incidence, sensor placement may become more difficult and the separation between modes and wavelengths becomes less apparent. Having the diffraction grating placed at an angle of around 20°-30° is fairly standard. As a medium, we have chosen to place the diffraction grating at an angle of 25° relative to the normal (i.e. $\theta_i = 25^\circ$).

3.2.7.2 Grating Selection

There are a variety of grating types that are used in spectroscopy in order to split apart different wavelengths for detection. The main two types are transmissive gratings and reflective gratings. Transmission gratings usually work at normal incidence and diffract light on the opposite side of the grating. They are simple to

align and easy to use but come at the cost of efficiency. Additionally, at normal incident specular reflection becomes a problem because that decreases the resolution of the spectrometer and the intensity of higher order diffracted beams. Reflective gratings on the other hand do not have transmission losses and therefore are generally more efficient. Specular reflection can also be avoided by using ruled or blazed gratings. Reflective diffraction gratings are standard in fluorescence spectroscopy, which is why they are the preferred grating of choice for the design of this spectrometer. Therefore, the grating choices are all reflective diffraction gratings most of which are blazed. Blazed diffraction gratings have small, angled grooves on the surface, which means that light incident onto the grating always hits the grating at some sort of angle which further increases the efficiency of the grating and decreases the amount of specular reflection and enhances the visibility of the higher order modes. Blazed diffraction gratings are also optimized for one specific wavelength, which will impact efficiency at other wavelengths.

Table 11. Comparison table between transmissive gratings and reflective gratings, with our choice in type of grating being highlighted in yellow.

Transmissive Gratings	Reflective Gratings
Lots of absorption losses	No losses due to absorption
Lower efficiency with higher groove density	Higher groove density often improves efficiency
Additional losses due to specular reflection	Standard in spectroscopy
Easier alignment	Higher resolving power

For the different grating options, similar to the options for the aspheric condensers, our main options were from Edmund Optics and Thorlabs. There are also some options from eBay which may have worked, but there are some specifications missing such as groove density and efficiency charts which are important for this spectrometer. One of the first options that we considered was a blazed reflection diffraction grating from Edmund Optics, part #41-028. This grating has a groove density of 600 grooves/mm, 25 mm x 25 mm, and a blaze wavelength of 750 nm. The size of the grating is close to the size of the aspheric condenser, which is good for uniformity and minimizes the amount of light that will miss the grating. The blaze wavelength, as mentioned above, is the wavelength to which the grating is optimized for. This is relatively close to our target wavelengths of 650 nm and 680 nm. Unfortunately, Edmund Optics does not have a blaze wavelength anywhere from 600-700 nm. The second closest blaze wavelength would be 500 nm which is further from 650nm than 750 nm is, which means worse efficiency. Given these diffraction grating parameters, the wavelengths of 650 nm and 680 nm (using the first order diffracted modes), using equation (8) would diffract at angles of

$$\theta_{1,650} = -1.87^\circ \text{ and } \theta_{1,680} = -0.84^\circ$$

The negative angles mean that the first order mode, with 600 grooves/mm, are on the same side as the incident light which is not ideal for our setup.

The second option, also from Edmund Optics, is part #43-210. This diffraction grating has similar parameters to the one mentioned above, at 25 mm x 25 mm, blaze angle of 750 nm, but it has a groove density of 1200 grooves/mm. Using this diffraction grating, the angles for the first order diffracted modes are

$$\theta_{1,650} = 20.94^\circ \text{ and } \theta_{1,680} = 23.17^\circ$$

These angles for the first order mode are relatively large, and the two wavelengths have about a 3° separation which are much easier to deal with than the grating with 600 grooves/mm.

Our third grating option that we considered was from Thor Labs is grating #GR25-068. It has the same specifications as the first option from Edmund optics. That is, 600 grooves/mm, blaze wavelength at 750 nm, and dimensions 25 mm x 25 mm. The benefit of this option from Thorlabs compared to Edmund Optics is that it is cheaper by about \$10.00. There is a fourth Thorlabs options part #GR25-1208. It is cheaper than Edmund optics by about \$10.00 and has the same parameters as #43-210 that has 1200 grooves/mm.

There were two more options from eBay, both are large gratings. One is 45 mm x 45 mm, and another is 2.25 inches x 2.25 inches. The first one already comes in a diffraction grating holder and is only \$71.28. There are many problems with this option, however, despite its good size, pre-mounted option and cheap price. It does not come with a datasheet, which means there is no way of knowing the efficiency at our select wavelengths, and there is no label or mention of groove density. Without the groove density, it is not possible to calculate the angular locations/positions of the first order spectrum. The same applies to the second eBay option. This problem can potentially be alleviated by contacting the seller and asking for more information, but unfortunately there is no actual way of verifying whether the information that the seller provides is true without actually testing the grating directly.

Each grating will also have a particular efficiency at different wavelengths, as aforementioned, and both Thorlabs and Edmund Optics provide graphs to determine this efficiency.

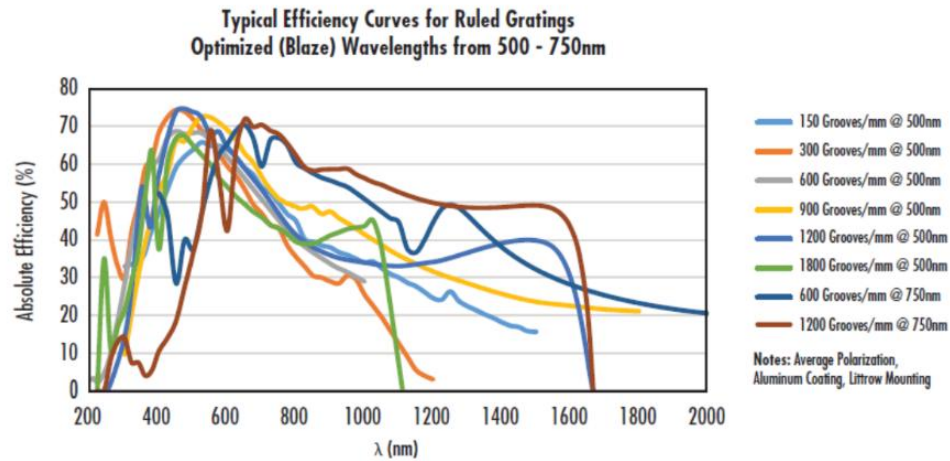


Figure 12. Efficiency Curves for Edmund Optics Gratings at different blaze wavelengths and groove densities [11]

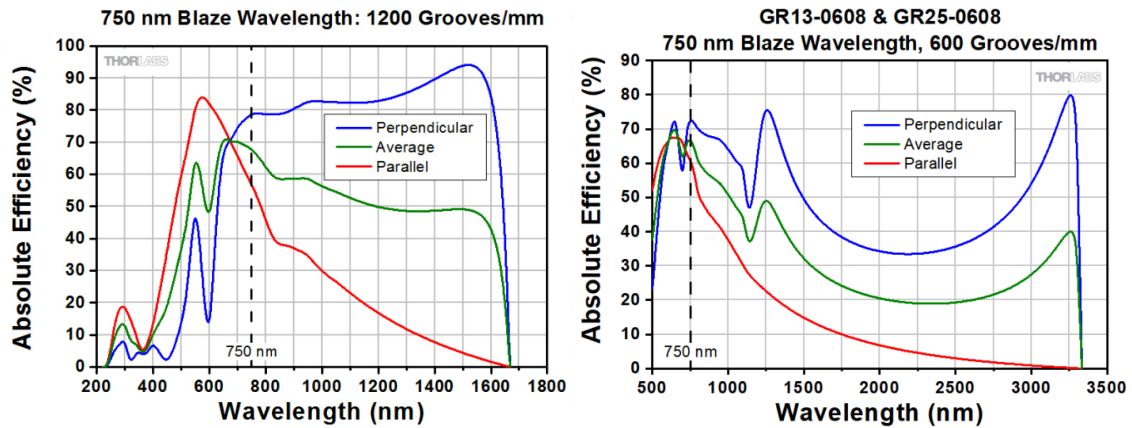


Figure 13. Efficiency Curves for Thorlabs Gratings at different blaze wavelengths and groove densities [13]

From all of the above graphs, it can be seen that the grating efficiencies are greatest not only using a 750nm blaze wavelength, but also at wavelengths of 1200 grooves/mm. Therefore, the two options that we considered are Edmund Optics #43-210, or Thorlabs #GR25-1208. The Thorlabs option seems to have had a higher efficiency at 650 nm and 680 nm – that is just a visual observation. The efficiency for both gratings seems pretty similar at our desired wavelengths which is about 70% efficiency.

Table 12. Parts Comparison Table for ruled, blazed reflective diffraction gratings with our choice highlighted in yellow

Supplier	Groove Density	Blaze Wavelength	Efficiency @ 650 nm & 680 nm	Diffraction Angles (650 nm, 680 nm)	Size	Cost
Edmund	600 grooves/mm	750 nm	~70%	-1.87, -0.84	25mm x 25mm	\$130.00
Edmund	1200 grooves/mm	750 nm	~70%	20.94, 23.17	25mm x 25mm	\$157.20
Thorlabs	600 grooves/mm	750 nm	~70%	-1.87, -0.84	25mm x 25mm	\$138.11
Thorlabs	1200 grooves/mm	750 nm	~70%	20.94, 23.17	25mm x 25mm	\$153.10
eBay	1200 grooves/mm	750 nm	~90%	20.94, 23.17	25mm x 25mm	\$64.42
eBay	N/A	N/A	N/A	N/A	45mm x 45mm	\$78.21
eBay	N/A	N/A	N/A	N/A	2.25" x 2.25"	\$58.66

Ultimately, the diffraction grating chosen for the L.U.C.I.D system is the Thor Labs GR25-1208. It has the same specifications as its counterpart from Edmund Optics, but it is slightly cheaper which is why it is the option that we chose, considering all other parameters are very similar. The first eBay option in the table above seems like the perfect option, but it is a product that seems “too good to be true”. The shipping plus delivery time was about 2-3 weeks as well, which is not ideal.

3.3 Sensor Comparison & Selection

The primary objective for the initial design of this spectrometer was a cost-effective and portable cyanobacteria detection system that's capable of analyzing the fluorescence spectrum of cyanobacteria. To achieve this, we considered the best hardware and software to balance performance, accuracy, power consumption, and cost.

The decision-making process involved identifying the key requirements for the L.U.C.I.D detection system. Overall, there will be two sensors. The first sensor served as a reference as to how much power is going to the cyanobacteria sample, which is an important parameter in inducing fluorescence. This photodetector needs to be optimized for our 405 nm laser. The second detector was in the

spectrometer chamber, which will require a very sensitive detector in order to detect the fluorescent light from the diffraction grating. The design needs to be capable of detecting weak fluorescent signals from the cyanobacteria. These fluorescent signals emit light in the 650-680 nm range. The optical sensor must have the capability of handling a highly sensitive signal with a dynamic range to be able to distinguish between the variations in the intensity of the fluorescence. The system must also support real-time data acquisition and wireless data transfer to a web app.

The first sensor will be detecting about 10 mW of UV laser power. Therefore, we must pick a photodiode that is not only cost-effective but also has a decent responsivity. The responsivity is defined as¹²

$$\mathfrak{R} = \frac{I_{ph}}{P_{op}} \left(\frac{A}{W} \right) \quad (14)$$

Where I_{ph} is the photocurrent generated and P_{op} is the incident optical power. It is also important to note that responsivity is dependent on wavelength. The photodiode was in a circuit, covered in section 3.3, and the voltage across a load resistor was measured to solve for optical power

$$P_{op} = V_{out} / (\mathfrak{R} * R_L) \quad (15)$$

The software conversion from output voltage to optical power is covered in section 3.5.

3.3.1 Photodiode Comparison

For the fluorescence chamber, a standard photodiode sufficed. Our UV laser had sufficient optical power to detect, and thus did not require more sensitive optoelectronics for photodetection such as phototransistors or avalanche photodiodes.

3.3.1.1 Hamamatsu S9195 Si PIN Photodiode

The Hamamatsu silicon photodiode comes housed in a TO-5 packaging, with a 3 pin connection. This photodiode is designed to have enhanced responsivity in the UV emission wavelength range, boasting a generous 0.28 A/W at 405 nm, perfect for our design specifications. The active photosensitive area is also very large at a massive 25 mm². However, the Hamamatsu website does not provide pricing or shipping information so it is safe to assume that, despite the fact that this photodiode would perform best for our needs, we have to exclude it from our decision.

3.3.1.2 Thorlabs FDS010 Si Photodiode

The spectral response of the FDS010 silicon photodiode is very low for the application we want to use it for. Again, we are just taking a baseline power measurement to monitor optimal operation of our device, so the low responsivity of this photodiode at 405 nm. However, the tiny photosensitive active area is a measly 1 mm², so it can easily be excluded from the options.

3.3.1.3 Thorlabs FDS100 Si Photodiode

The FDS100 silicon photodiode from Thorlabs is the most balanced of our 3 options. While it still has a small responsivity, at only roughly 0.05 A/W, it has the second best photosensitive active area to be able to get a voltage reading compared to the other options at 13 mm².

3.3.1.4 Photodiode Selection

Below is a table comparing our options for the photodiode in the fluorescence chamber used to measure the baseline reference power. Despite its low responsivity at our required laser wavelength, the large active surface area of the FDS100 silicon photodiode prevails over its competition.

Table 13. Comparison & Selection of Photodiode

Photodiode	Responsivity @ 405 nm	Active Area	Price
Hamamatsu S9195	0.28 A/W	25 mm ²	Unknown
Thorlabs FDS010	~0.05 A/W	1 mm ²	\$52.86
Thorlabs FDS100	~0.075 A/W	13 mm ²	\$16.40

3.3.2 Spectrometer Sensor

The second sensor our group focused on selecting was for the spectrometer. The spectrometer needed to have a sensitive detector. Also, we must consider the fact that the light coming off the diffraction grating will be diffracting at two different angles.

One option was to put two photodiodes (an array) at the location where each diffracted wavelength is. This was doable but leaves very little room for error because of a small misalignment or slightly different wavelength than expected and the detector will detect practically nothing.

An alternative option is to use a CMOS or CCD sensor. They are also arrays but have significantly more pixels like a camera and are easier to align because we do not have to worry about carefully placing the location of a singular photodiode – a CCD is long enough to pick up diffracted light and also incredibly sensitive.

However, during our testing covered in chapter 9, we discovered that the CCD required a precise clock signal to drive, and also a different MCU with a fast enough ADC to properly read the output. Consequently, this forced us to pivot to using a CMOS sensor. A CMOS sensor provides adequate enough resolution and active sensor area. Although not as sensitive as a CCD, it still sufficed for this project.

We compare photodiodes, the ILX511B CCD sensor, and CMOS camera for detecting cyanobacteria in table 12 below.

Table 14. Spectrometer Sensor Options

Spectrometer Sensor Options					
Technology	Features	Pros	Cons	Price	Selected
Photodiode Array	Simple, Low-cost	Easy to integrate and low power consumption	Limited Spectral Resolution, Poor sensitivity for weak signals	~\$16	✗
CCD Sensor (Sony ILX511B)	2048-Pixel Resolution	High Sensitivity, High Dynamic Range	Requires prices sampling rates for the ADC	~\$40	✗
CMOS Camera Attachment	Low power, integrated ADC	Fast readout, integrated processing	Less sensitive to low-light signals, low dynamic range	~\$18	✓

Additionally, by using a CMOS sensor, we were able to easily integrate into our system using a Raspberry Pi 5. The optical alignment was also significantly easier to trouble shoot. The table below shows our CMOS sensor as well as alternative options that we considered:

Table 15. CMOS Sensor Options

CMOS	Sensitivity @650 and 680 nm	Resolution	Sensor Dimensions	Cost
Arducam OV5647 5MP	Sensitive to visible light	5 Megapixels	36 x 36 mm	\$25.00
Edmund Optics IDS-U3	Sensitive to visible light –	2.1 Megapixels	36 x 36 mm	\$350.00

	low light conditions			
Onsemi NOIP3SE5000A-QTI	Can detect visible light – works best with infrared	2 Megapixels	18.8 x 18.8 mm	\$186.00

As shown in the above table, highlighted in yellow, we chose the Arducam OV5647 with 5 MegaPixels. This was the most budget friendly options, and does not require any external software; the Arducam can easily be integrated with the Raspberry Pi 5.

3.4 Controller Unit Technology

Choosing the right controller unit for a project depends on several key factors, including performance requirements, power consumption, cost, and design complexity. We have several different types of Controller Unit Technology to compare and choose to use for our project. When looking for controller unit technology we are choosing to compare Microcontroller units, Field Programmable Gateway Arrays, and Complex Programmable Logic Devices. These controllers all have their pros and cons we need to decide on what will suit our project best based on our skill set. Ultimately, choosing the right controller technology involves balancing computational needs, development complexity, and budget constraints to ensure the best fit for our project.

3.4.1 Microcontroller Unit (MCU)

A microcontroller (MCU) is a small device made to be the brain of an embedded system, integrating essential components such as a processor (CPU), memory, and input/output (I/O) ports into a single chip. It is widely used in embedded system applications, where efficiency, low power consumption, and real-time processing are crucial. Common CPU architecture includes 8-bit, 16-bit, and 32-bit, with higher-bit architecture offering greater processing power and memory addressing capabilities. MCUs utilizes random access memory (RAM) for temporary data storage and read-only memory (ROM) or flash memory to store firmware and program instructions. Most MCUs feature general-purpose I/O (GPIO) pins, enabling them to send and receive signals for interfacing with external components like sensors, displays, and actuators. Internal clocks are included to regulate timing and ensure precise execution of instructions, while pulse-width modulation (PWM) functionality allows for the control of devices such as LEDs and motors. To facilitate communication with other devices, MCUs support multiple protocols, including UART (Universal Asynchronous Receiver-Transmitter), SPI (Serial Peripheral Interface), I²C (Inter-Integrated Circuit), and CAN (Controller Area Network), enabling seamless data exchange in various applications. Additionally, many MCUs integrate Analog-to-Digital Converters (ADC) and Digital-to-Analog Converters (DAC) to process analog signals, making them essential for

applications requiring sensor data interpretation and signal conversion just like we need for our project.

3.4.2 Field Programmable Gateway Array (FPGA)

FPGAs (Field-Programmable Gate Arrays) are highly flexible digital computing devices made up of configurable logic blocks, registers, and digital signal processing (DSP) cores, which can be interconnected to create virtually any digital circuit. Unlike microcontrollers, which execute instructions sequentially, FPGAs operate with true parallelism, allowing multiple computations to run simultaneously, making them ideal for high-speed and real-time applications. They are programmed using Hardware Description Languages (HDL) like Verilog or VHDL, which describe the hardware at varying levels of abstraction, from low-level logic gates to complex digital systems.

One of their key advantages is their extensive input/output (I/O) capabilities and the ability to tailor the internal architecture to fit specific design needs, making them well-suited for tasks requiring unique bit-widths or custom logic. Their reprogram ability allows for easy updates, bug fixes, and modifications without requiring new hardware, and under certain conditions, they can also be optimized for power efficiency.

However, FPGAs have several drawbacks, including higher costs—often tens to hundreds of dollars—compared to microcontrollers, and a more complex development process that requires a deep understanding of digital circuit design and parallel programming. Debugging is also more challenging, as errors in parallel execution can be difficult to isolate and fix. Additionally, FPGAs are generally slower than microcontrollers and ASICs at the same technology node due to the way their internal signals are routed. Another limitation is the lack of built-in Analog-to-Digital (ADC) and Digital-to-Analog (DAC) converters in most FPGAs, requiring external circuitry for analog signal processing, which adds complexity and potential failure points.

Despite these challenges, FPGAs remain a powerful choice for applications requiring high-performance computing, AI acceleration, custom digital processing, and adaptable hardware solutions.

3.4.3 Complex Programmable Logic Devices (CPLD)

A Complex Programmable Logic Device (CPLD) is a programmable logic device that is the in-between of simpler programmable logic devices (PLDs) and more powerful Field-Programmable Gate Arrays (FPGAs). CPLDs are made from macrocells, which are groups of logic gates, flip-flops (which store information), and multiplexers (which control data flow). These macrocells are connected through a fixed routing system that ensures predictable timing and low delays, making CPLDs ideal for tasks that need precise control.

One of the key features of CPLDs is that they use non-volatile memory, meaning they keep their configuration even after being powered off, so they don't need to be reprogrammed when powered up again. This makes them instant-on devices, which are useful in applications where fast startup times are needed. CPLDs are typically programmed using Hardware Description Languages (HDL) like VHDL or Verilog, which describe how the circuit should work. While CPLDs are flexible, they have less processing power than FPGAs, making them better suited for simpler tasks. The fixed connection layout in CPLDs limits their ability to handle very complex or parallel designs, unlike FPGAs. However, CPLDs are cost-effective and efficient for applications like connecting different digital components (glue logic), designing control systems (state machines), adding more input/output ports (I/O expansion), or handling security functions. Their low power consumption, easy programmability, and fast execution make them a good choice for many embedded system projects, though FPGAs may be needed for more complex designs.

Table 16. Comparison of controller unit technologies

Category	MCU	FPGA	CPLD
Architecture	Consists of CPU, RAM, ROM, and I/O peripherals	Consists of programmable logic blocks, interconnects, and memory	Made of macrocells
Speed	High (1Mhz - 1GHz)	Moderate (10KHz - 500MHz, parallel processing)	Moderate (2MHz - 22MHz)
Programming	Uses software (C, C++, Assembly)	Uses Hardware Level Languages (HDL) like VHDL or Verilog	Uses HDL (VHDL, Verilog)
Reconfigurability	Unconfigurable until flashing firmware with new code	Fully reconfigurable during operation	Limited reconfiguration (typically one-time programmable)
Power Consumption	Low (1mW – 1W)	High (50mW – 100W)	Moderate (10mW – 5W)
Complexity	Low Complexity due to it being Easier to program and then integrate into a system	High complexity due to parallelism and HDL	Moderate Complexity needing to use HDL
Cost	Low (\$5 - \$100)	High (\$5 - \$10,000)	Moderate (\$5 - \$200)

Applications	Embedded systems, consumer electronics, and robotics.	High-speed computing, AI acceleration, and video processing	Simple control systems and Glue logic
Examples	ARM Cortex-M, AVR, PIC, ESP32	Xilinx Zynq, Intel Stratix, Lattice iCE40	Xilinx CoolRunner-II, Altera MAX series

For our project we have decided to use an MCU. The cost effectiveness with the low complexity are our main factors in choosing it. It does exactly what we need to do with being very simple to use and program as we need for our project. The other two options were too expensive to make or were too complicated to code in the given time period for our project.

3.4.4 MCU Selection

For our MCU Selection we need to choose something that can work well with powering lower power devices and are still smart enough to make the calculations we need for results from our optical parts. We would like something cheaper that is easy for our CPE to work with and can be easily used and adjusted for each test. For this we have focused on a selection of five ESP32-S3, Raspberry Pi Pico W, STM32H7, Nordic nRF5340, and TI CC3220.

3.4.4.1 ESP32-S3

The ESP32-S3 is a highly capable and cost-efficient microcontroller, ideally suited embedded systems requiring Wi-Fi and Bluetooth connectivity. Its dual-core Xtensa LX7 processor delivers excellent performance, while the 512KB SRAM and optional PSRAM facilitate the handling of complex tasks such as image processing and neural networks. The integrated 2.4GHz Wi-Fi and Bluetooth 5 (BLE + Mesh) make it ideal for wireless communication and industrial monitoring. Additionally, the USB OTG support allows direct interfacing with peripherals like keyboards, storage devices, and cameras, enhancing its versatility. It also comes with 36 GPIO pins. Priced between \$5 and \$15, it offers an affordable alternative to high-end microcontrollers. Its compatibility with C, C++, and Arduino ensures broad support among developers. Whether for AI-driven projects or low-power embedded solutions, the ESP32-S3 stands out as a comprehensive and feature-rich option.

3.4.4.2 Raspberry Pi Pico W

The Raspberry Pi Pico W is an economical and adaptable microcontroller that integrates Wi-Fi capabilities, rendering it an excellent option for low-power projects and wireless applications. It operates on the RP2040 microcontroller, which

includes a dual-core ARM Cortex-M0+ processor, offering an optimal combination of performance and energy efficiency suitable for various tasks. With 2MB of flash memory, it provides ample storage for code and data, and its 26 GPIO pins enable the connection of numerous peripherals and sensors, thereby enhancing its versatility. Its cost-effectiveness priced between \$5 - \$10 makes it particularly well-suited for prototyping and hobbyist endeavors. Additionally, the extensive support for programming in C and C++ ensures accessibility for both novice and seasoned developers. Whether designing a straightforward sensor-based project, a smart home device, or experimenting with wireless communication, the Raspberry Pi Pico W presents a powerful yet budget-friendly solution for a wide array of applications.

3.4.4.3 STM32H7

The STM32H7 is a high-performance microcontroller series built around the ARM Cortex-M7 core. This microcontroller excels in applications requiring substantial processing power, precision, and real-time control. With clock speeds reaching up to 480 MHz and 1MB of RAM, the STM32H7 is particularly suitable for demanding tasks such as audio processing, industrial control systems, and complex signal processing. It features an extensive set of peripherals, including advanced timers, high-speed ADCs, DACs, and communication interfaces like SPI, I2C, and CAN. These attributes make it an excellent choice for applications in robotics, automation, and automotive systems. Although the STM32H7 does not incorporate built-in Wi-Fi or Bluetooth, it offers robust support for external wireless modules, facilitating seamless integration into connected devices. The cost is between \$10 - \$30. The amount of GPIO varies a lot depending on which model one gets. The microcontroller's high level of precision, low latency, and extensive peripheral options render ideal for industrial and real-time embedded applications where performance and reliability are needed. Additionally, its comprehensive ecosystem and compatibility with C/C++ development ensure a smooth experience for both prototyping and large-scale production.

3.4.4.4 Nordic nRF5340

The Nordic nRF5340 is a microcontroller designed for applications focused on Bluetooth Low Energy (BLE) and ultra-low power performance. It features a dual-core ARM Cortex-M33 processor, providing both high-performance and energy efficiency, suitable for battery-powered devices such as wearables, medical sensors, and smart home applications. Its Bluetooth 5.3 support includes enhanced range, data throughput, and mesh networking capabilities for modern wireless communication systems. The device has 512KB of RAM and 1MB flash storage, offering sufficient resources for complex applications. Additionally, its hardware-based security features provide robust encryption and secure communication, which are essential for wearable devices. The nRF5340's low power consumption makes it suitable for applications requiring long battery life without compromising performance. This device costs around \$40 - \$50, making it

more expensive than most. It also has 48 GPIO pins. It is supported by Nordic Semiconductor's SDK and development tools, facilitating rapid development with languages like C/C++. This makes it applicable for creating Bluetooth-connected accessories or low-power wireless sensors, offering a foundation for secure and energy-efficient solutions.

3.4.4.5 TI CC3220

The TI CC3220 is a highly capable microcontroller designed for secure and reliable Wi-Fi connectivity, making it an optimal choice for industrial and enterprise-level applications. Equipped with an ARM Cortex-M4 processor, the CC3220 delivers robust performance for data processing tasks while maintaining low power consumption. Its integrated Wi-Fi capabilities eliminate the need for external modules, thereby simplifying the development process for connected devices. Furthermore, the built-in security features—including hardware-based encryption, secure boot, and key management—guarantee that communications are protected from cyber threats, which is crucial for applications such as smart cities, industrial automation, and health monitoring. The microcontroller also provides 256KB of RAM and 1MB of flash memory, offering sufficient storage for firmware and data. Additionally, TI's support for cloud services through their Simple Link ecosystem allows for seamless integration with cloud platforms for remote device management and data analytics. With its emphasis on security, low power, and dependable Wi-Fi connectivity, the CC3220 represents a reliable solution for developing secure, connected products in demanding environments.

Table 17. Comparison of MCUs

Category	ESP32-S3	Raspberry Pi Pico 2 W	STM32H7	Nordic nRF5340	TI CC3220
CPU	Dual-core Xtensa LX7 at 240MHz	Dual-core ARM Cortex-M0+ at 133MHz	ARM Cortex-M7 at 480MHz	Dual-core ARM Cortex-M33 at 128MHz	ARM Cortex-M4 at 80MHz
Flash Storage	Up to 16MB	2MB	Up to 2MB	1MB	1MB
Ram	512KB	264KB	1MB	512KB	256KB
Bluetooth	Yes, BLE and Bluetooth 5	Yes, Bluetooth 5.2	No	Yes, BLE and Bluetooth 5.3	No
WIFI	Yes 2.4GHz	Yes 2.4GHz	No	No	Yes 2.4GHz
USB OTG	Yes	No	Yes	No	Yes
GPIO pins	36	26	Varies	48	24
Power Consumption	Moderate	Low	High	Very Low	Low

Programming languages	C, C++, Arduino	C, C++	C, C++	C, C++	C, C++
Cost	\$5-\$15	\$5-\$10	\$10-\$30	\$40-\$50	\$40-\$50

The ESP32 is a system all of our group members have worked with before and all have coded at least once. This system is familiar to use and comes with a lot of familiarity with working on it and we have all coded on it before. It also comes with Bluetooth and Wi-Fi which we will need for our software side. It has more than enough storage and memory for what we need and should be perfect for our project's MCU.

3.5 Voltage Regulators

Voltage Regulators are used to control the output voltage of a system regardless of changes from the input voltage or load conditions. They are essential for our project to deliver the required voltage to our MCU and optical components. They can also limit the amount of current going through them, however there are better ways to limit current rather than using a voltage regulator. One of the main things we need to also make sure is that they can handle environmental constraints such as heat buildup since we are using a laser that can get hot. The two main types of voltage regulators are linear and switching regulators. These both have their own advantages and disadvantages. When we are learning about these, we can use them to our advantage in our project, deciding what is best for our project needs.

3.5.1 Linear Regulators

Linear voltage regulators dissipate the excess energy as heat to ensure a stable output voltage that is lower or equal to the input voltage. This works by using a higher input voltage and using a transistor to regulate the voltage then converting all excess energy to heat. Linear voltage regulators are also simple circuits because they don't require extra components because they are built to only put one voltage output. They also are low noise and minimal electromagnetic interference making them good for sensitive electronics. They can also require use of heatsinks and other heat dissipation mechanisms for higher current applications. These regulators are used in systems where clean and stable voltage is vital.

3.5.2 Switching Regulators

Switching voltage Regulators are a more efficient regulator that instead of converting the excess voltage as heat converts the excess voltage into a high frequency switching signal that can be adjusted to the desired output voltage using capacitors and inductors. This means that the switching regulators don't waste as much of the input voltage are more efficient than linear regulators. There are several types of switching regulators which consist of buck regulators (step down voltage regulators), boost regulators (step up voltage regulators), and buck-boost

regulators (step up or step down voltage regulators). With these advantages of higher efficiency and lower heat generation makes them more efficient for high power systems. They do however make more electromagnetic interference.

Table 18. Comparison of Voltage Regulators

Category	Linear Voltage Regulators	Switching Voltage Regulators
Complexity	Easy to implement and easy design	More complex design with capacitors and inductors
Heat Generation	High because all excess energy is converted to heat	Low because all excess energy is converted to a high frequency switching signal
Efficiency	Low (40% - 60%)	High (80%-90%)
Size	Small circuit only requires the regulator most of the time	Moderate with extra components
Noise	Low noise with clean and stable dc output	High noise due to constant high-frequency switching
Cost	Low (\$.10 - \$2)	Moderate (\$1-\$10)
Input-Output Voltage Difference	Input must be higher than output voltage	Can step up, step down, or invert input voltage
Applications	Low power electronics, battery-powered devices	High power electronics, energy-efficient electronics

For our project we have decided to focus on using linear regulators for their simple design and most of our parts are very low power, making heat dissipation not really an issue for our project. Their simple design and effectiveness for low power electronics makes them preferable for our project seeing how we aren't looking for high energy efficiency.

3.6 Signal Amplifiers

Signal Amplifiers are designed to increase the amplitude of the signal without distorting its waveform. These devices are designed to perform certain roles that are based on configuration, purpose, and frequency range. For our project we are focused on current a current response from our optics side into a readable voltage for our MCU. So, we will be looking into a transimpedance amplifier, current feedback amplifier, and logarithmic amplifier. Out of these options we will find the best one that suits our project

3.6.1 Transimpedance Amplifier (TIA)

Transimpedance amplifiers convert input current into a proportional output voltage through an op-amp. The is made by a simple circuit with an op-amp with a feedback resistor coming from negative feedback of a current response. A TIA has

a high input impedance preventing it from loading the signal source. A TIA also has low noise which is needed for weak current response like from most photodiodes. It also a linear response making it ideal for precise measurements for a signal.

3.6.2 Current Feedback Amplifier (CFA)

Current feedback amplifiers are designed to have quick response times and high slew with feedback that affects the input current. They are used mostly for voltage amplification and are configured where the current feedback controls the circuits' performance. These amplifiers have feedback based on the input current and the gain is based on the external feedback network.

3.6.3 Logarithmic Amplifier (Log Amp)

Logarithmic Amplifiers convert their input signals into a logarithmic form that produces a voltage output proportional to the logarithm of the input signal. For current to voltage in this system would be more like a logarithmic function rather than a linear one. This amplifier has access to a wider dynamic range that compresses the input signal's dynamic range. These are also ideal for very weak or very strong signals with how much gain or compression the amplifier can handle.

Table 19. Comparison of Signal Amplifiers

Category	Transimpedance Amplifier	Current Feedback Amplifier	Logarithmic Amplifier
Primary Function	Converts current to proportional voltage	Amplifies signals with current feedback	Converts input signal to a logarithmic output
Input Signal Type	Current input	Voltage input with current feedback	Current or voltage input, but compressed logarithmically
Bandwidth	Typically low to moderate, made for DC to low-frequency signals	High-speed, wide bandwidth, and fast response	High-speed, wide bandwidth, and fast response
Noise Performance	Low for weak current signals	Moderate prioritizes speed over noise	High noise due to logarithmic processing
Complexity	Low needs a feedback resistor mostly	Moderate needs more complex circuitry	High due to needing logarithmic conversion circuitry

Power Consumption	Moderate (10mW – 500mW)	High (50mW – 500mW)	Low (10mW – 200mW)
Cost	Low to moderate (\$1 - \$100)	Low to High (\$2 - \$200)	Low to high (\$5 - \$200)
Key Applications	Photodiodes and sensors	RF and high-speed signal processing	Signal Compression and optical power meters

For our project we have decided to use a TIA for its simple circuitry, low noise, and its bad width made for weaker signals. The TIA is exactly what we need from our signal amplifier to boost our current response into a readable voltage for our MCU. For our test we think we will get around 1mA of optical power.

$$I_1 + \left(\frac{V_{out} - 0}{R_L} \right) = 0 \quad (16)$$

$$V_{out} = -R_L I_1 \quad (17)$$

Based on this formula above we find that we are going to need a 3.3k ohm resistor to get the voltage required to be a digital high on our esp32, which is what we are aiming for in this current test. Below in the figure we made this is a simple circuit example of what our TIA will look like on LTSpice.

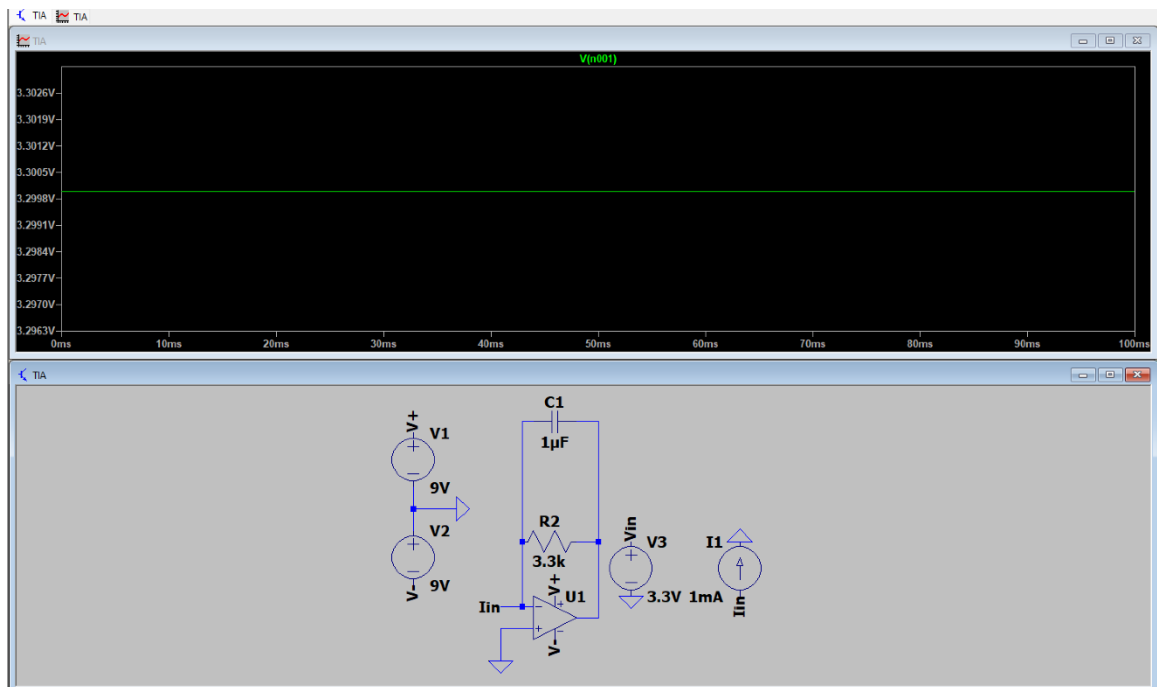


Figure 14. TIA Example

For our project we decided that this circuit was unnecessary for our photodiode because on the datasheet we got provide a voltage readout from the application circuit on a load resistor that we used to calculate the laser power.

3.7 Motors

For our project we are going to need a motor to control a staging platform for our diffraction grating that will give different angular positions for fluorescing the cyanobacteria. For this we need a motor that we can easily control the speed of and can control from our MCU. We need a good level of precision with this while easily being able to stop the motor also. We have a selection of four different motor technologies to choose from for our project. These being the DC motor, Stepper Motor, Servo Motor, and Brushless DC motor. These are all good selections for motors, but we need to find which of these will best fit our project.

3.7.1 DC Motors

DC motors are probably the simplest and easy to use of all motors. The speed of a DC motor is controlled by varying the voltage and direction is controlled by swapping the polarity of the voltage. These motors are efficient but can wear out overtime faster than others. They provide good speed and torque that makes them a great option for our project seeing how we are only making a prototype and not a full product.

3.7.2 Stepper Motors

Stepper Motors offer precise control over the position and speed of the motor making them ideal for need precision. They are controlled by pulse signals from stepper driver or MCU. They are not power efficient and even draw more power than others when idle. They require no feedback system and can hold positions without power. They can be nosier at high speeds and can overheat if used for too long. This precision that stepper motors provide could be extremely useful for our project and make them a great choice.

3.7.3 Servo Motors

Servo Motors require PWM for precise position and speed controls. This also requires an integrated feedback system, which is normally an encoder, that allows the motor to adjust positions. These motors are high efficiency power motors that provide controlled movement and great torque at low speeds. However, they usually have a limited rotation range of 180 °. These motors are also usually easy to integrate into projects. This motor could be good for our project with the high efficiency and precision levels it brings to the table.

3.7.4 Brushless DC Motors

Brushless DC motors are probably the most complex out of all these more and ever require and Electronic Speed Controller (ESC) for proper operation. The speed of the motor is based on PWM signals and will provide smooth control over a wide range of speeds. These motors are extremely high efficiency and don't have brushes as the name implies so they have less wear and tear than other motors. They have probably the best Torque-to-weight ratio and higher speeds compared to the other motors. They also can be quiet for a motor. These would probably be the best option for us if they were not so complex as they are and required more components than a regular motor.

Table 20. Comparison of motor technologies

Category	DC Motor	Stepper Motor	Servo Motor	BLDC
Control	Simple controlled through voltage and polarity	Uses a stepper driver or MCU to send pulse signals	Controlled through a PWM	Requires and ESC
Speed	High and can run continuously	Limited due to the step increments	Limited due to the 180 ° rotation	Very high with excellent range of speeds
Efficiency	Moderate brush friction reduces efficiency	Low and has constant current draw even when idle	High and the system optimizes power use with a feedback system	Very high with no brushes and very low energy loss
Torque	Good at low speeds and lower at higher speeds	High at low speeds but rapidly decreases as it goes faster	High in limited rotation range	High torque-to-weight ratio and good at all speeds
Precision	Low can use external encoders for accuracy	High in precise steps with feedback	Very high built-in feedback for accurate positioning	High precision with external encoders
Feedback System	External encoder required for	Open-loop control and loses position	Closed-loop and uses built-in	Can be open-loop or closed-loop with sensors

	closed-loop control	without feedback	feedback for positioning	
Heat Generation	Moderate due to brush friction	High due to constant power draw	Low to moderate depending on speed and load	Low most effective design for load and speed
Power Consumption	Moderate (1W – 500W)	High draws when idle (5W – 100W)	Moderate (5W – 500W)	Moderate (5W – 500W)
Longevity	Limited because brushes wear out over time	High with no brushes and only mechanical wear over time	High gears will wear over time, but the feedback system will reduce the load	Very high with no brushes and minimal wear and tear
Complexity	Low doesn't need anything but power and polarity	Moderate requires stepper driver	Moderate requires PWM to control the signal and has built-in feedback	High requires an ESC
Holding Torque	No holding torque	High can hold position without power	Moderate depends on the servo	Low to moderate depends on the ESC settings
Maintenance	Moderate Requires maintenance because brushes wear out	Low no brushes but can overheat to lower lifespan	Low mechanical gears can wear down and need replacement	Very low because of brushless design
Noise Level	Moderate due to brushes	High, especially at high speeds	Low due to built in feedback system	Very low because of no brushes
Weight and Size	Small and light	Bigger and heavier	Can be big or small depending on applications	Very small and lightweight for output power
Cost	Low (\$5 - \$100)	Moderate (\$5 - \$200)	Moderate (\$10 - \$500)	High (\$10 - \$1000)

Applications	Toys and fans	CNC machines and 3D printers	RC vehicles and automation	Drones and Electric vehicles
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We don't need a super complex motor for our project, just something that can spin with some precision and uses very minimal power. A DC motor is also very cheap, making it another reason to choose it. We think that a DC motor should be more than enough for the staging platform for our project. Since this is also a prototype, we shouldn't have to worry about it wearing out before the project is over from overuse. This motor can be easily controlled by our MCU and can be precise enough to get accurate reading from our photodiode to evaluate whether there are cyanobacteria.

After doing more research we decided when didn't really need a motor for this project to work and was just add unnecessary complexity that could make the project suffer so we decided that we would be using fresh samples with a little bit of shaking before putting it into the system for testing that would provide similar results to using a motor.

3.8 Software Technology Comparison

The though process for the software selection for this stage primarily focused on ease of use and ease of integration. This firmware is what we will be using for programming the microcontroller.

Table 21. Comparison of software technologies

Software	Features	Pros	Cons
ESP-IDF	Low-level control, optimized for ESP32	High performance, Precise timing	Harder to debug and get started
Arduino IDE	Easy to initialize, fast prototyping	Extensive Libraries	Limited low-level control
MicroPython	Simple Scripting	Extensive Libraries	High overhead, not ideal for real-time processing

We are considering selecting ESP-IDF as the primary development environment because of its ability to provide low-level access to the ESP32's hardware feature. This will allow us to generate accurate time signal needed for the ILX511B CCD sensor if we decide to use it in the future.

3.9 Data Processing

In our development plan we are using the same strategy for data processing for both the photodiode-based design and the potential ILX511B sensor design.

3.9.1 Photodiode Data Processing

In the LUCID design using photodiode, we are using the photodiode to detect the light intensity coming from the UV laser. The analog output from the photodiode will then be fed directly into the ESP32 on its 12-bit ADC pins. This will then allow us to capture the signal intensity with a sufficient resolution for prototyping and initial testing.

The Analog to Digital Converter (ADC) will continuously convert analog signals in the 0V to 3.3V range and translate it to a digital value between 0 and 4095. After collecting full test's worth of digital signals, we will then apply a simple filter like a moving average filter to reduce the electrical noise and smoothen rapid spikes.

The data obtained from the FDS100 photodiode will be used to calculate optical power using Equation 10. After applying a simple filter, the signal will then be processed on the ESP32 to compute the metrics such as UV laser power, peak fluorescent intensity and relative light unit value, and also . These values will then be transmitted to a web app that will be accessed through the user's phone via Wi-Fi.

3.9.2 Data Visualization

The processed data will be sent over Wi-Fi to a website or mobile application. The website will have a python script that will translate the data to a visual to show the light intensity in real-time. For future development with the integration of the FDS100 and Raspberry Pi, we are planning to modify the website to display the full fluorescence spectrum from the test. This will require software firmware like Python with the use of libraries like Matplotlib to provide the basic analysis and graphing. In the future we may also use MATLAB to provide more accurate and visually detailed spectral analysis.

3.10 Communication Protocol for Microcontroller

The communication protocol will need to provide a reliable and efficient way to transfer the data between the system components and the web app. The choice of protocol depends on many different factors like range, data transfer speed, power consumption, and ease of implementation.

3.10.1 Internal Communication

For internal communication, the primary role is to facilitate data transfer between the microcontroller and the peripherals, such as the sensors and ADC. There are three protocols we are considering for this: Serial Peripheral Interface (SPI), Inter-

Integrated Circuit (I2C) and Universal Asynchronous Receiver-Transmitter (UART). Each one of these protocols has been considered for its suitability for the LUCID systems requirements.

SPI is a high-speed, full-duplex protocol that is commonly used for short-distance communications. This method has the benefit of having high data transfer speeds and is simple to implement. For these reasons, there is a wide support of SPI communication for many microcontrollers. However, this functionality requires more pins and has the major issue of range. SPI requires MOSI, MISO, SCK, and CS pins. I2C on the other hand is a two-wire protocol that supports communication between multiple devices on the same bus. It requires fewer pins than SPI (SDA and SCL) and offers a moderate transfer speed. This is slower than SPI and communication is limited by the bus capacitance and the pull-up resistors. UART is a simple, asynchronous protocol for point-to-point communication. This communication protocol also only requires two wires (TX and RX), but it has the lowest data transfer speeds and is very limited by its point-to-point protocol.

After considering these 3 communication methods, we decided to use SPI as our communication protocol to facilitate communication between the microcontroller and the CCD linear sensor. The CCD sensor will capture a lot of data, which will need to be transmitted and received quickly to work effectively. SPI's high transfer speed and full-duplex capabilities make it ideal for transferring large amounts of data from the CCD sensor efficiently. We have also considered using I2C for our external ADC for its simplicity and support across devices. This makes it better suited for connection with the ADC.

3.10.2 External Communication

For the external communication protocol, we will need to be able to have wireless data transfer between the microcontroller and the web/mobile app. There are several protocols that can be used for external communication. Some of these methods are Wi-Fi, Bluetooth, LoRa, ESP-NOW, Zigbee, etc. Each of these protocols are assessed by data transfer speed, range, power consumption, and ease of use.

The first communication we looked into was Wi-Fi. Wi-Fi is a high-speed wireless protocol used in many IoT applications. This method provides high data transfer speed and a long range. These two capabilities make it ideal for real-time data transfer and visualization. Unfortunately, this protocol consumes much more power compared to most other protocols. Bluetooth is a low power protocol that is suitable for short range communication. While it is easy to implement, its low data transfer speed and limited range makes it difficult for the use case of the LUCID system. LoRa is a communication protocol that offers a very long range, with very low power consumption. However, its low data transfer speeds and complex implementation make it less ideal for real-time data transfer. Another protocol is ZigBee. This is a low powered protocol that uses mesh networking for multiple

device support. While this protocol offers low power consumption and mesh networking, it has very slow transfer speeds and limited range.

After evaluating all these options, we saw that Wi-Fi would be the best protocol to use for our use case. With its high transfer speed and long range, it makes it ideal for real-time data visualization and transfer to our web app. Although this option consumes more power than Bluetooth or LoRa, the ESP32-S3's built-in Wi-Fi module greatly simplifies the implementation and reduces the overall system cost.

3.10.3 Protocol Implementation

For the internal communication, the microcontroller will generate the clock signals (ϕ CLK and ϕ ROG) and read the pixel data from the ILX511B through the SPI. The microcontroller will also need to read the digitized data from the ADC via I2C. For external communication, the microcontroller will transmit the processed data to the web app via Wi-Fi using the MQTT protocol. MQTT is a light messaging protocol that ensures efficient data transfer with minimal overhead.

3.11 Programming Languages

The LUCID system has two separate software components. First would be for the ESP32-S3 and the second would be for the mobile/web app development. For both these areas, we have considered the programming language and framework we will be using to accomplish each task. Arduino's software will primarily act as the main unit for capturing and transmitting the incoming data from the LUCID system and the web app will be used to display, store, and analyze the results gathered.

3.11.1 Firmware Development

For firmware development, the Arduino Nano is tasked with handling data acquisition, signal processing, and wireless communication on the LUCID system. For this task we have evaluated three options for programming languages. We researched C/C++ with the Arduino IDE, C/C++ with the ESP-IDF, and MicroPython.

We have found that C/C++ with the Arduino IDE is a popular choice for programming microcontrollers. It offers a beginner friendly environment for C/C++ development and has a large community for support. This environment provides the advantage of having extensive libraries, a low barrier of entry for developers and provides a useful debugging tool. The downside is that it's difficult to implement more complex tasks, due to its low-level control. Alternatively, C/C++ with ESP-IDF, offers great performance with precise control over the hardware. This support also includes real-time operating systems. This allows for greater performance and efficiency but has a much larger learning curve for the development effort. Lastly, we investigated the MicroPython language. It provides a lightweight method to implement python into our microcontrollers. This language

has the benefit of being easy to learn and enables fast development cycles and makes it ideal for prototyping. Unfortunately, this ease of implementation comes at the cost of higher overhead and is not well-suited for real-time processing.

Table 22. Comparison of programming languages

Language	Pros	Cons	Selected
Python (Raspberry Pi)	Beginner-friendly, large community, Extensive libraries, Quick development	Slow, No lower-level control	✓
C/C++ (Arduino IDE)	Beginner-friendly, large community, Extensive libraries, easy debugging	Limited low-level control, less efficient for complex tasks	✓
C/C++ (ESP-IDF)	High performance, Precise hardware control, Suited for Real-Time tasks	Steep learning curve, more development effort	✗
MicroPython	Easy to learn, fast development cycle ideal for prototyping	Higher overhead, slower, not ideal for real-time processing	✗

After evaluating all these options, C/C++ through the Arduino IDE and python for the Raspberry Pis appeared to be the best options for this early development cycle for the firmware. It provides a beginner-friendly interface, large community for support, and a large collection of libraries. These benefits make it an ideal choice for the LUCID system. While it offers less low-level control than the ESP-IDF, it's sufficient for the current requirements and ensures a smoother development process.

3.12 Mobile/Web App Development

For mobile and web app development, the stretch goal is to create a user interface for real-time data visualization, storage, hardware monitoring & hardware control. For this we researched three different languages options: JavaScript for web apps, Flutter for mobile apps, and native development for iOS and Android through Swift and Kotlin.

After looking into JavaScript / React, we found that it is a widely used language for web development that offers high performance, cross-platform compatibility, and a large ecosystem and libraries. React is also a popular framework used when developing dynamic user interfaces. With that being said, it will require more experience in web development with React. This would be our first time using React as a framework. Flutter is a cross-platform framework, specifically for mobile

app development. It uses the Dart programming language. This framework allows developers to create apps for both iOS and Android in a single codebase. With this benefit of cross-platform compatibility, it offers a fast development time and has a large library of UI components. The downside of this language is that it requires knowledge in Dart. Swift for iOS and Kotlin for Android are the native programming languages for each device. With these languages being the native programming language, it provides high performance and full access to platform specific features between the two devices. This boost in performance is a great benefit, but they require separate code bases between iOS and Android. This can cause major issues with compatibility across different devices and greatly increases the development time and effort.

Table 23. Comparison of mobile/web programming languages

Language	Pros	Cons	Selected
JavaScript	High performance, cross-platform compatibility, large ecosystem	Less familiar with JS language, less optimized for mobile	✓
Flutter / Dart	Single codebase for iOS / android, fast development, versatile UI components.	Requires learning Dart, less mature than native frameworks	✗
Swift (iOS) / Kotlin (android)	High performance, full access to platform-specific features.	Separate code bases for iOS / Android, longer development time	✗

After careful consideration between these options, JavaScript with a React framework appeared the most promising for the web app. For the mobile app development, Flutter also seemed like the better option. JavaScript's performance and cross-platform compatibility make it ideal for web development, while Flutter's use of a single code base for both platforms allow for a streamline development. This combination balances performance with ease of use and cross-platform support.

3.13 Mobile Software Research

The mobile software will be a core component of the LUCID system, providing a user-friendly interface for real-time data visualization, system control, and data storage. The app will enable the users to have the ability to monitor the fluorescent intensity of an algae sample in real-time, adjust system settings, and review historical data, all in intuitive and responsive design. To ensure compatibility across platforms and streamline development, the app will most likely be built through Flutter. This is the framework discussed previously in the programming languages section. It allows for the creation of iOS and Android apps using the

same codebase. This section will outline the design, functionality and development strategy for the mobile software.

3.13.1 User Interface (UI) Design

The UI design will primarily focus on intuitiveness, simplicity, and real-time feedback to ensure a smooth user experience. The app will consist of four main screens: Login Screen, Home Screen, Historical Data, Settings Screen. The Login Screen will be the first page the user will navigate to. This is where the user will login and connect to the LUCID device. The Home Screen will serve as the primary interface where the user will be able to navigate to the Historical Data and Setting Screens, view real-time data acquisition, and begin test for samples. A control panel will allow the user to start and stop the data acquisition process. When tests are running, there will be status indicators such as notifications that will provide immediate feedback to the user on the status of the system, such as if it's connected or actively acquiring data.

The Historical Data Screen will be the page that provides access to all the previous diagnostics run and allows the user to review past measurements in either a list or graph format. On this page there will also be an export option that will allow the user to save the data and share it to other devices or apps to conduct further analysis. The Setting Screen will also enable the user to configure the system and calibrate the spectrometer built in the LUCID system. The Setting Screen will also be the place where settings like connection to a wireless network, notifications and sensor calibrations.

3.13.2 Functionality

The mobile app will offer many features to improve the functionality and usability of the device. Real-time data visualization will be achieved through the graphs that will be constructed through the incoming data received from the microcontroller. Users will need to be able to control the system through an intuitive interface, start and stop diagnostics and adjust settings according to their use case and sample. The app will also provide real-time feedback on the system's status, ensuring the user is always informed during the diagnostic process.

The historical data logging functionality will allow the users to store and review the past data from previous samples and provide timestamps of when the samples were taken. With this, the mobile app will need to have access to the CSVs of all the previous diagnostics to transfer the results to another device or application like google for further storage options. Alerts and notifications will inform users of important events, such as the detection of cyanobacteria or urgent system errors. These features collectively will ensure that the user can monitor the system effectively and respond to events promptly.

3.13.3 Development Framework

The mobile app will be developed using Flutter as the main framework and Dart as the programming language. Dart is optimized for UI development and asynchronous programming. For the backend functionality, our team has done research on Firebase to manage all the real-time data storage authentication, and notifications. Firebase also provides scalability and is known to be easy to integrate with Flutter. The communication between the app and the microcontroller will be facilitated using MQTT. This is a lightweight messaging protocol design for IoT applications. MQTT ensures efficient data transfer with low overhead.

3.13.4 Implementation Strategy

The development of the mobile app will follow a structured roadmap to ensure nothing is missed in the execution of the mobile app. The first step will involve creating a mockup through Figma to visualize the app's screen and components. The Flutter also has a library for widgets that we will use to implement the UI. The backend will be set up using Firebase for quick and efficient data storage and notifications, while MQTT will be implemented to facilitate communication between the app and the ESP32-S3.

Responsive data visualization will be achieved using Flutter's charting libraries like "chart_flutter". These libraries will create interactive graphs that will update dynamically as new data is received. System control functionality will be implemented through buttons and controls that will allow the user to start and stop the data collection.

3.14 Data Transfer Research

Efficient and stable data transfer will also be a critical aspect of the accuracy of our results. There needs to be a secure and reliable connection to ensure that the fluorescent data collected by the Raspberry Pi camera was properly transmitting that data. The data transfer process will be divided into two main categories: internal data flow from sensors to the microcontroller and external data flow from the microcontroller to the web application. We are researching the optimal way to approach both these data transfer processes to ensure data integrity, reliability, and security.

3.15 Data Storage Research

Our LUCID system will need to support both temporary and persistent data storage to ensure users can retain the fluorescent data for further analysis & reporting. The data storage strategy will require real-time processing from the photodiodes and the CCD sensor. We are planning to store three data types, the captured

fluorescence digital values in a csv, the timestamp and metadata and the image generated from the csv.

Table 24. Comparison of Data Storage

Storage Type	Pros	Cons
Local Storage (SD, MCU Memory)	Simple integration	Prone to corruption
Cloud Storage (AWS, Firebase)	Scalability, Remote Access	Requires internet
Mobile App Local Storage	Integrated into user device	Device-Dependent

For usage of the LUCID system on the field, the SD card on the Raspberry Pi will store all fluorescent data captured during the test locally. This will ensure that there will be no data lost due to weak or no Wi-Fi availability. The LUCID system will have a hybrid storage approach by utilizing the local SD card storage during field test and use Mobile App storage when transmitted to the device via Wi-Fi.

Chapter 4: Design Constraints & Standards

As with any technological product, there are natural constraints and standards that are followed to properly keep up with modern technology. The development of the L.U.C.I.D. project is constrained due to time, environmental, technological, and money constraints. Additionally, modern technology has well established standards that must be met in order to comply with modern safety standards to prevent and hazards from harming people while using the device.

4.1 Time Constraints

The time constraint is one of the most significant challenges our team will have in the development of the LUCID system. Due to the scheduling of senior design, the first semester is mainly focused on the research and planning of the design and the second semester is where the development begins. This split scheduling between the design period and development period presents an additional challenge because of the semester our team began senior design. Since our team started designing the LUCID system in the spring semester our timeline for development has been compressed further due to its being held in the summer term.

The compressed development timeline has had a great impact on the design process. With only 12 weeks to complete this senior design project, the team has focused on essential features and functionalities, often at the expense of advanced optimizations and innovative features. This prioritization allows for the core objectives of the LUCID system to be met within the given deadline. However, this has meant that some features like advance data analytic tests and predictive paralytics may be pushed to stretch goals and deferred to future iterations during development. Our primary goal is to deliver a functional prototype within our own deadlines so that we will have time to create multiple iterations of our system and further optimize it.

To accomplish our goals and meet our deadlines, the team will heavily rely on smart prototyping techniques such as scaling up our PCB design for our first iteration of the LUCID system. By increasing the size of our first PCB it will help greatly with the debugging process for the PCB design and prevent issues that can arise when surface mount soldering. Another example is the enclosure for our system will simply be 3D printed, which allows for quicker iteration and adjustments without the need for expensive tooling and manufacturing. Similarly, fully designed microcontrollers like the ESP32-S3 have reduced the development time significantly. While this approach will significantly accelerate the prototyping phase, it will also limit the customization and optimization of the system. Despite this trade-off, these quick development techniques are essential for our team to meet its projected timeline and ensure that the system will be functional and ready for testing.

Another one of the key challenges posed by the time constraint will be the limited opportunity for testing and debugging. Comprehensive testing will be essential for identifying and resolving issues. As a result of this, the system and extensions of the system like the mobile app and website may have unresolved issues that will require update to get pushed toward the end of the development cycle. Another example of this will be seen in the reliability of this system. While the core functionality of the fluorescent detection and data transmission will be tested, more nuanced aspects, such as long-term reliability and performance under extreme conditions, will not be fully evaluated. This limitation demonstrates the importance of designing a robust and modular system that is easily modified and updated through development.

Another significant challenge will be in component availability. Delays in sourcing and important items going out of stock during our summer semester will play a major role in the development timeline. Additionally with geopolitical issues rising, components will become more expensive and difficult to transport to the United States. To mitigate this risk, the team has begun to identify alternative suppliers and components for the early design process. For example, if Arduino's ESP32 were to go out of stock or be unavailable, our team has created an additional parts list of alternative parts with similar specifications and manufactures to replace this component. This proactive approach to procurement helps ensure that the LUCID system will remain on track despite potential supply chain disruptions.

As discussed earlier, the compressed schedule will limit the opportunity to iterate our design of the LUCID system. Typically, a project would have multiple design iterations, with each cycle incorporating improvements and optimizations. Unfortunately, due to the 12-week timeline, it only allocates enough time for a few iterations. This increases the risk of design flaws and will result in worse performance. To address this, the team is relying on simple simulations and theoretical models to predict the system's performance and identify potential issues before constructing our first physical iteration. This approach may reduce the risk of some major design flaws; it will not find all the flaws.

The time constraint will be a significant challenge for the LUCID system development, requiring careful planning, constant communication, and proactive risk management. By adopting a modular design, creating effective design decisions and delegating the development across team functions, our team will be able mitigate issues and maintain our tight schedule. While the limited timeline will necessitate trade-offs, focusing on delivering the core functionality of the LUCID system will allow our team to navigate this constraint.

4.2 Economic Constraints

Economic constraints are a critical factor that has influenced the design of the LUCID system. One of our goals is for this project to be cost-effective to ensure

accessibility for researchers and stay within our allocated budget. This constraint has affected every aspect of the design, from component selection to custom manufacturing. Due to the small budget for our design, it requires careful balance between the cost, performance, and functionality. The goal of the LUCID system is to deliver a product that meets the technical requirements without exceeding the allocated budget to ensure the system remains affordable and accessible to the targeted user base.

The economic constraint has had a significant impact on the component selection for this system. To keep costs low, the team has prioritized cost-effective components, such as off the shelf focal lenses, microcontrollers like the ESP32-S3, low-cost photodiodes, and the CCD linear sensor. These components are widely available, well documented, and relatively inexpensive compared to their custom-made variations. This makes them ideal for staying within the budget. Photodiodes alone can cost anywhere between \$0.50 to thousands of dollars. With such a drastic difference between pricing for each one of these components, it has become incredibly important to carefully select comparable parts that stay within the budget. However, this approach involves tradeoffs with performance and functionality. For example, the ESP32-S3 is an affordable microcontroller that has many features that make it versatile for many use cases, but it does not offer the same processing power and precision as other more expensive microcontrollers like the MAVRIC-IIB ATmega128 AVR microcontroller or the Arduino Mega ADK. Similarly, the photodiodes we have selected will have the tradeoff of lower sensitivity and higher noise levels compared to the premium options. Regardless of these lower specifications, the use of cost-effective components ensures that the system remains within our budget while still meeting its core objectives.

To address the economic constraint, the team has employed several strategies. Each component has been evaluated through a cost to performance analysis to ensure the best value for money. This process involves finding the exact requirements needed for the LUCID system to operate, then going through components and comparing their cost, performance, and reliability. By directly outlining our use case for each component, the team can carefully select components that fit the performance specifications. For example, the ESP32-S3 was chosen over more expensive microcontrollers because it provides sufficient performance for the system's needs at a fraction of the cost. Selecting a microcontroller with the strongest performance specifications may be important for operating projects that require a lot of processing power, but it will not be necessary for the LUCID system. Also purchasing components in bulk has also been beneficial to reduce the price per unit. This approach not only lowers the overall cost but also ensures that the team has spare components for testing and replacement for repair.

The economic constraint has required careful budgeting and diligent cost analysis to deliver a functional and affordable system. By prioritizing cost-effective components, simplifying functionality, and outlining the design requirements, the

team can stay within budget and meet the LUCID systems performance requirements. While the economic constraint has necessitated the need to make decisions between performance and functionality, the primary focus has been to ensure the LUCID system remained accessible to its target user.

4.3 Environmental Constraints

As the LUCID system is designed for outdoor use, the environment has been an important constraint to address. The system must be able to withstand conditions like humidity, dust, and temperature fluctuations while producing reliable results and performance. This constraint has mainly influenced the design of the enclosure and some of the component selection to ensure the system remains durable in real-world conditions. One of the core design goals is to create a device that will be able to operate and provide reliable results in diverse environmental conditions, like humid lakeshores and sunny beaches.

Thermal management has been a key consideration in the design of the LUCID system. As the temperature fluctuates there can be a significant impact on the performance of the circuitry inside the enclosure. With consistent extreme temperatures, issues like thermal throttling can occur and ruin the performance or even invalidate the results of a test. Thermal throttling occurs when a component like the CPU reduces performance to prevent overheating and potential damage when reaching beyond its temperature threshold. The system will include components like heat sinks, and thermal pads to prevent overheating in high-temperature environments. These heat dinks are going to be used to dissipate heat from critical components, to sure that the system remains safe from its maximum thermal range. Also, selecting low power components that do not produce a vast amount of heat has the potential to greatly improve the overall thermal management of the system. An example of one of these components is the ESP32-S3. This microcontroller does not produce much heat underload and has a temperature range of -40°C to 85°C, making it suitable even under extreme conditions.

4.4 Additional Design Constraints

In addition to the primary functional requirements of the system, several secondary but critical design constraints must be considered to ensure reliable operation in real-world environments. These additional design constraints span power compatibility, timing precision, environmental protection, and mechanical limitations. Each of these aspects plays a key role in the portability, durability, and accuracy of the device, especially given its intended use in outdoor field conditions for cyanobacteria detection. The following subsections outline these considerations in greater detail.

4.4.1 Power Constraints

The ESP32-S3 operates on a voltage of 3.3V with the laser and FDS100, but an external component like the Raspberry Pi operates at 5V. This requires the implementation of voltage regulators to ensure safe and reliable performance between each device.

The system also must be battery-powered to allow for portability and field use. This necessitates careful placement and selection of batteries to prevent thermal issues.

4.4.2 Timing Constraints

The photodiode design will have minimal timing constraints, but the future CCD implementation will require hardware PWM and dedicated timers from the microcontroller for signal generation. The ILX511B CCD sensor requires a 2MHz ϕ CLK clock and accurate synchronization with the ROG (Reset Gate) signal. T

4.4.3 Physical & More Environmental Constraints

The device will be deployed outdoors near bodies of water for cyanobacteria testing. This requires the enclosure to be IP-rated as defined by the International Electrotechnical Commission (IEC) to protect against moisture, dust, and dynamic temperatures.

Another environmental constraint for the project is how the amount of cyanobacteria, and what is considered dangerous by the World Health Organization, changes from country to country, state to state, county to county, and per species of cyanobacteria. This makes it difficult to pinpoint how much cyanobacteria is dangerous and it is difficult to consider all of these factors in the timespan that we have for this project. This constrains our project to fairly loose regulations and constrains our project to more or less what is in Orlando, Florida.

4.4.4 Mechanical Constraints

- **PCB Form Factor:** The PCB will be designed to fit within a portable enclosure.
- **Thermal:** Wireless modules like the ESP32-S3 generate heat during Wi-Fi transmission. The enclosure must be able to allow for thermal dissipation without introducing condensation or moisture.

4.5 Standards & Compliance

To ensure the LUCID system is safe, reliable, and compatible with industry practices, adherence to established standards and regulations is essential throughout the design and implementation process. These standards cover a broad range of considerations, including laser safety, optical component handling, electrical compliance, wireless communication protocols, and signal integrity. Following these guidelines not only promotes user safety and environmental responsibility but also enhances device performance, interoperability, and

durability in field conditions. The following subsections detail the specific standards and compliance measures integrated into the system's design.

4.5.1 Laser Safety Standards and Regulations

There are a variety of types of lasers depending on power and how much harm they can cause to the human eye and skin. As a result, the danger of a laser is divided into several classes: Class 1, Class 2, Class 2M, Class 3R, Class 3B, and Class 4.⁸

Class 1 Lasers are safe to operate under all conditions and pose no risk to the human skin or eye. They typically operate at powers less than 1 mW. Laser printers and the lasers found in CD players are examples of class 1 lasers. Class 1 lasers also include high power lasers that are within an enclosure and cannot be opened without the laser being switched off.⁸

Class 2 lasers are safe: they do not damage or burn the skin in any way, and also if they make contact with the human eye – the human reaction time is fast enough (~0.25 seconds) to not cause any permanent damage. This class of lasers only applies to visible light lasers which range from 400-700 nm and typically are on the order of 1-5 mW.⁸

Class 3R lasers are generally safe when handled carefully. They are limited to 5 mW of power and should not be viewed directly class 3B laser, on the other hand, are hazardous if viewed directly even if for a very short amount of time. They are limited to 500 mW (0.5 W) and change range from about 400 nm to the far infrared region of light.⁸

Class 4 lasers are extremely dangerous and pose great risk for eye and skin damage. They are, in the most general sense, any laser more powerful than class 3R laser, so any power above 0.5 W. They must usually also be operated with an experienced operator or laser technician in the same room. This policy may vary from location to location.⁸

In the case of this project, our UV laser is a Class 3B laser. For the utmost precautions, safety goggles will be used in order to decrease the damage the beam will cause should it come into direct contact with anybody's eye. Another precaution that will be put in place when aligning our laser system is not to wear any reflective accessories such as watches, wristbands, rings, etc. Any stray beam of light may reflect off the surface of such accessories and caused serious eye damage, especially since 405 nm laser high energy and a 20 mW UV laser will cause retinal damage. The device will also be enclosed and will be appropriately labeled to reduce the chances of the device being accidentally opened while the laser is on. Perhaps, even an LED feature may be added to indicate when the laser is on for further enhanced safety.⁸

4.5.2 Lens Handling Standards

The physical optics of our system such as the beam splitter, focusing lens, collimating lens, collection lens, and diffraction grating are all sensitive pieces of optical equipment. Most, if not all of these optics, have some sort of coating on them to enhance performance, such as anti-reflection coatings. Therefore, there are standards in place for proper handling and cleaning of all of these pieces of optical equipment.

First and foremost, all reflective, diffractive, transmissive, and refractive optics must be handled by the edges.²⁴ This ensures no fingerprints damage the coating on the lens and also prevents a loss in transmissive, reflection, or diffraction efficiencies. It therefore follows that no types of these optical equipment shall be stored faced down under any circumstance, unless face down in a soft foam or similar type of storage to ensure no smudging or damage of the coating ensues. Shall a piece of optical equipment become contaminated, with a fingerprint or dust, there are cleaning standards. Wiping an optic with dust or some other debris will ruin the coating – thus it is recommended to use a blow duster. For more severe contaminations, such as with the coating, it may have to be sent back to the manufacturer for repair if it is at all offered. For more stubborn pieces of debris on uncoated items, it is recommended to use optical grade tissues such as Kim-Tec wipes. These tissues are soft on glass and are made specifically for the purpose of cleaning optical glass - light pressure should be applied. Consequently, all optics when not used for experiments or in the actual device shall be stored properly in a tight container restricting its movement with a piece of foam or optical grade tissue. Ideally, storing it in the same packaging the manufacturer sent the item in will work well enough.

4.5.3 Electrical & Safety Standards

- **Restriction of Hazardous Substances Compliance:** All components must comply with the RoHS standards to reduce environmental harm.
- **ESD Protection:**

4.5.4 Wireless Standards

The wireless communication component of the LUCID system is enabled through the ESP32-S3, which supports the IEEE 802.11 b/g/n Wi-Fi standards in 2.4 GHz ISM band. These standards dictate how the device will establish connections, handle data packets, ensure reliability and manage the error correction across wireless networks. By adhering to the standards, the ESP32 will ensure interoperability with common routers, mobile hotspots, and mobile devices in both the field and laboratory [28].

A key element of the 802.11 standard is media access control (MAC) protocol. This protocol facilitates how the ESP32 shares through a wireless medium with other devices. In an environment with many Wi-Fi devices, it is important to have an

efficient handling of contention and transmission to ensure that the LUCID systems data uploads are not lost or incur significant delays. Additionally, the 802.11 standard incorporates adaptive data rate modulation (e.g., BPSK, QPSK, OFDM) to maintain reliable links even when the signal strength varies, such as when using the system in outdoor settings or near metal obstructions.

The ESP32 also supports advanced security protocols, including WPA2-PSK and WPA3. These protocols ensure that the sensor data is securely encrypted before being transmitted to Firebase. For example, AES encryption and handshake authentication mechanisms prevent unauthorized interception or injection of malicious data. From a design perspective, the antenna layout and the RF trace design must follow the guidelines in the Espressif hardware design manual to ensure signal strength, efficiency, and compliance with regional RF emission standards [29].

4.5.5 Signal Integrity Standards

Signal integrity refers to the quality of the electrical signals as they propagate through the system's PCB traces and cables. In high-speed designs and analog-sensitive designs like L.U.C.I.D., poor signal integrity can result in inaccurate sensor readings, timing mismatches, and transmission failures. This can be incredibly important with sampling like the ILX511 Linear CCD Sensor. The timing signals like ROG, SHSW, CLK and analog voltages must stay stable. And clean across thousands of clock cycles per scan.

One important concept in the signal's integrity is impedance matching. To ensure that the signal traces are routed with a consistent impedance. Without this, reflections at the load or source can corrupt the signal, cause distortion or multiple edges per clock pulse. Other practices like using shorts, direct traces, and integrating decoupling capacitors to the power pins can maintain clean ground planes. All these techniques can reduce the EMI coupling and voltage bouncing [30].

For analog signals like the ILX511 output, it's important to isolate the analog section of the PCB from noisy digital lines. Guard rings, analog ground zones, and low-pass filter circuits are all useful techniques to help preserve the signals fidelity before it reaches the ADC on the ESP32. Additionally, the ADC performance can be verified using differential non-linearity and integral non-linearity measurements to ensure minimal distortion across the input voltage [31]. These practices follow standard PCB and signal integrity design guidelines used in sensors and RF systems.

4.6 PCB Standards

Printed Circuit Boards (PCBs) are the bodies of electronic devices, and adhering to industry standards ensures their reliability, quality, and safety. Various organizations or companies set PCB standards to regulate design, manufacturing, and testing processes. This ensures that our PCBs can be safely made while being easily used in industry and provide a good general standard for the entire industry.

These standards also make it possible to reduce our failure rates of circuits and increase our performance.

Key PCB Standards:

- 1) **IPC Standards:** The IPC (Association Connecting Electronics Industries) is the primary organization for PCB standards and the Key IPC standards include:
 - a. **IPC-2221:** General design standards for PCBs.
 - b. **IPC-A-600:** Acceptability of printed boards.
 - c. **IPC-6012:** Qualification and performance for rigid PCBs.
 - d. **IPC-7351:** Standard for surface mount component footprints.
- 2) **RoHS and REACH:**
 - a. **RoHS (Restriction of Hazardous Substances)** limits the use of hazardous materials.
 - b. **REACH (Registration, Evaluation, Authorization, and Restriction of Chemicals)** regulates chemical substances in electronic products.
- 3) **ISO Standards:** The International Organization for Standardization provides guidelines for quality management systems in PCB manufacturing:
 - a. **ISO 9001:** Ensures quality control in production processes.
 - b. **ISO 14001:** Focuses on environmental management in PCB production.
- 4) **UL Standards:** Underwriters Laboratories (UL) ensures safety and flammability resistance of PCBs:
 - a. **UL 796:** Safety standards for printed wiring boards.
 - b. **UL 94:** Flammability rating for PCB materials.

Importance of PCB Standards:

- **High Reliability:** Reduces failure rates and enhances performance.
- **Global Compliance:** Facilitates international trade and compatibility.
- **Environmental Responsibility:** Ensures safe disposal and minimal environmental impact.
- **Cost Efficiency:** Reduces design errors and manufacturing defects.

Following these standards help manufacturers produce durable, safe, and efficient PCBs suitable for various industries, from consumer electronics to Industrial applications. The standards as a whole make PCBs better in every possible way

should be continually improved to keep the industry being better and better without trying to take shortcuts.

4.7 Soldering Standards

Soldering is an important process in electronics manufacturing, ensuring secure and stable connections between components and printed circuit boards (PCBs). The solder needs to be able to safely connect a component to the PCB with solder paste or solder wire without worrying about solder bridges or disconnects. Adhering to industry soldering standards guarantees consistency, durability, and safety in industry and consumer soldering.

Key Soldering Standards:

- 1) **IPC Standards:** The IPC (Association Connecting Electronics Industries) Makes several key soldering standards:
 - a. **IPC J-STD-001:** Requirements for soldered electrical and electronic assemblies, focusing on materials, methods, and acceptance criteria.
 - b. **IPC-A-610:** Acceptability of electronic assemblies, covering workmanship standards for solder joints.
 - c. **IPC-7711/21:** Rules for repair, rework, and modification of electronic assemblies.
 - d. **IPC-7530:** Rules for temperature profiling in reflow soldering processes.
- 2) **RoHS and REACH:**
 - a. **RoHS (Restriction of Hazardous Substances):** Limits the use of hazardous materials in solder.
 - b. **REACH (Registration, Evaluation, Authorization, and Restriction of Chemicals):** Regulates chemicals used in soldering materials.
- 3) **ISO Standards:**
 - a. **ISO 9001:** Ensures quality management in soldering processes.
 - b. **ISO 14001:** Focuses on environmental impact and sustainable soldering practices.

Importance of Soldering Standards:

- **Consistent Quality:** Standardized procedures lead to repeatable, high-quality results.
- **Environmental and Safety Compliance:** Meets regulations for sustainable soldering practices.

- **Reliable Connections:** Reduces defects such as solder bridges, cold joints, and voids.
- **Enhanced Durability:** Ensures long-lasting electronic assemblies by preventing premature failures.

Following these standards helps manufacturers produce high-performance, safe, and environmentally responsible electronic assemblies, crucial for every electrical industry. These standards are needed to make sure that all PCBs are going to work properly constantly without breaking any of the constraints set by these standards. To keep our PCB working for as long as possible without risk of failure.

4.8 USB Standards

Universal Serial Bus (USB) standards are designed specifications for connectors, cables, and the protocols used for communication, connection, and power supply between computers and electronic devices. Adhering to USB standards guarantees interoperability speed, and reliability.

Key USB Standards

1. USB Versions:

- **USB 1.1:** Introduced low (1.5 Mbps) and full-speed (12 Mbps) data rates.
- **USB 2.0:** Enhanced speed up to 480 Mbps (Hi-Speed USB).
- **USB 3.0/3.1/3.2:** Introduced SuperSpeed USB (5 Gbps, 10 Gbps, and up to 20 Gbps).
- **USB4:** Offers data transfer rates up to 40 Gbps, leveraging Thunderbolt 3 compatibility.

2. Connector Types:

- **USB Type-A:** Standard rectangular connector.
- **USB Type-B:** Square-shaped connector, typically for printers and peripherals.
- **Mini-USB and Micro-USB:** Smaller connectors for portable devices.
- **USB Type-C:** Reversible connector supporting high-speed data, video, and power delivery.

3. Power Delivery (PD) Standards:

- **USB PD 2.0/3.0:** Provides flexible, higher power levels (up to 100W) for charging a wide range of devices.
- **USB PD 3.1:** Extends power delivery up to 240W for larger electronics.

4. Backward Compatibility:

- USB standards emphasize backward compatibility, ensuring that newer devices can communicate with older standards where possible.

Importance of USB Standards

Adhering to USB standards ensures:

- **Universal Compatibility:** Enables seamless communication between a wide variety of devices.
- **Optimal Performance:** Guarantees data transfer speeds and power delivery rates meet industry expectations.
- **User Safety:** Ensures safe voltage and current levels to prevent device damage.
- **Product Reliability:** Certified devices are tested for durability and interoperability.

Following these standards ensures that USB-connected devices perform reliably, safely, and efficiently across industries ranging from consumer electronics to industrial applications. USB standards are essential for the seamless operation, safety, and performance of contemporary electronic devices. By adhering to these established specifications, manufacturers and developers can produce products that are universally compatible, have optimal performance, safe, and reliable. This adherence supports the rapid advancement and innovation of technology across various sectors.

Chapter 5: Comparison of ChatGPT and other Generative AI

The use of artificial intelligence (AI) is at an all-time high and consequently plays an important role in modern-day technological development. This chapter focuses on the use of ChatGPT, its benefits as well as its drawbacks in relation to the L.U.C.I.D. project.

5.1 Generative AI

Generative AI is a new and explosive technology based on large datasets to generate new content such as images, text, music, and videos. Generative AI analyzes patterns in these datasets to answer questions and summarize information quickly. Generative AI creates an output that aims to mimic humans' creativity and reasoning based on the operators' inputs.

The biggest con of generative AI is the ethical concerns that arise with its continued use. Some educational institutions consider the use of LLMs and other generative AI platforms to be cheating and unethical. As scientists we don't want to spread misinformation and bias unnecessarily. But as with any research, we must also verify that where our information is coming from is a reliable source also. In this section, we weigh the pros and cons of the most popular LLMs.

5.2 ChatGPT (OpenAI)

OpenAI, an artificial intelligence research company, released ChatGPT in November of 2022 [15]. The GPT, which stands for Generative Pre-trained Transformer, is a free to use large language model trained to produce text. ChatGPT's user interface is very simple, you can quickly and easily start a chat with the LLM by initiating interaction; or you can select one of the many other generative GPTs created by the ChatGPT team and the community. Its optimization for dialogue is enabled by a method that requires human input and preference to guide the LLM toward a desired outcome.

Large datasets, written by humans, from the internet are used to train ChatGPT, giving it a very human like tone to its responses. ChatGPT is not connected to the internet and can produce incorrect answers and the datasets it learns from are very limited after 2021 as stated on their website [16]. The training of the language model is ongoing as it is fed the content that you also create making it good at iterative problem-solving. OpenAI has also created an AI text-to-art generator named Dall-E. Fitting for the abstract nature of some of its outputs.

A limiting factor of ChatGPT is that the language model cannot cite its sources accurately for the responses it generates. The LLM can produce some sources

that are real and relevant, but the outcome is never guaranteed. Not only can it not properly cite sources, but it also goes a step further and can generate responses that are not true. This severely limits the amount of trust we can place in using an LLM like ChatGPT without having to do extensive fact checking.

As demonstrated in Prompt 1 (Appendix B), ChatGPT's latest model has the capability of interpreting the context of a question and searches through to find information on the given request and provide a response. In the example above, ChatGPT was able to recognize the ILX511B CCD Linear Sensor and evaluate that the microcontroller would need to have clock speed of above 2 MHz, which is the correct specification from what's found in the ILX511B datasheet. It then was able to find boards that met these requirements, such as the ESP32-S3 for the low-price range and NVIDIA Jetson Nano for the high-end quickly.

One of the most significant drawbacks of LLMs such as ChatGPT has been their tendency to assume or imagine specifications. For example, while searching for more niche microcontrollers to fit our specific use case, ChatGPT began to assume the resolutions and sampling speed of the ADC and assumed that is only had a 10-bit resolution, when in the datasheet it stated that the MCU had a 16-bit resolution ADC. Another issue is that ChatGPT lacks current data like price fluctuations and doesn't account for shipping costs so often the pricing on these boards was incorrect.

ChatGPT is a great troubleshooting tool. For instance, if you are having trouble coding something in Python or MATLAB, ChatGPT is excellent at identifying issues based on errors and can even help to build a basis for your code given a few direct commands. However, when it comes to optical design in Zemax, ChatGPT falls short. It has the basic knowledge of the capabilities of the optics studio, as seen in prompt 2. But in prompt 3, when asked how to model a beamsplitter, the instructions that ChatGPT returned were incorrect and incomplete.

5.3 Gemini (Google)

Gemini is afforded the luxury of access to the world's most popular search engine: Google Search. This luxury provides access to real-time information and greater factual accuracy, unlike ChatGPT. Gemini also has extensive versatility with its multimodal capabilities and can create a wide range of data types like text, code, images, audio, PDFs, and videos [17]. This multimodal capability allows Gemini to generate a more complete answer to fit the context of the users' input.

ChatGPT does have the edge on Gemini in the conversational category, with Gemini being less refined and not as creative with its output responses. Google intentionally limits the conversational context so in long-form conversation the context must be re-established. Gemini also tends to over-explain simple topics, having difficulty adapting to a natural conversational flow. Gemini is still evolving

compared to some of its competitors as its relatively new still and doesn't have the same level of adoption in industry practice [17].

5.4 Claude (Anthropic)

Designed with AI safety in mind, Claude focuses strongly on ethics and safety and is less likely to generate harmful or biased content. Claude can handle extended conversations better than many LLMs, making it useful for summarization, research, legal, and technical documentation. Its response generation is balanced, clear, logical, and well-reasoned emulating academic and professional settings.

Its cons include aspects like less creativity, limited real-world integrations, and speed. Claude is not as strong in imaginative storytelling and can feel unnecessarily rigid when asked for jokes or brainstorming. Claude also doesn't possess the support of other LLMs in that it isn't connected to the internet or support plugins. Claude is a little slow on the upswing, response generation can be significantly slower for more complex and larger queries.

5.5 Llama (Meta)

Meta released its latest model LLaMA 3.3 in December 2024 featuring the ability to be able to process both images and text for a thorough analysis and subsequent response generation. This includes interpreting charts, maps, and translating text in an image. This new model improves on the old with an optimized transformer architecture. LLaMA 3 is open-source and flexible for the customization needs of a business.

LLaMA does require some technical prowess to utilize it. This LLM is not casual user-friendly, and requires some extensive setup, tuning, and hosting. This is in part due to another con; LLaMA lacks consumer facing deployment and isn't widely deployed like other AI tools. And like other underutilized LLMs there is no real-time internet connection, so information is limited to pre-trained knowledge and requires external tools for updates.

5.6 Comparison

Table 25. Comparison of Ai Services

	ChatGPT	Gemini	Claude	Llama
Pros	• Conversational abilities • Iterative problem solving	• Real-time information • Multimodal Capabilities	• Prioritizes safety and ethics • Long-context documents • Summarization and	• Open-source and customizable • Reasoning abilities structure

	Integrates with other tools	Factual Accuracy	research tasks	Competitive
Cons	- Lacks real-time internet access - Occasional misinformation	- Inconsistent in long-form - Not as creative - Less refined casual conversation	- Struggles with highly creative content - Limited real-world integrations - Too cautious	- Requires technical expertise - Not widely used in consumer-facing applications - Less polished in general conversation

5.7 Using AI for component research and part selection

When researching electronic components, using AI can be an incredibly efficient and powerful tool to streamline the process. AI can help you quickly find and compare similar parts across various manufacturers and distributors, saving you time during the design and research phase.

A highly effective approach is to begin by telling the AI about your project or providing details about what you're building, your system requirements, and any constraints you're working with (such as voltage ranges, size limitations, or budget). This context allows the AI to generate more relevant and targeted suggestions for the types of components you might need. You can start by asking straightforward prompts, such as which components are recommended for a specific function, or by requesting a list of alternatives for a given part number. Beyond simple recommendations, you can also ask the AI to generate detailed comparisons between components. These comparisons can be provided in a variety of formats, such as paragraphs summarizing pros and cons, organized bullet points highlighting key features, or structured tables that lay out the full specifications, such as voltage ratings, tolerances, power consumption, operating temperatures, and more for side-by-side for easier evaluation. Once the AI provides a list of suggested components or comparisons, it's important to take the next step yourself: verifying the information.

Always cross-reference the AI's recommendations with the official datasheets of the products and reputable sources. This helps ensure that the data is accurate and up to date. Additionally, you should carefully assess how well each suggested component fits within your specific design requirements and system architecture. Even a component with excellent specifications may have subtle incompatibilities

with your project that the AI might not fully account for. Doing this follow-up research strengthens the reliability of your design and helps you make more informed engineering decisions.

Chapter 6: Hardware Design

The L.U.C.I.D. system will be an entirely enclosed device consisting mainly of two parts: the fluorescence chamber and the spectrometer. Each optical component has its own housing, and there are appropriate spaces for the electronics to be housed. The housing does not transmit any potentially harmful light radiation, such as stray light rays that may come from the UV laser. For further safety, an LED is included on top, indicating the activation of the laser diode.

6.1 Overall Housing Schematic

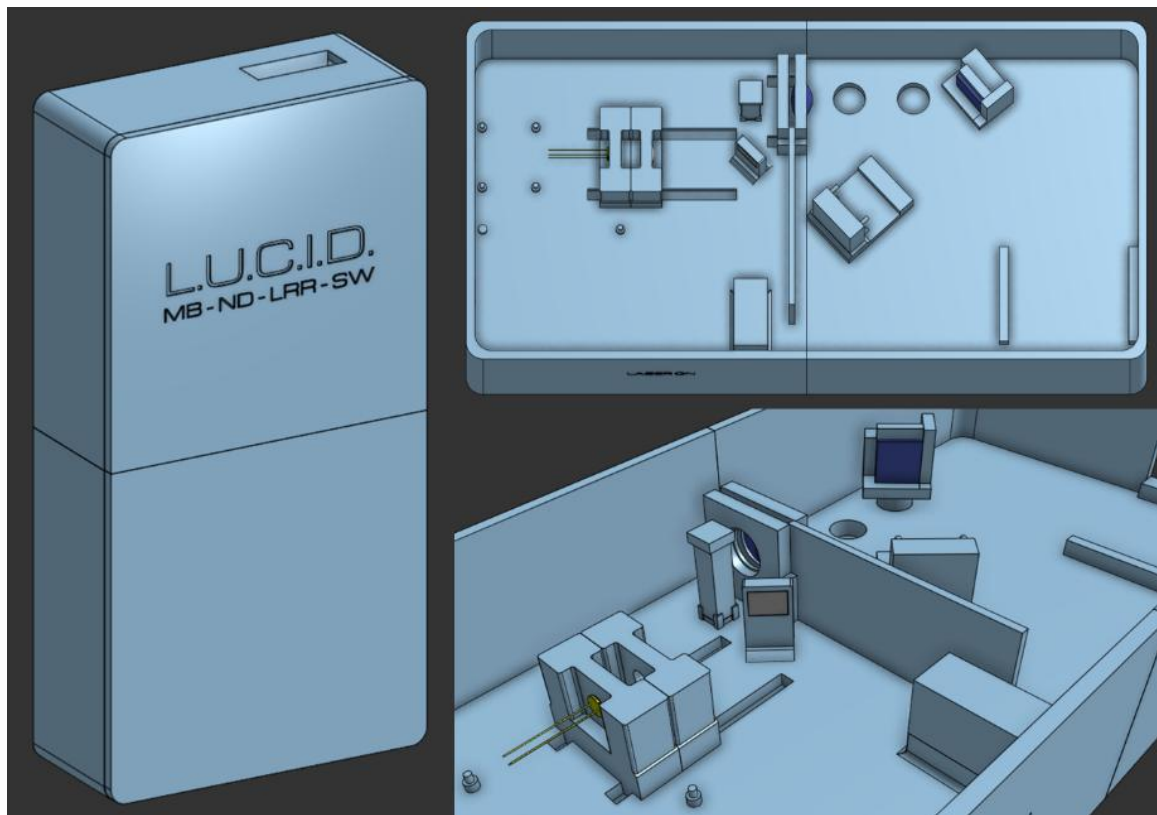


Figure 15. L.U.C.I.D. 3D CAD Modeling

As shown in the above figure, the L.U.C.I.D. device will be entirely enclosed. First for laser safety and secondly for more accurate spectroscopy. The housing was 3D printed entirely from PLA. We considered other materials like wood and aluminum as they are both cheap materials, but the overall flexibility to design the system the way we desired and then have multiple iterations led us to 3D printing everything.

The housing was printed in 4 main pieces, 2 bases and 2 lids. The housing was designed this way in order to accommodate for the printable volume of the 3D printer. Each optical mount was meticulously measured and toleranced to have

secure installation of each of the optics and also some degree of flexibility for alignment. Space on the interior of the housing was also allocated to also mount our PCBs and Raspberry Pi, but the optical hardware takes up most of the space inside of the housing.

6.2 Optical Hardware

The hardware for the optical components consists of several lenses, a beam splitter, sensors, and a diffraction grating. All of the components will be mounted, and their respective mounts will be 3D printed. The CAD work for these 3D printed mounts is still being designed. To understand the reasonings behind the 3D printed layouts, it is important to first simulate the optics to know the approximate dimensions of the L.U.C.I.D. system.

6.2.1 Fluorescence Chamber Dimensions and Zemax Simulation

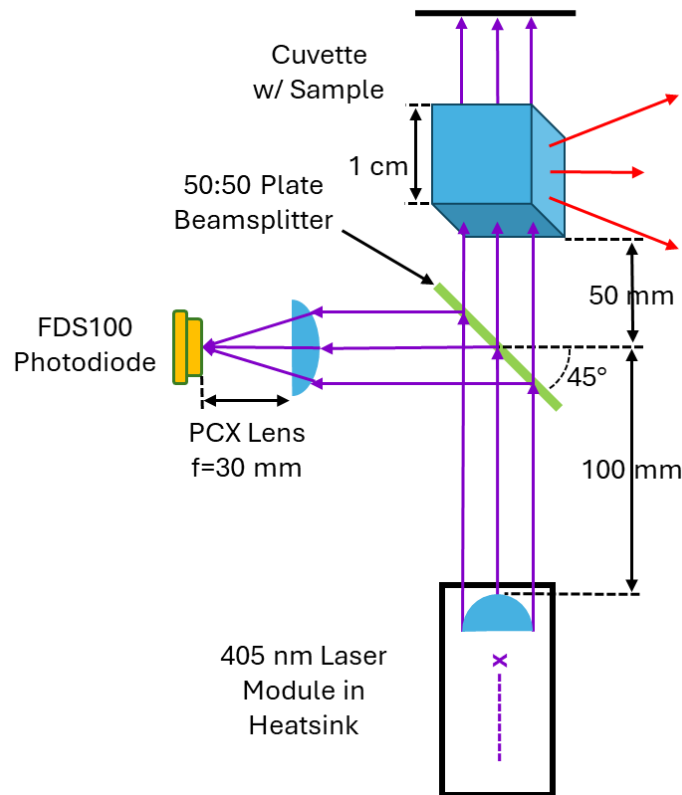


Figure 16. Fluorescence Chamber Schematic

The above figure displays the dimensions of the fluorescence chamber hardware. The measurements of the following are included in the schematic: Laser to beamsplitter, beamsplitter to focusing lens, focusing lens to photodiode, beamsplitter to cuvette, necessary angle of rotation from the incident beam in order for the beamsplitter to function at 50:50 split ratio.

Then, we used Zemax to simulate the calculated values in the design of our plano-convex focusing lens in the figures below. Using an incident beam of 3 mm in diameter, chosen because it is smaller than the width of the FDS100 photodiode, the simulation of the desired lens was performed using some common values obtained from the specifications sheet of some of the lenses available to consumers. In figure 15 it shows the prescription of the plano-convex lens design. Specifically, we used the radius of curvature that we calculated using the Lensmaker formula in Chapter 3, and the thickness and diameter of the lens taken from the specification sheets.

	Surface Type	Comment	Radius	Thickness	Material	Coating	Clear Semi-Dia
0	OBJECT Standard ▾		Infinity	Infinity			0.000
1	STOP Standard ▾		Infinity	10.000			1.500
2	(aper) Standard ▾	front of lens	15.600	3.200	N-BK7		6.350 U
3	Standard ▾	back of lens	Infinity	27.279 V			1.395
4	IMAGE Standard ▾		Infinity	-			1.342E-03

Figure 17. Lens design prescription

Figure 16 shows the tight focus that we can achieve with this prescription.

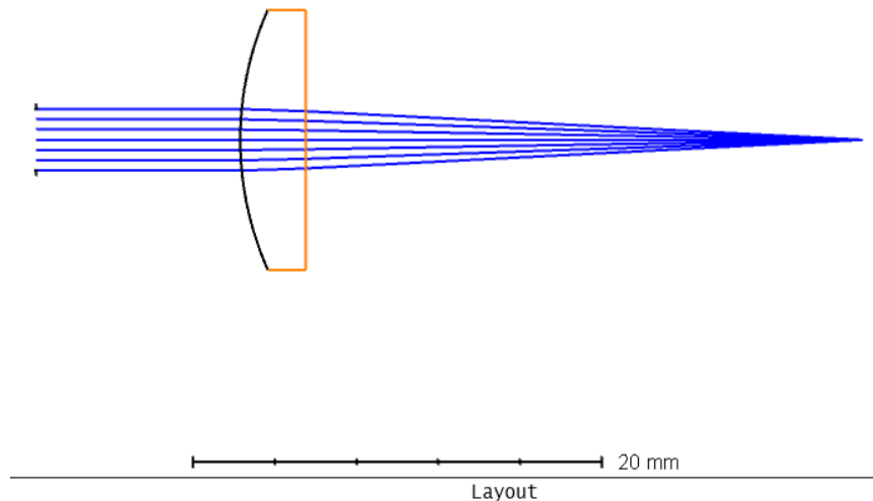


Figure 18. Plano-convex lens simulation

Below, in figure 17 we examined the spot size that the lens created to determine if this design would focus onto a space smaller than the active area of our photodiode. The geometric radius of the gaussian spot produced by our ideal lens is $1.34 \mu\text{m}$, far smaller than the width of the FDS100 photodiode at 3.6 mm.

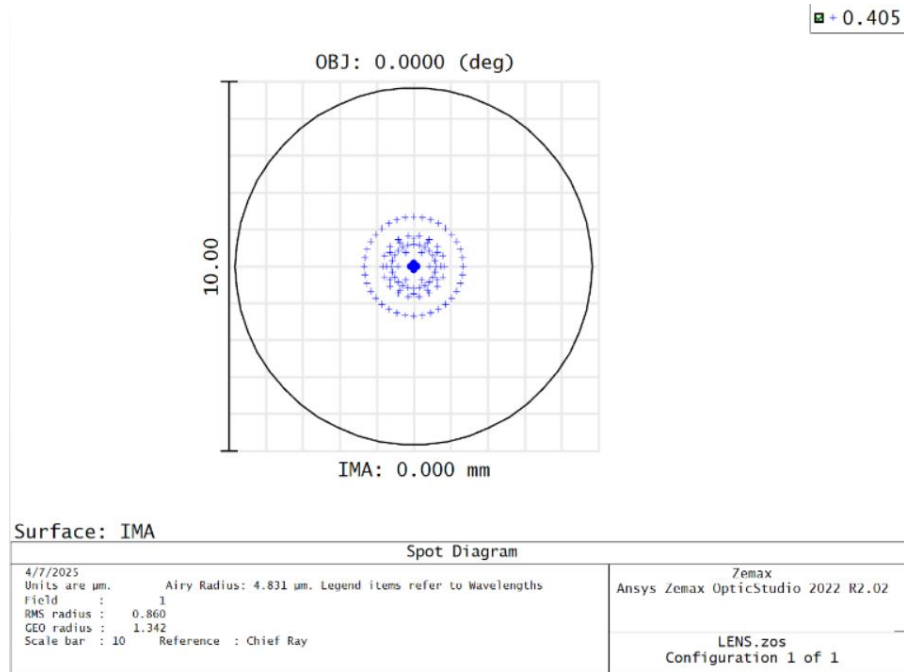


Figure 19. Plano-convex lens spot diagram

We then simulated the chosen LA1289 Plano-convex lens that we planned to purchase from Thorlabs. We downloaded the Zemax file from the Thorlabs website and simulated our chosen lens in Zemax to again observe the spot size of this lens in figure 18.

	Surface Type	Comment	Radius	Thickness	Material	Coating	Clear Semi-Dia
0	OBJECT Standard ▾		Infini...	Infinity			0.000
1	STOP (aper) Standard ▾	LA1289	15.450	3.160	N-BK7		6.350 U
2	(aper) Standard ▾		Infini...	26.274			6.350 U
3	IMAGE Standard ▾		Infini...	-			0.084

Figure 20. Thorlabs LA1289 lens prescription

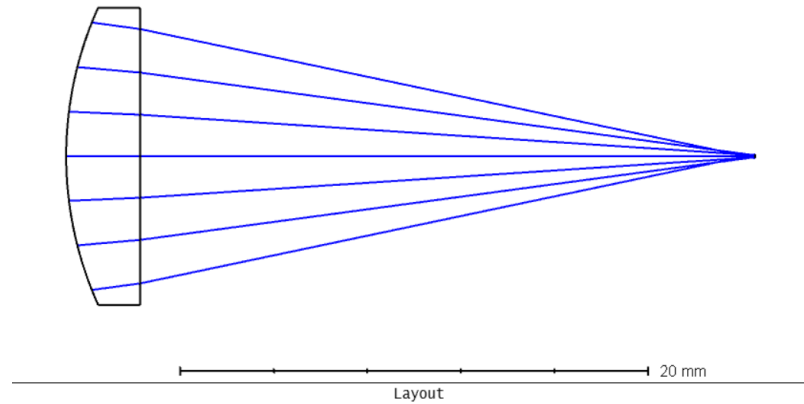


Figure 21. Thorlabs LA1289 Zemax simulation

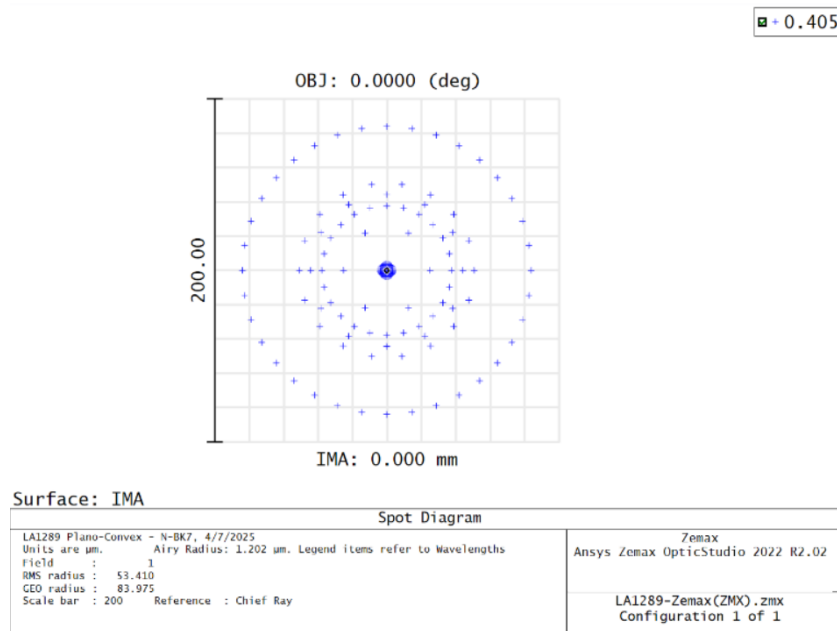


Figure 22. Thorlabs LA1289 spot diagram

We then attempted to model a beamsplitter in Zemax as well. The selected Thorlabs LA1289 was also included into the simulation to get a better understanding of the overall system. The prescription in figure 21 was very difficult to understand as it's not a true beam splitting simulation. The simulation in Zemax is basically a superposition of two configurations: the first acts as a reflective surface, and the second acts as a transmissive surface as seen in figures 22 and 23.

	Surface Type	Comment	Radius	Thickness	Material	Coating	Clear Semi-Dia
0	OBJECT	Standard ▾	Infinity	100.0...			0.000
1	STOP (aper)	Standard ▾	Infinity	100.0...	BK7		12.000 U
2	(aper)	Standard ▾	Infinity	20.000	BK7	AR	6.250 U
3	(aper)	Tilted ▾		0.000	BK7	1.50	6.250 U
4	Coordinate Break ▾			0.000	-		0.000
5	(aper)	Tilted ▾		20.000	BK7		6.250 U
6	(aper)	Standard ▾	Infinity	10.000		AR	6.250 U
7	(aper)	Standard ▾	15.450 V	3.160	BK7		6.350 U
8	(aper)	Standard ▾	Infinity V	31.650			6.350 U
9	IMAGE	Standard ▾	Infinity	-			0.016

Figure 23. Beamsplitter Zemax simulation prescription

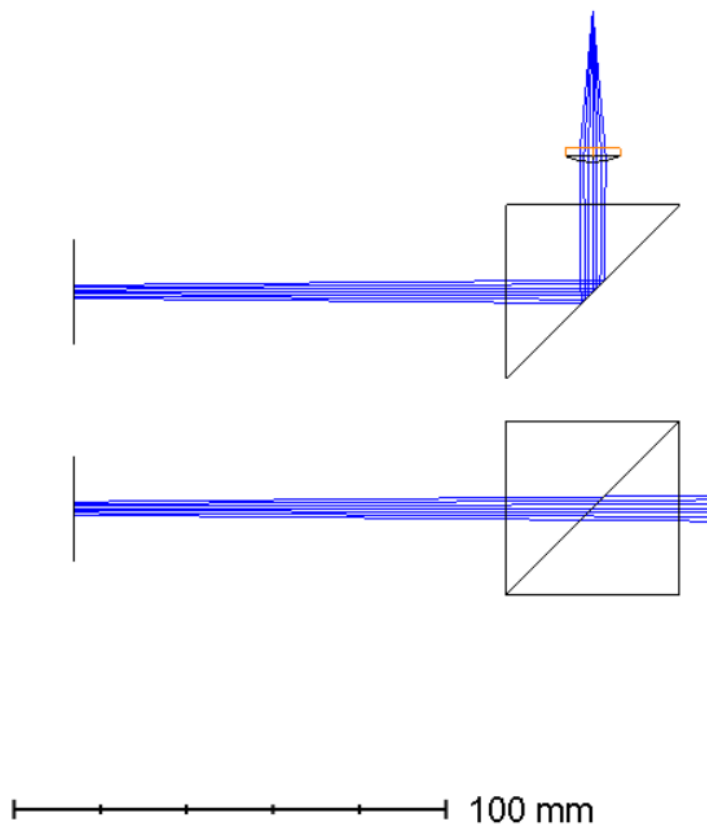


Figure 24. Beamsplitter two-configuration simulation: reflective and transmissive

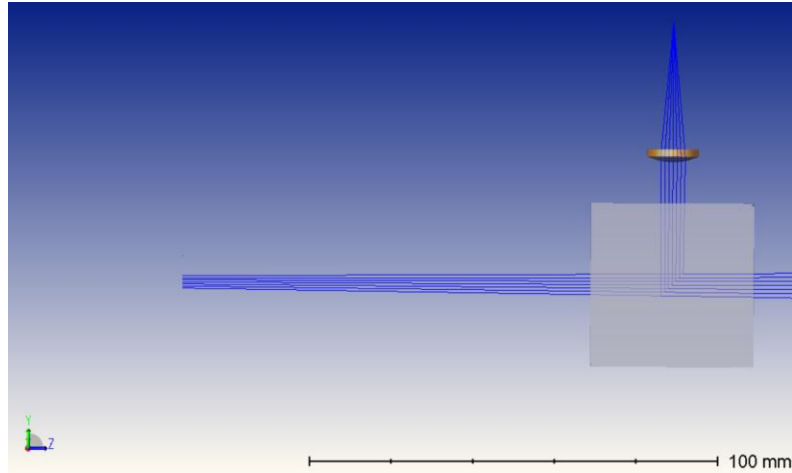


Figure 25. Beamsplitter Zemax superposition simulation

6.2.2 Spectrometer Dimensions and Zemax Simulation

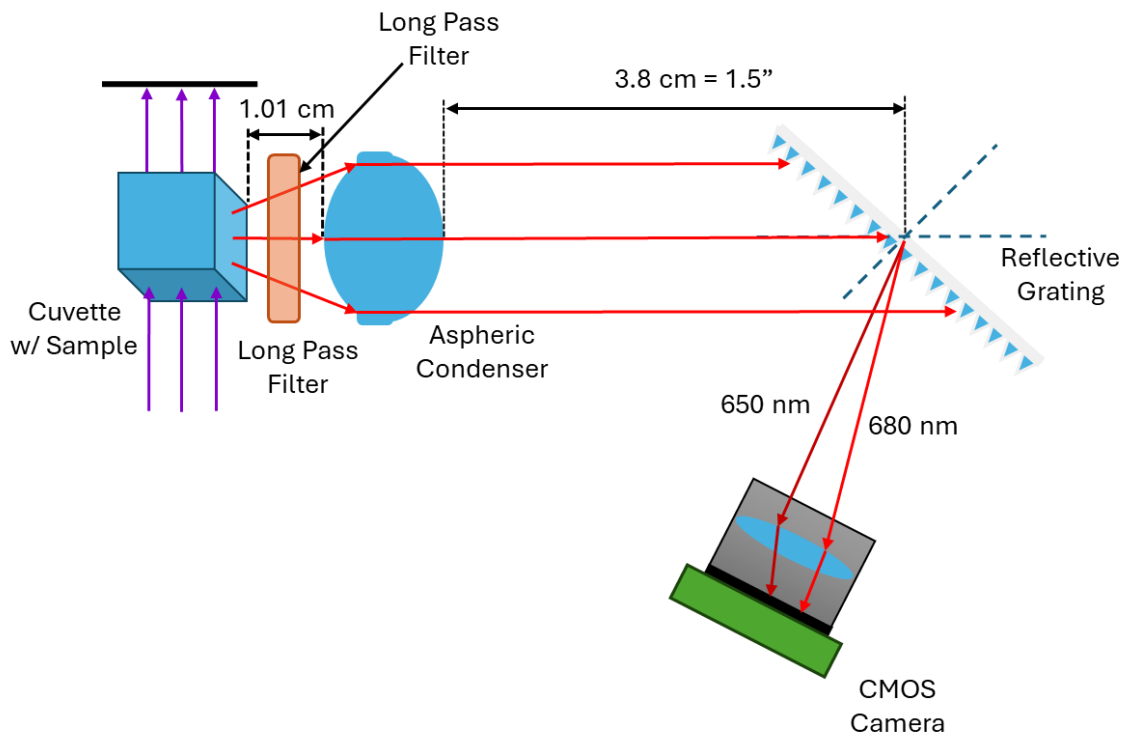


Figure 26. Spectrometer Schematic

The above figure showcases the dimensions of the spectrometer hardware design. This includes the distances from the sample to collection lens, distance from collection lens to the grating, and the distance from the grating to the CMOS sensor.

Next, we simulate the collection lens at a variety of fields and with lots of incoming rays to most accurately simulate the randomly emitting fluorescence light coming from the cyanobacteria sample

The lens prescription of our chosen asphere is entered into Zemax as shown below in figure 14 with the corresponding results

	Surface Type	Comment	Radius	Thickness	Material	Coating
0	OBJECT Standard ▾		Infinity	10.400		
1	STOP (aper) Standard ▾		24.000	14.000	LB2000	EO_NIRI_785
2	(aper) Even Asphere ▾	15895	-12.790	202.811		EO_NIRI_785
3	Standard ▾		Infinity	10.000		
4	IMAGE Standard ▾		Infinity	-		

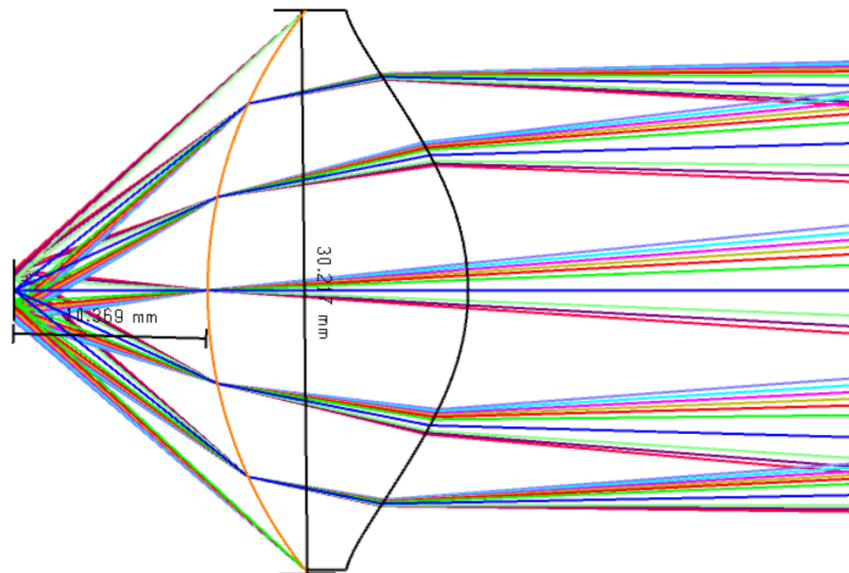


Figure 27. Zemax simulation with different fields of light (i.e. angles of incidence)

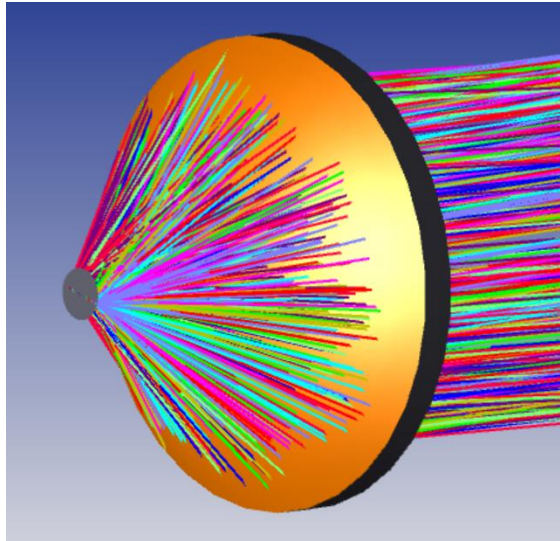


Figure 28. Zemax model of aspheric condenser lens

Next, shown below in figure 28, we simulate the entire spectrometer which includes the diffraction grating

	Surface Type	Comment	Radius	Thickness	Material	Coating	Clear Semi-Dia
0	OBJECT	Standard	Infinity	10.400			0.000
1	STOP (	Standard	24.000	14.000	LB2000	EO_NIRI_785	12.000 U
2	(aper)	Even Asphere	15895	-12.790		EO_NIRI_785	12.000 U
3		Coordinate Break		0.000	-		0.000
4	(aper)	Diffraction Grating	Infinity	0.000	MIRROR		12.500 U
5		Coordinate Break		-32.000	-		0.000
6	IMAGE	Standard	Infinity	-			20.000 U

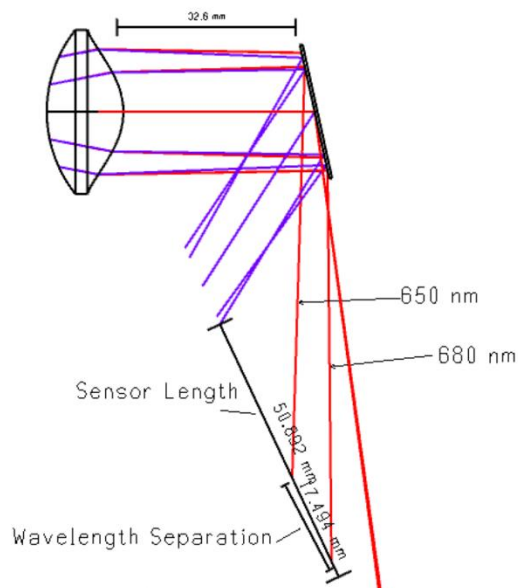


Figure 29. Zemax spectrometer simulation

In the spectrometer above, the diffraction grating is placed at an incidence angle of 25 degrees relative to the normal of the aspheric condenser lens. We are simulating some 405 nm light scattering into the spectrometer, 650 nm which simulates the diffraction of the light coming from the cyanobacteria, and 680 nm which simulates the diffraction of the light coming from green algae. As shown in figure 16, all the wavelengths have ample enough separation from each other to detect. The sensor is about 5 cm, which is more than enough to cover the 17.5 mm or 1.75 cm separation between the wavelengths when the CCD sensor is at a distance of 32 mm (1.25") away from the diffraction grating.

6.3 Electrical Hardware

For our project, LUCID, it is essential that we successfully connect and supply power to all the major electrical hardware components within our system to ensure smooth and reliable operation. Each component plays a crucial role in the overall functionality of the system, and providing the correct type and amount of power to each one is critical for their optimal performance. In addition to power management, we must also establish robust communication pathways between the various major components. Effective data transmission is necessary for the system to operate cohesively, allowing different modules to work together seamlessly. Our design must ensure that data flows efficiently and accurately between sensors, processors, and output devices, minimizing latency and preventing data loss or corruption. The LUCID system is intended to operate continuously, constantly gathering and updating data from all its subsystems. This real-time data acquisition is vital for monitoring system health and performance. Our program must be capable of analyzing incoming data streams, comparing information from different components, and cross-referencing these inputs to ensure that everything is functioning as intended. A significant objective of the system is to accurately determine concentrations of cyanobacteria in the environment we are monitoring. To achieve this, the data collected must be processed and interpreted correctly, requiring not only precise sensor readings but also a highly reliable system that can detect anomalies or failures early. Below is a basic block diagram that outlines the structure of our system, illustrating the main components and how they are interconnected to achieve these goals.

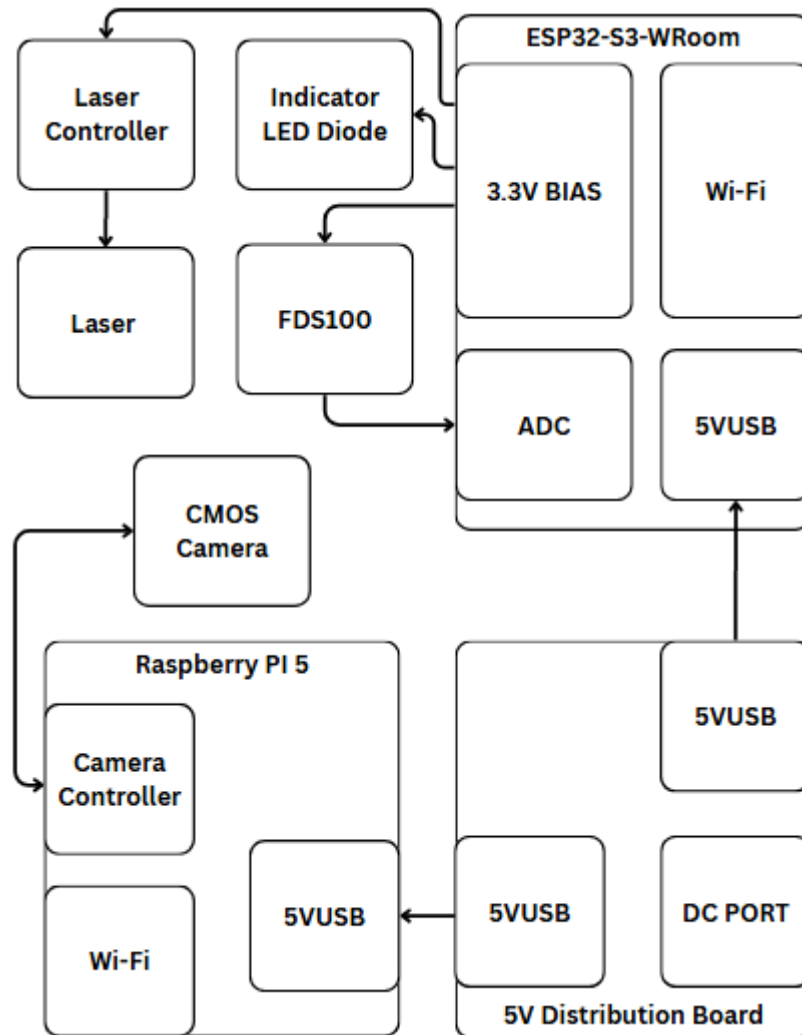


Figure 30. Electrical Hardware Block Diagram

This figure provides an overview of the fundamental electrical connections required to supply power to the ESP32 microcontroller, as well as to the other major components integrated into our system. It highlights the essential links needed to ensure that each part receives the appropriate power necessary for proper functionality. While this initial figure presents a simplified view of the system's power distribution and component connections, a more detailed representation can be found in the schematic that follows. In the schematic, we expand upon the basic connections and provide a comprehensive view of how all the major hardware elements are interconnected. The schematic illustrates how the wires are used to establish electrical connections between the various components, with labeled nets showing where connections are made electronically even if they are not physically continuous. Each of the primary system elements is clearly identified with large, easily readable green text labels, making it easy to locate and understand the role of each part within the system. By studying the schematic, one

can gain a clear understanding of the complete electrical layout, including power distribution, data lines, and control signals to the intended design specifications.

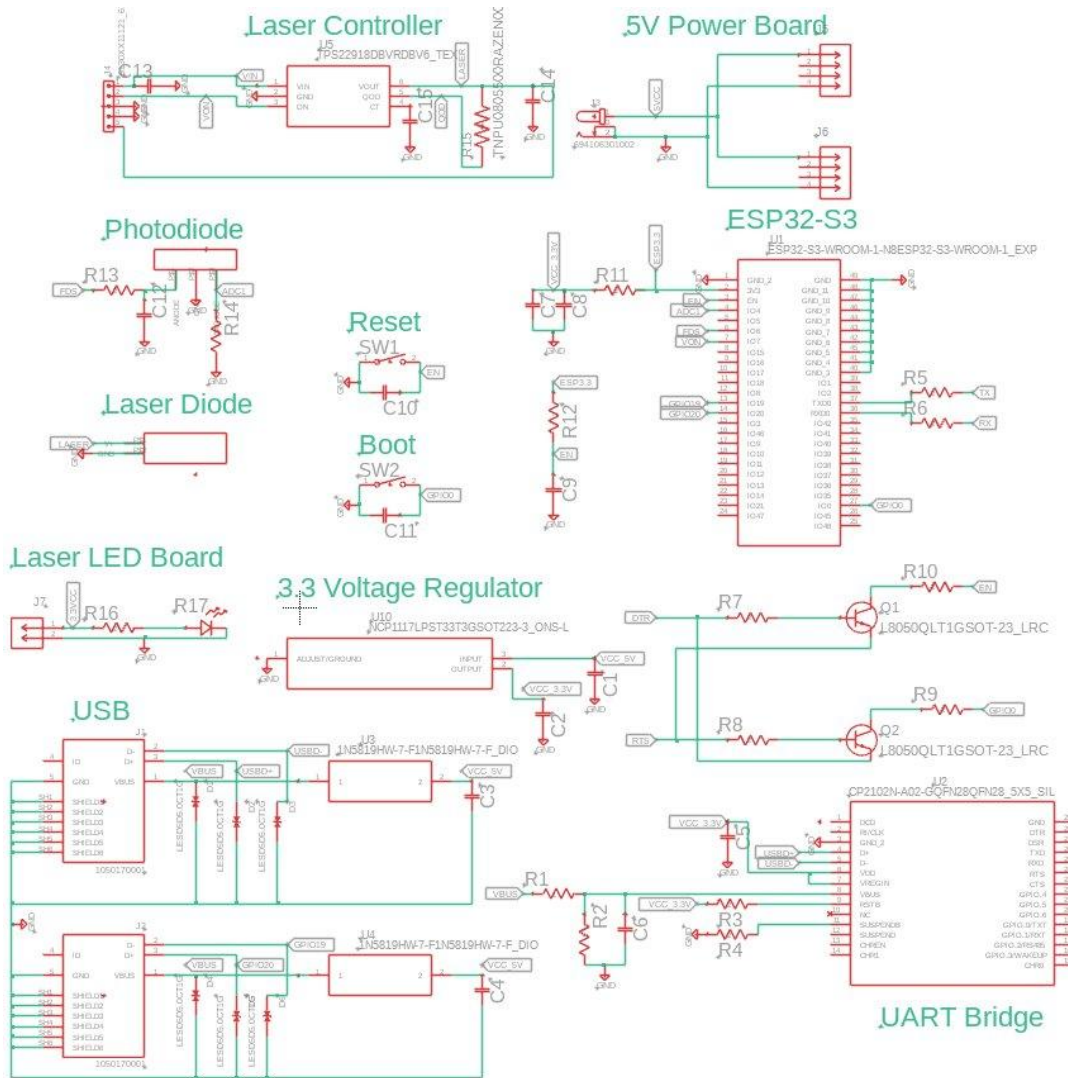


Figure 31. Overall Schematic

After reviewing the overall schematic presented above, we can identify three major electrical hardware sections that form the foundation of the LUCID system: power distribution, the UART bridge, and the ESP32 microcontroller. Each of these subsystems plays a critical role in ensuring that the project operates reliably and meets its design objectives. The power distribution system is responsible for delivering the correct voltage and current levels to all major components, maintaining stability and protecting sensitive hardware. The UART bridge enables essential communication between the ESP32 and a host computer by translating USB signals into UART serial communication, allowing for programming, real-time data logging, and system debugging. Meanwhile, the ESP32 serves as the central processing unit of the project, managing data acquisition from sensors, controlling

the laser diode, and coordinating system operations through its GPIO header pins. By closely examining each of these major sections, we gain a deeper understanding of how the LUCID system integrates its hardware components to accurately detect and quantify cyanobacteria concentrations in real time.

6.3.1 Power Distribution

For the LUCID project, our power management strategy involves utilizing two distinct voltage levels: 5V and 3.3V. These voltage levels are sufficient to meet the requirements of all the major components in our system while keeping the design simple and efficient. To supply 5V to the system, we plan to either use a dedicated wall outlet adapter that is specifically designed to output a regulated 5V up to 6A of current directly from our power distribution board. This will be used to power both the ESP32 and the Raspberry Pi. Once we have obtained the 5V supply whether from the USB port. We will then use a linear voltage regulator to step down the 5V to 3.3V. This regulated 3.3V output is essential for powering components that require lower operating voltages without compromising stability or performance. In terms of specific component connections, the 5V source will be used primarily to power the ESP32 and the Raspberry Pi, which operates at that voltage level. Meanwhile, the regulated 3.3V output will be used to power the ESP32 microcontroller, which is designed to operate at 3.3V for both its logic and core functions. Furthermore, additional components that require 3.3V operation, such as our laser diode and the FDS100 photodiode sensor, will be powered directly through the 3.3V output available from the ESP32. This setup helps streamline the overall wiring and minimizes the number of separate voltage regulation circuits needed within the system.

While the electrical hardware block diagram provides a high-level overview of how the major components of the LUCID system are interconnected and powered, it simplifies many of the detailed electrical connections necessary for the system to function properly. To fully capture the complexity and specific wiring of the project, we move from the block diagram to a more detailed schematic. By examining the schematic, we can better understand how the 5V and 3.3V power rails are distributed throughout the system. This detailed view is essential for ensuring that the entire system operates safely, efficiently, and according to our design requirements.

3.3 Voltage Regulator

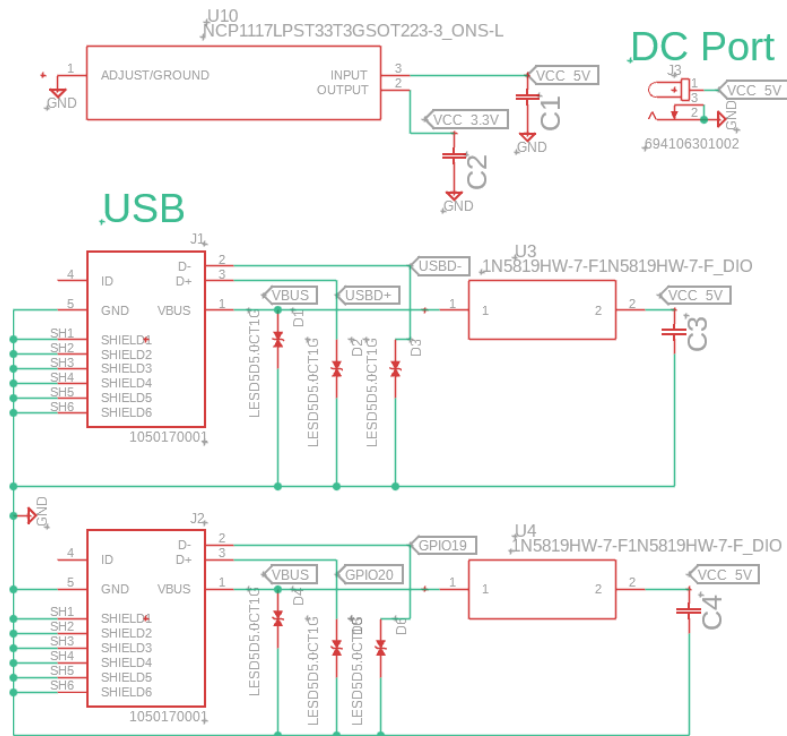


Figure 32. Power Distribution Schematic

6.3.2 CP2102 UART Bridge Design

In our LUCID project, we use a CP2102 USB-to-UART bridge to facilitate reliable communication between the ESP32 microcontroller and a host computer through a standard USB connection. The CP2102 plays a critical role in our system by converting USB signals — which are the native communication protocol for computers and other high-level devices — into UART (Universal Asynchronous Receiver/Transmitter) serial signals that the ESP32 can natively understand and process. This conversion is essential because the ESP32 operates using low-level UART communication, while modern computers exclusively use USB for peripheral communication. By integrating the CP2102, we enable the LUCID system to send and receive data between the ESP32 and a computer for multiple purposes, including programming, firmware updates, real-time data logging, debugging, and system control. The bridge essentially "tricks" the computer into recognizing the connection as a standard serial COM port, allowing software tools such as serial monitors and integrated development environments (IDEs) to interface seamlessly with the ESP32, despite the physical connection being USB. Our design also ensures that the CP2102 provides a clean, regulated interface between the USB and the ESP32's UART pins, protecting both devices and ensuring data integrity during transfers. Below is the schematic for our CP2102 circuit, which details all the necessary connections, such as TX, RX, power, and

ground lines, as well as additional components like pull-up resistors and capacitors that are required to ensure stable and reliable operation in conjunction with our ESP32 microcontroller.

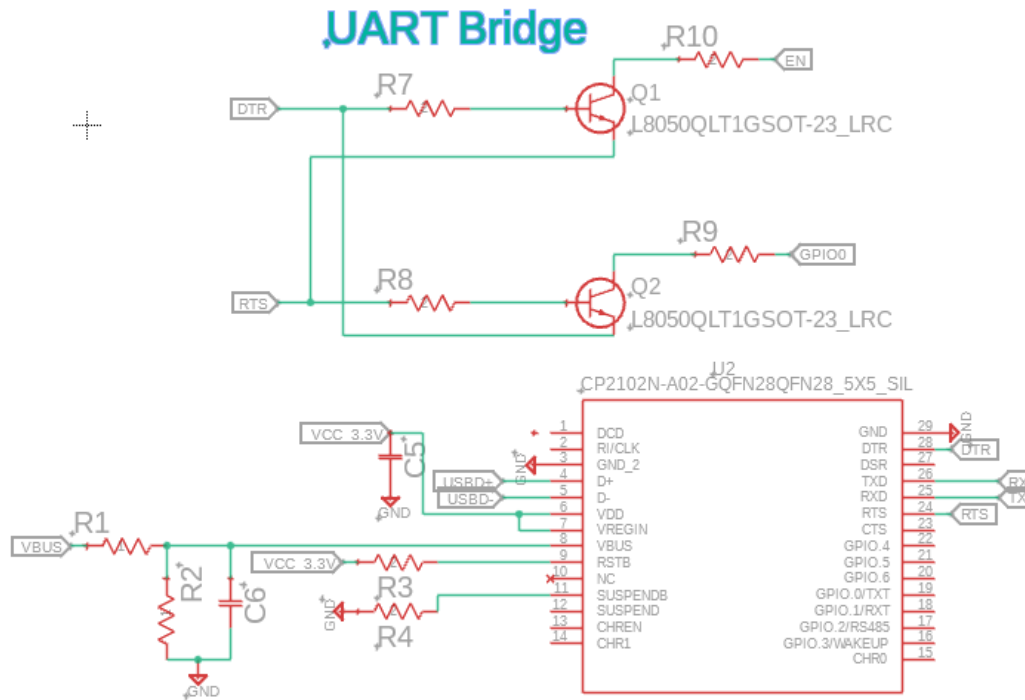


Figure 33. UART Bridge Schematic

6.3.3 ESP32-S3 Design

In the LUCID project, the ESP32 microcontroller acts as the central processing unit responsible for managing data collection, analysis, and control across all major hardware components. Its primary role is to receive and send raw signals from the system's sensors specifically the FDS100 photodiode and the Laser Controller and to process this information in real time to detect and quantify concentrations of cyanobacteria. Additionally, the ESP32 controls the operation of the laser diode, which serves as the light source necessary for the system's optical measurements. The connection between the ESP32 and each major component is made through the use of the microcontroller's header pins, which provide convenient access to its general-purpose input/output (GPIO) lines, as well as to its specialized communication interfaces such as SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), ADC (Analog-to-Digital Converters), and PWM (Pulse Width Modulation). We will also make use of Wi-Fi or Bluetooth to connect LUCID to a website and apps to compare the data to a database constantly in real time. In the schematic below you can see the 44 header pins will use in our circuit the two button switches for reset and boot and can see the two pins using for transmitting and receiving for the cp2102.

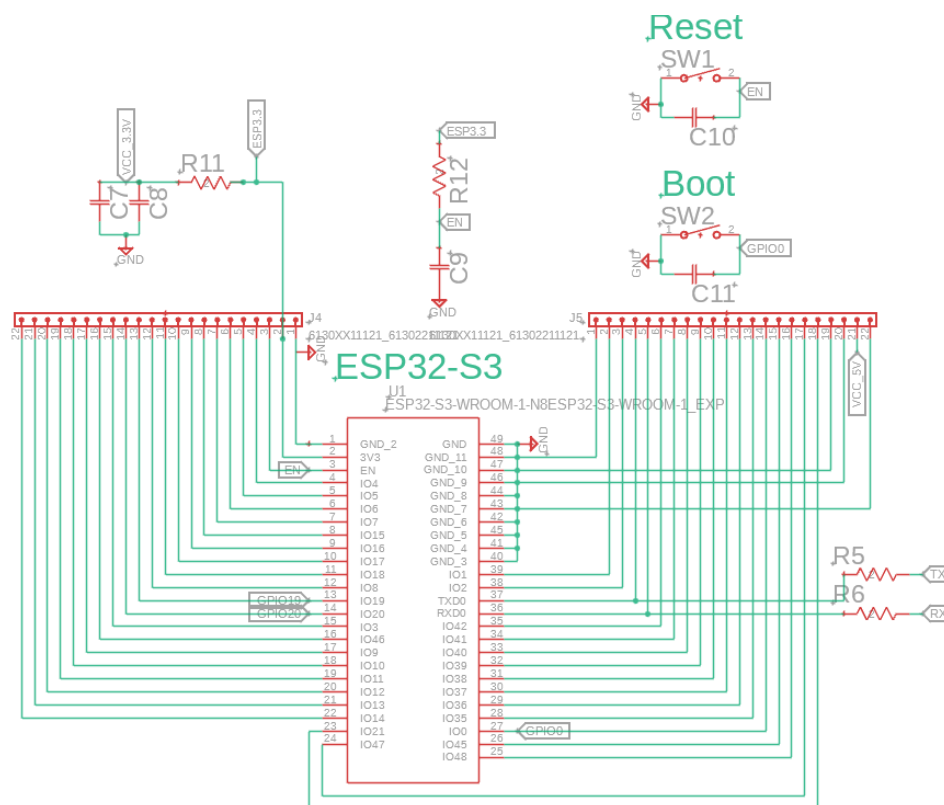


Figure 34. ESP32-23 Schematic

6.3.4 Raspberry Pi

In addition to the PCBs designed for LUCID, other electronic components are also used. The LUCID system includes a Raspberry Pi 5 and a CMOS image sensor to facilitate high-level processing and optical data acquisition for cyanobacteria sample analysis. The Raspberry Pi 5 functions as the central high-level processing unit in the LUCID project, serving as an intermediary between data acquisition and user-facing analysis. The ESP32 is responsible for low-level analog readings and sensor control, while the Raspberry Pi manages more computationally intensive tasks that a microcontroller cannot perform efficiently. The Pi can conduct image analysis to identify the peak wavelength and intensity of fluorescence signals, providing information used to detect and quantify cyanobacteria concentrations. Additionally, the Raspberry Pi manages network communication and cloud integration by connecting to a Firebase server, where it checks for incoming commands such as starting measurement cycles or modifying capture settings. After analysis, the Pi packages and uploads results to the project's website, making data available for monitoring and interpretation. This configuration allows for remote operation and real-time visualization, which can support environmental monitoring and water quality assessment. The Raspberry Pi's processing capabilities, Linux-based operating system, GPIO adaptability, and networking features contribute to its role within the LUCID system.

Chapter 7: Software Design

The L.U.C.I.D system will need to utilize embedded firmware integration, wireless communication, and a web-based interface to enable real-time monitoring and analysis of cyanobacteria in water samples. To accomplish this, the software will need to seamlessly coordinate multiple operations like processing analog data from the sensors and output the acquired data into a database that will be connected to the web application design for monitoring the data acquired from the ESP32. This chapter will outline the software architecture, key workflows, and display system states that make up the LUCID system platform. By structuring the software platform in logical layers, the design will have the benefit of modularity. This modularity will allow for each component, such as sensor control, wireless communication, and data visualization, to be developed and improved independently.

7.1 System Architecture Overview

The software framework of the L.U.C.I.D. system will be built on a three-tier architecture that consists of an embedded layer, a communication layer, and a web application layer. This structure will allow the system to collect the environmental data at the sensor level, transmit the data collected reliably, and displayed on the responsive web application.

The embedded system layer will be powered by an ESP32 module. This dual-core microcontroller is equipped with integrated Wi-Fi capabilities and has many peripheral interfaces to use. This microcontroller will be responsible for executing firmware that directly interfaces with the sensors like the ILX511 CCD sensor and the FSD100 photodiode sensor. The firmware will control the timing signals needed for the CCD to operate. These signals are the clock, readout gate, and the sample and hold switch. The CLK signal will need to operate at a frequency of 2MHz for the IXL511. The communication layer will utilize the ESP32s Wi-Fi integration to transmit data over a secure internet connection. The communication layer will be utilizing Firebase to provide quick and reliable real-time services using the `Firebase_ESP_Client` library. The final tier will be the application layer, which will provide the interface through which the users will interact with the L.U.C.I.D. system. This web application will be developed using a modern JavaScript stack, that will utilize React.js to provide a responsive web design for many different devices.

7.2 System Software Use Case

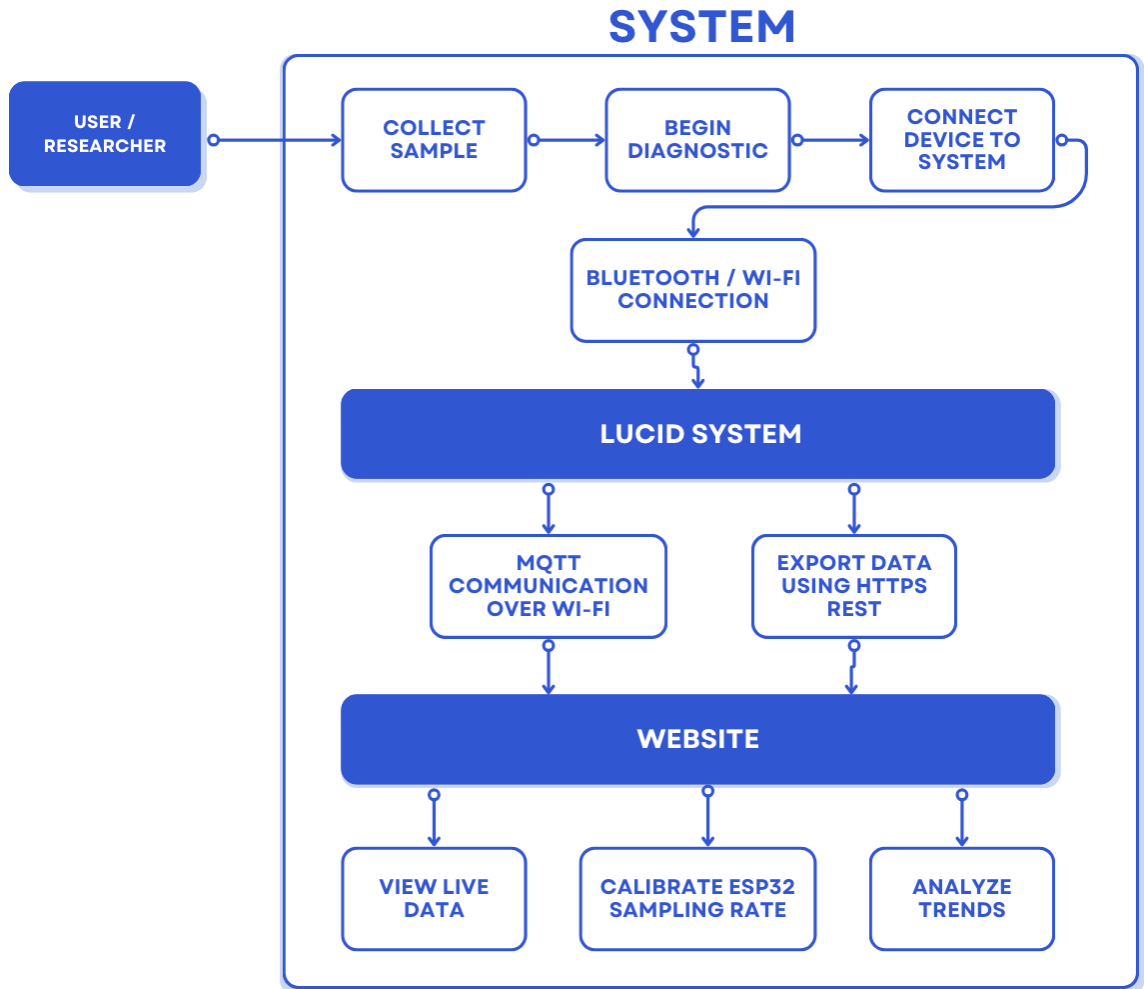


Figure 35. System Software Flowchart of L.U.C.I.D. device

The LUCID system software workflow demonstrated in Figure 34, outlines the end-to-end process from sample collection to data visualization. During power-up the system will be requesting to connect to a device to begin diagnostic. Once connected the data acquisition will begin and the LUCID system will run the diagnostic on the given water sample. The ESP32 firmware will collect and process the sensor reading coming from the FDS100 through the ADC pins and filter out noise and outliers through a data filtering algorithm. The processed data will then be stored locally until the system establishes and connection with MQTT via Wi-Fi.

7.3 Software State

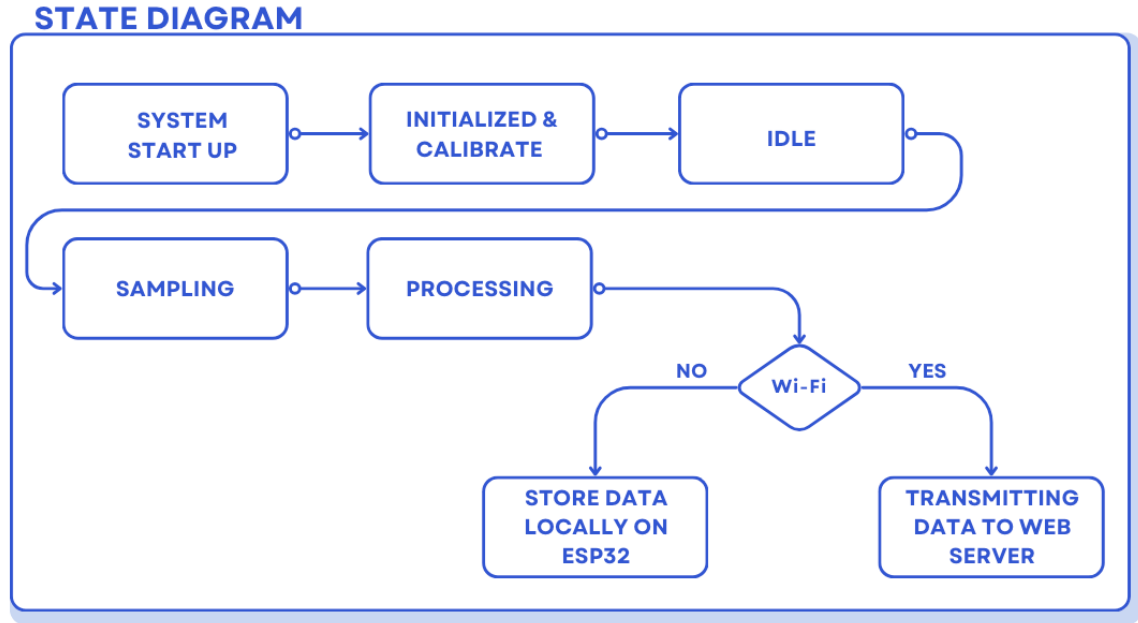


Figure 36. Software State Diagram of L.U.C.I.D. device

The LUCID system software state diagram defines the different operational modes the system will use when going through a full diagnostic. The system will need to go through a calibration state when starting up to ensure accuracy and reliability of the system. When self-calibration is complete, the system will go into an idle state. In the idle state, the system will await user input and water sample. When the user is ready to begin the user will enable the state change to sampling to begin data acquisition, where the ESP32 will continually collect optical power data and process it. Once a diagnostic is complete, the ESP32 will check if there is Wi-Fi connectivity and make the decision on whether to store the data locally or transmit it to the web server, depending on the connectivity status.

7.4 Responsive Web Design for Real-Time Visualization

In place of a mobile application, the L.U.C.I.D. system will use a web interface with a responsive design. This transition allows for the system to be more stable and provides larger accessibility between different devices. A responsive website design will be developed to ensure consistent performance and a clean user experience across a wide range of screen sizes and desktop monitors. By having a development plan for the mobile portion, the interface will be able to adapt fluidly to the dimensions of the device being used. This ensures that the field technicians or researchers in remote areas can view real-time results on multiple devices. The previous development thought process of creating a web app and a mobile app

would require too much development time and would face issues with compatibility with more niche devices. This strategy avoids this problem all together.

To accomplish this, the front-end will be built using React.js. This is a component-based JavaScript library that facilitates modular, reusable UI elements. By using React.js, it will allow the computer engineer to implement dynamic views, such as a dashboard that automatically updates when new sensor readings are uploaded to the database. For styling, Tailwind CSS will also be implemented. Tailwind's utility focused framework offers tools that will assist in making the responsive design, enable quick implementation of layouts that scale with tables, phones, and desktops without the need for complex CSS.

The visual display of real-time data will be a critical part of the user interface. Libraries like Recharts and Chart.js will need to be used to create simple, visually appealing, and interactive charts that will reflect the changing values of fluorescent intensity and cyanobacteria concentration. These charts will be automatically updated using event listeners tied to the Firebase real-time database. Additionally, there will be visual indicators such as threshold bars and light indicators that will help the user quickly interpret the quality of the water sample being evaluated.

To ensure accessibility and long-term maintainability, the entire web app will most likely be deployed using Firebase Hosting service. Firebase Hosting provides a fast setup by using CDN, free HTTPS by default, simple commands deployment. This setup greatly simplifies the development time and provides secure and stable hosting for the web application. The use of Firebase will also be helpful in the future implementation of authentication. Firebase will provide security through its authentication so that it can restrict sensitive information to authorized users only.

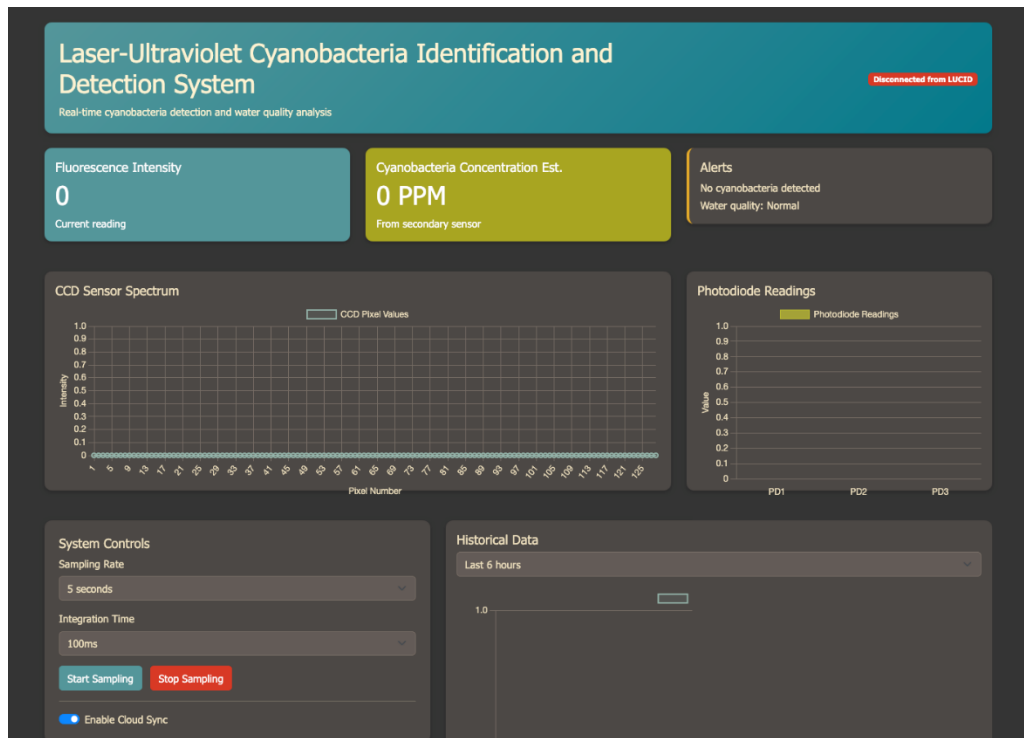


Figure 37. Prototype Web Design

7.5 Firebase Integration with ESP32-S3

In order to collect and deliver the sensor data from the ESP32 microcontroller to the web application in real-time, the strategy moving forward has been to integrate Firebase as the database. Firebase is a cloud-hosted backend solution provided by Google. This backend solution enables a seamless connection between the embedded hardware that is collecting the data and the website that is visualizing and interpreting the data. The core Firebase service being utilized is Firebase's Real-Time Database. This database allows for low-latency synchronization of data between connected clients and backend services and fortunately has libraries that will assist in the integration of the ESP32.

Firebase Real-Time Database stores information in a hierarchical NoSQL JSON format, which makes it easy to structure the data according to the session, reading, timestamp, and other metadata collected during each diagnostic. Whenever a new data point is uploaded by the ESP32, listening clients like the web application will be instantly notified of the update and can refresh the values its displaying accordingly. This allows for true real-time interaction and monitoring. This will play a crucial role in the accuracy and responsiveness of the L.U.C.I.D. system [32]. In addition to Firebase's Real-Time Database, the project will most likely make use of Firebase's Hosting to serve the front-end web application. Our team may also implement Firebase's Authentication to secure the data uploads and readings between multiple users. Authentication can also ensure that only authorized

devices like the ESP32 can push data to the database and only registered users can view or download the historical measurements collected. This adds an additional layer of data integrity and security for the system.

Firebase's REST API also enables compatibility between microcontrollers like the ESP32, which do not support the full JavaScript SDK used in most web development applications. The ESP32 can connect to Wi-Fi, then serialize the data into JSON format, and send HTTP POST or PATCH requests to the Firebase REST endpoint. This flexibility allows for real-time data transmission to be done with minimal overhead, and the ESP32 can be easily reconfigured or updated without needing to change the Firebase Database schema.

7.6 ESP32-S3 Data Upload Implementation

The ESP32 serves as the component that will deal with the digital processing and communication between the LUCID hardware. It interfaces directly with the FSD100 photodiode sensor to convert the analog fluorescent measurements into digital signals. This microcontroller is equipped with a 12-bit ADC that enables high-resolution sampling of voltages values that correspond to the fluorescent intensity. This resolution is slightly lower than what would be expected for the sensitive CCD linear sensor like the ILX511, but will provide a baseline to indicate if we would need to acquire an external ADC, with a higher sampling rate and larger resolution. These values captured by the ADC, along with the metadata such as the timestamp and optical power, will be collected and prepared for transmission to Firebase.

The data upload process begins when the ESP32 connects to a local Wi-Fi network. The device is then programmed to automatically attempt reconnection if disconnected and will retry uploads in a case of network failure. Once online, the ESP32 will format the collected data into a JSON object and perform an HTTP POST request to the Firebase Real-Time Database's REST endpoint. This data is then sent under a unique key to allow the frontend application to organize and track the sessions and sets of readings acquired.

The JSON format will need to allow for flexibility and structured querying later on to be useful to the web application. For instance, sessions should be able to be queried by the data or by the average intensity captured during a diagnostic to identify the trends in algae growth and analyze the system overall performance.

To implement this system, the ESP32s firmware will include libraries such as:

- `#include WiFi.h` to facilitate the ESP32s connection to Wi-Fi.
- `#include HTTPClient.h` for sending HTTP requests
- `#include ArduinoJson.h` for building and formatting JSON payloads to be sent to the database.
- `#include <Firebase_ESP_Client.h>` to connect the ESP32 to the Firebase seamlessly

- `#include "addons/TokenHelper.h"` for authentication
- `#include "addons/RTDBHelper.h"` for authentication

This approach will ensure an efficient, lightweight, and modular design that can be modified and maintained easily. Future improvements may include automated calibration routines and diagnostic data of the system's health and condition.

7.7 Clock Signal Implementation (S/H Mode)

Precise control and timing signals will be important in operating the Sony ILX511 linear CCD sensor. In the L.U.C.I.D. system, the ILX511 contains digital control lines such as the Readout Gate, the input clock pulse, and the sample and hold switch that all require strict timing sequence to transfer the pixel charge from the CCD to the analog output. This voltage output is then sampled by the ESP32's ADC. Each control signal is mapped to a dedicated GPIO pin on the ESP32 and initialized in the `setup()` function in the ESP32's software.

The ILX511 includes a Sample and Hold Switch input that will enable the Sample and Hold Mode. In this mode, an external SHSW pulse will be able to temporarily stop the current analog charge level in each pixel before the readout begins. This will prevent the additional charge accumulation during the readout and improve the timing and accuracy of the system.

For the L.U.C.I.D. system, the Sample and Hold mode will be highly beneficial because it ensures the captured fluorescent intensity will accurately represent the instantaneous optical state of the sample at the moment of the SHSW being triggered. Without the Sample and Hold enabled, charge could continue to accumulate or leak during the readout process. This has the potential to introduce artifacts or cause the smearing of the data. S/H mode also allows for more precise comparisons across samples and time intervals. This could be useful for detecting subtle differences in the cyanobacteria's fluorescent levels. In figure 33 you can see the difference between the voltage output when using the sample and hold mode and not while having the same clock timing.

Clock Timing Diagram (With S/H mode)

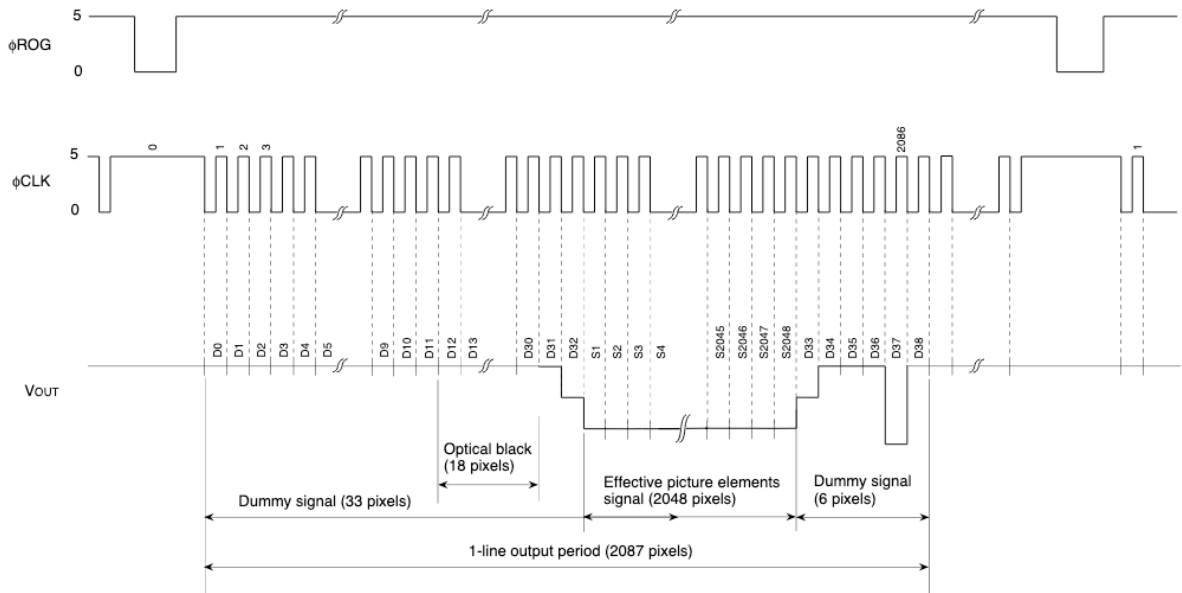


Figure 38. Clock Timing Diagram (With Sample-Hold Mode)

Clock Timing Diagram (Without S/H mode)

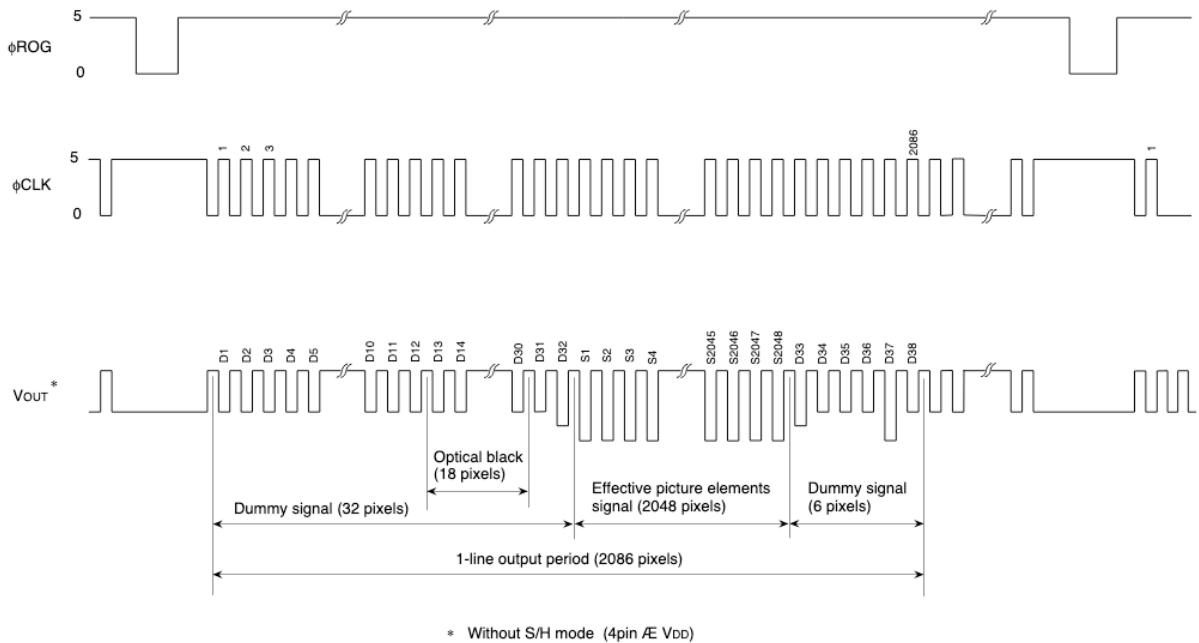


Figure 39. Clock Timing Diagram (Without Sample-Hold Mode)

The logical flow for the CCD capture begins with asserting the Readout gate to begin the scan. This is followed by a SHSW pulse to sample the current

fluorescence exposure. Afterward the CLK will begin to toggle continuously while the ESP32 samples the analog V_{out} signal in sync with the clock cycle. After 2048 clock cycles, the scan will be complete, and the CCD will be ready for the next cycle. The ESP32 will toggle the SHSWQ line immediately after the Readout gate is triggered and before the clock starts. This ensures that all pixels are sampled simultaneously and that the subsequent clocked readout reflects a consistent and stable charge distribution across the pixel array.

Chapter 8 System Fabrication

During the prototyping phase of the LUCID project, careful PCB design was critical to transforming our initial concept into a reliable, working system. While breadboarding helped us validate individual circuits early on, a custom-designed PCB allowed us to create a more organized, compact, and robust platform for integrating all major components. By designing a PCB specifically for our ESP32-S3, CP2102 bridge, FDS100 photodiode, Indicator LED, Laser Controller and power distribution board, we were able to reduce electrical noise, ensure consistent signal connections, and improve the mechanical stability of the system. This step was essential not only for making the system easier to assemble and troubleshoot but also for preparing the project for future testing and potential field deployment.

8.1 PCB Layout/Design

The PCB layout and design are critical elements of the LUCID project, directly impacting on the system's reliability, performance, and overall success. A well-designed PCB is not just about connecting components together; it plays a fundamental role in ensuring that the electrical and mechanical aspects of the system work harmoniously, especially in a precision-driven application like LUCID where sensitive analog signals and real-time data collection are involved.

First and foremost, proper PCB layout ensures electrical integrity. By carefully planning the routing of power lines, ground connections, and signal traces, we minimize issues such as voltage drops, noise, and electromagnetic interference (EMI). For LUCID, where we are measuring extremely small changes in optical signals from the FDS100 photodiode, even minor electrical noise or poor grounding could significantly degrade the accuracy of our readings. A clean and organized layout helps preserve signal fidelity and ensures that the ESP32 receives clean, accurate data for processing.

Secondly, PCB design is crucial for mechanical stability and compactness. Transitioning from a breadboard or prototype wiring setup to a professionally manufactured PCB allows us to secure all components in a fixed, durable structure. This is important for LUCID because we aim to create a portable and robust system that can operate consistently without the risk of loose connections, vibration-related failures, or physical wear on the wiring. Another important consideration is thermal management. Power-regulating components like the linear voltage regulator may generate heat during operation. A good PCB layout takes thermal dissipation into account by placing components carefully, providing thermal vias or copper planes to spread heat, and ensuring that temperature-sensitive components are not adversely affected.

Additionally, PCB design impacts power distribution efficiency. By creating short, direct paths for high-current traces and properly decoupling power supplies with capacitors placed close to sensitive ICs like the ESP32, we ensure stable voltage levels even under dynamic load conditions, further improving the system's reliability.

8.1.1 ESP32-S3 PCB

The LUCID project's custom PCB is specifically designed to integrate and support the major electrical subsystems, with a strong focus on reliability, expandability, and efficient data communication. At the heart of our PCB is the ESP32-S3 microcontroller, a powerful and versatile device that offers built-in Wi-Fi, Bluetooth, extensive GPIO capabilities, and support for high-speed data transfer. The ESP32-S3 is directly mounted onto the PCB and carefully routed to interface with all other major components and external connectors. One of the key features of our PCB is the inclusion of two 22 header pins connected directly to the ESP32-S3's available GPIO lines. These headers provide flexible access to almost all of the ESP32's digital I/O, analog inputs, power outputs (3.3V and GND), and communication buses such as UART, I2C, and SPI. This extensive breakout allows for future expansion, prototyping, and easy interfacing with peripheral sensors and actuators. It also simplifies development and debugging by making it straightforward to connect external modules without modifying the main board.

For communication with external devices such as computers, the PCB includes a CP2102 USB-to-UART bridge. The CP2102 chip is responsible for translating USB signals from a computer into UART serial signals that the ESP32-S3 can understand. This enables seamless programming, real-time data monitoring, and system debugging. On the PCB, the CP2102 is connected to one of the two MicroUSB ports, which we specifically designate as the COM USB port. This port acts as the primary interface for flashing firmware to the ESP32 and for serial communication during development and system operation.

In addition to the COM USB, the PCB also features a second MicroUSB port dedicated to USB OTG (On-The-Go) functionality. The ESP32-S3 natively supports USB OTG, allowing it to operate either as a USB host or a USB device. Separating the COM and OTG USB ports avoids interference between programming/debugging activities and USB-based device communications, improving overall reliability and user experience. To power the board and its connected components, we include a 5V DC barrel jack input on the PCB. This allows the system to be powered externally from a stable 5V DC supply capable of delivering higher currents than what is typically available through a USB port alone. After receiving 5V from either the barrel jack or the COM USB port, the PCB uses a linear voltage regulator to step the voltage down to 3.3V, which is then used to power the ESP32-S3, the laser diode, and the photodiode. Power routing is carefully managed with appropriate decoupling capacitors placed near key ICs, ensuring clean, stable voltage levels across the board.

Overall, the PCB design consolidates critical features — a powerful microcontroller (ESP32-S3), dual USB functionality (CP2102 COM and OTG ports), extensive header breakout (44 pins), and robust power management (5V DC input) into a compact, durable, and expandable platform that is well-suited to the LUCID project's data-driven, precision-focused objectives. In the figure below you can see the fully completed and mapped out ESP32-S3 with all the traces set and ready for the PCB to be built and ordered.

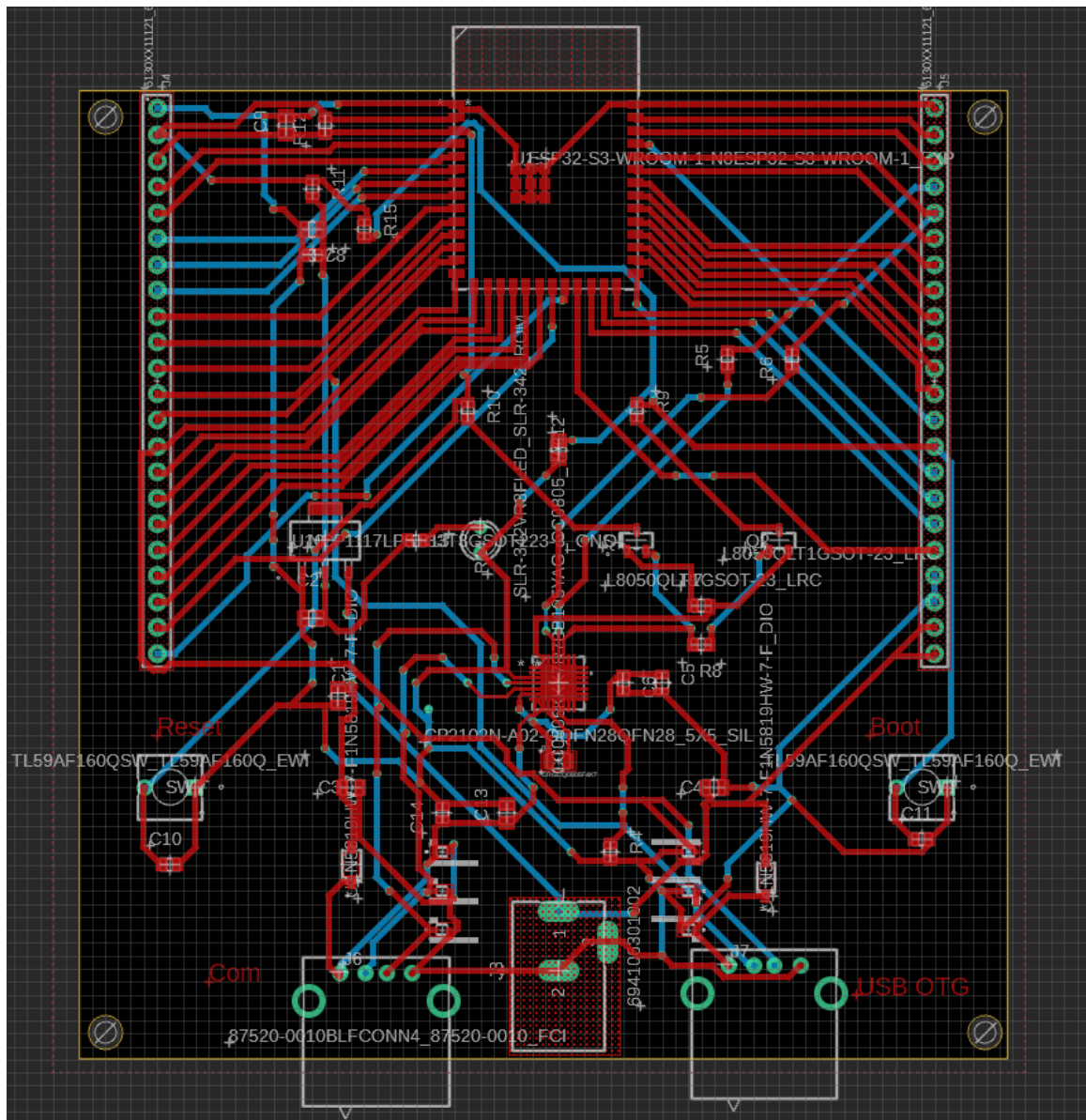


Figure 40. ESP32-S3 PCB

8.1.2 FDS100 PCB

The FDS100 photodiode in our LUCID project is connected to the PCB through a simple 3-pin header, which interfaces directly with the ESP32-S3. The three pins provide connections for 3.3V power, ground, and the analog output signal from the photodiode. To help reduce high-frequency noise and stabilize the signal, we added a basic noise filter circuit consisting of a resistor and a capacitor to ground placed close to the FDS100 output. This RC filter smooths out fluctuations and ensures cleaner data for the ESP32 to process. The photodiode's output current is converted into a readable voltage across a load resistor, and this voltage is then fed into one of the ESP32's ADC (Analog-to-Digital Converter) pins, allowing us to accurately monitor changes in light intensity as part of the system's laser power process.

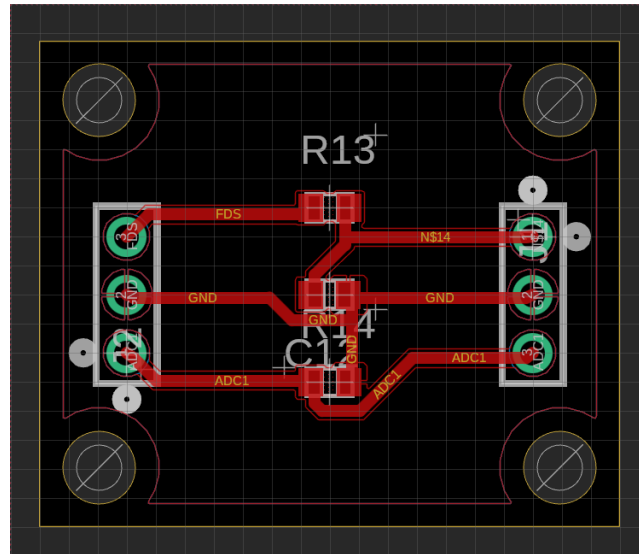


Figure 41. FDS100 PCB

8.1.3 Indicator LED PCB

The Indicator LED PCB designed for LUCID enables clear visual indication of the laser's operational status, enhancing safety by alerting personnel when the system is active. This PCB is installed on the side of the Lucid housing and operates in direct synchronization with the ESP32, corresponding precisely to the laser's on/off state.

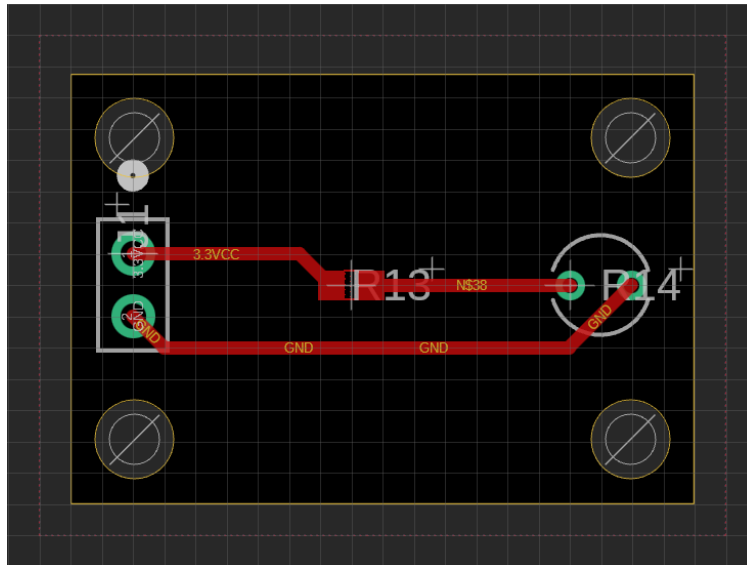


Figure 42. Indicator LED PCB

8.1.4 Laser Controller PCB

The laser controller PCB is designed to control the activation of the laser, helping to prevent overheating and excessive use. It receives power directly from the 3.3V output of the linear regulator on the ESP32, providing sufficient current for the laser. The PCB also uses one GND pin and one GPIO pin to enable or disable the laser as needed. Then the last two pins on the bottom are used to provide power to the laser.

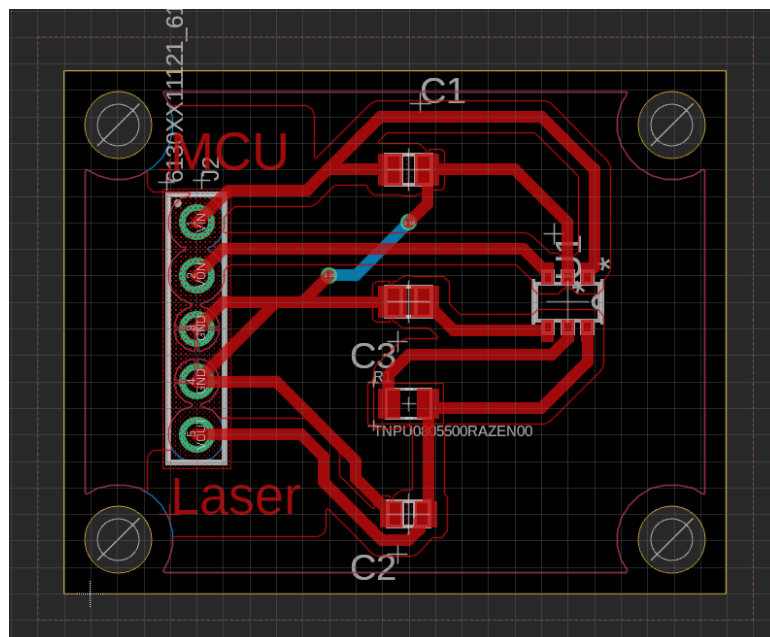


Figure 43. Laser Controller PCB

8.1.5 Power Distribution PCB

The power distribution PCB provides a stable 5V supply to both boards in use. LUCID utilizes a 5V 6A DC outlet plug to power the Raspberry Pi and ESP32. Approximately 5V 5A is required for the Raspberry Pi, while 5V 500mA is allocated to the ESP32 and its connected devices. The board delivers power from the outlet directly through USB to both components.

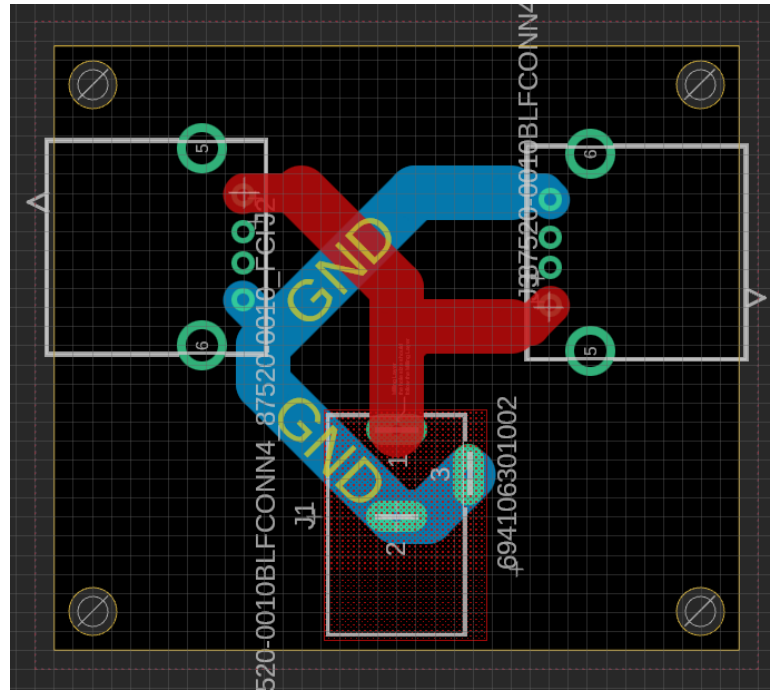


Figure 44. Power Distribution PCB

Chapter 9: Prototyping and Testing

Throughout the stages of development, the optoelectronics of the device were tested to ensure proper optical signals were being transmitted and received. The optoelectronics include the laser diode, FDS100 photodiode, and SONY ILX511B CCD sensor. This chapter focuses on the feasibility and testing of the optoelectronics and confirms the feasibility of the optoelectronic components necessary for the L.U.C.I.D. system. Additionally, this chapter explains how we measured different concentrations of cyanobacteria for initial testing purposes and confirms the fluorescing light thus solidifying the feasibility of the project. We also began to integrate our optoelectronics with our MCU (ESP32), which outputs to our software for data measurements and interpretation.

9.1 Optoelectronic Feasibility

The optoelectronics used for the project are: the laser diode module, the FDS100 silicon photodiode, and the SONY ILX511B CCD linear array sensor. This section focuses on the testing of these components since we are using these components for the excitation of the sample of cyanobacteria, baseline power detection, and for spectral measurements. They are the main components and therefore are critical for the success of the system, hence they were tested in a brief prototype way.

9.1.1 Laser Module Test

The test for the laser module is rather simple, especially considering the fact that it already comes with a pre-made laser driver inside. The diode should be biased voltage and contains two wires: one for positive voltage connection and another wire for ground. A standard bias voltage for laser modules is about 3.3 V – this is also the voltage that will be provided to the laser via the ESP32 so it is important that we can get lasing action at this voltage level.

At a forward bias voltage of 3.3 V, the laser diode module operates perfectly fine. The main problem was with the collimating lens that comes attached to the laser module – the laser output was not collimated. This issue was quickly resolved because luckily, the lens is adjustable. We adjusted it to be a bit further away from the laser module and got a good enough collimation such that the beam was no longer diverging or converging within a short distance. We used our FDS100 photodiode to measure the laser output power to be around 18-20 mW.

We also tested the laser module at different bias voltage: 2V and 5V biases. The reason we did this test was to see if there was an optimal bias voltage at which the laser likes to operate. Surprisingly, the 2V bias still provided enough forward bias to surpass the threshold of the laser diode and lasing was still induced. The main drawback at 2V is that the laser output power was lower which is expected. At 5V, the laser output was no different than at a bias of 3.3V – this is most likely because

the gain saturates past a certain level which we can infer is around the level of 3.3 V.

We also tested the performance of the laser diode module with and without the heatsink. Without the heatsink, the laser module would not only physically get hot, but the output power would decrease. The decreasing output power was observed visually because the laser spot on a surface kept getting dimmer and dimmer. With the heatsink, this issue was a lot less prevalent, and the laser was able to remain on for a longer period. This is a fairly standard result because the threshold current necessary for lasing increase with temperature, thus making it harder for the laser to function properly. Adding in a proper heatsink helps to alleviate this issue for some time, which increases the reliability of our project because we will not have to worry about burning out our laser module. This also saves us money in the long run because we will not have to worry about purchasing new laser diodes to replace a cheap one.

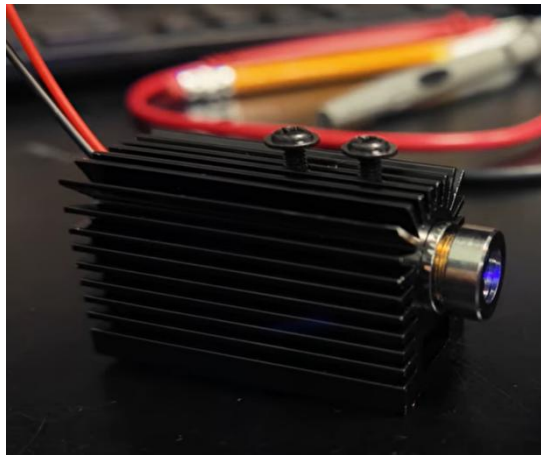


Figure 45. 405 nm laser diode module with heatsink.

9.1.2 FDS100 Photodiode Test

We purchased and tested the FDS100 photodiode to make sure it works properly, and that we were able to calculate optical power from measuring voltage across a load resistor.

To test the photodiode, we first constructed its corresponding circuit provided to us on the data sheet. The circuit consists of DC bias source, an RC noise filter, and a load resistor through which the photocurrent flows, and the output voltage can be measured as a result. The bias voltage was set to 3.3V because, similar to the laser diode module test, is the voltage that can be provided to the photodiode directly from the ESP32. Initially we tested with a small laser pointer, with an optical power less than 5 mW (exact power unknown). In the appropriate circuit, using a 1 kOhm load resistor and applying a 5V forward bias to the photodiode, we were able to measure an output voltage and using Equation 10 were able to calculate

an optical power of about 2.83 mW which is a logical reading. We tested the photodiode using a 405 nm laser pointer and a 532 nm laser. Tests with both laser wavelengths gave reasonable results.

Next, to more accurately test the function of the FDS100 photodiode, we used our laser module. As aforementioned in section 9.1.1, the laser module was reading about 18-20 mW of optical power – and this is before the beam splitter. This means that after the beam splitter, we should be getting an optical reading closer to 10 mW which is precisely what we need. Since we are using a 50:50 beam splitter, the reading on the FDS100 photodiode is equal to the amount of optical power going to our sample, and we have a goal of sending about 10 mW of optical power to the sample which we now know will be an easy target to meet – and this target is one of our demonstratable that we have planned for the final portions of the project.

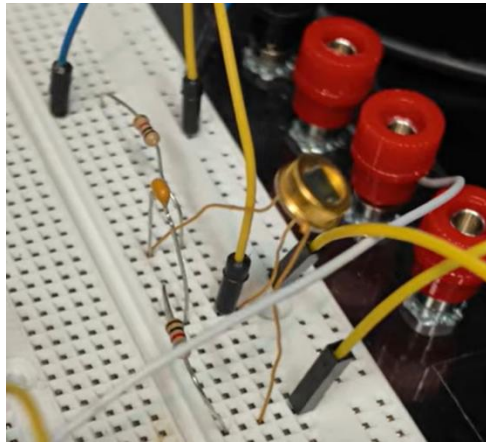


Figure 46. FDS100 Photodiode test on breadboard with appropriate circuitry

9.1.3 SONY ILX511B CCD Sensor Test

The last piece of optoelectronic equipment used in this project for our spectral measurements was the SONY ILX511B CCD linear array sensor, but was eventually replaced by the Raspberry Pi 5 with a CMOS camera. In the data sheet for the CCD sensor, it came with a recommended circuit consisting of capacitors, resistors, and a npn transistor. The main challenge when it came to testing the CCD sensor was finding the appropriate transistor. The recommended transistor was no longer selling anywhere (the 2SA1175), and it was difficult to find another similar transistor that had the same and/or similar specifications. Fortunately, a seller on eBay was selling the 2SA1175 transistor and we were able to order about 10 of them. This of course comes at risk because there is a possibility that they could not be working. Fortunately, it seemed to work in our experimental run.

The SONY ILX511B, built with the appropriate circuitry, does work in the sense that we are able to get an output voltage reading, but we are certain that this is not

quite the correct measurement. The output voltage is constant and does not really change with respect to how much light is incident on the sensor which is concerning because the CCD sensor is supposed to be sensitive. We speculate that our scaling on the voltage is wrong, and that we have to “zoom-in” to see more fluctuations in the intensity values across the pixels.

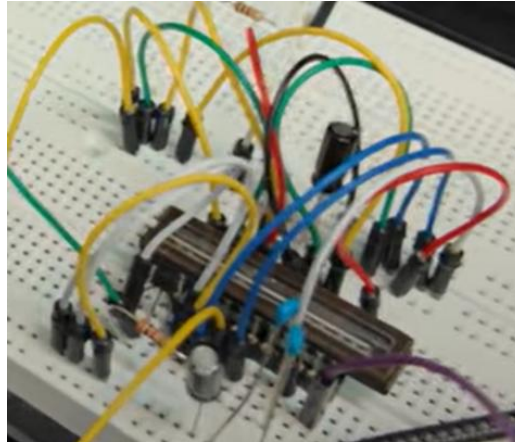


Figure 47. SONY ILX511B Breadboard Test

It is important to note that this was performed during the prototyping phase of the project. We did not end up using the CCD due to arising issues, and promptly switched to using the Arducam as aforementioned.

9.2 Fluorescence Testing

The spectrometer in our system will have to be calibrated in order to properly measure the cyanobacteria concentration. To do this, the sensor has to be calibrated with known concentrations of cyanobacteria, that way we can map the intensity on the ILX511B with a particular concentration and repeat that experimentally several times to build a Concentration vs Intensity curve with a line of best fit that can be referenced to make concentration measurements in real time.

There are multiple different species of cyanobacteria that produce different toxins, but we wanted to be on the safer side and use the spirulina strain of cyanobacteria. This will not make a difference spectrally because the spectrum of cyanobacteria excited by 405 nm UV light is the same since the spectrum depends on chlorophyll and phycocyanin contents, not the cyanotoxins produced. Furthermore, spirulina is easy to acquire because it is a common dietary supplement. So, there are ample amounts of this cyanobacteria strain for testing without the risks of touching or accidentally ingesting the cyanotoxins.

To measure the concentration of cyanobacteria, we have to first know our cuvette volume (i.e. how much water it can hold). The cuvette can hold approximately 4.5 mL of water, which means that if we want to have a concentration of 1 mg/mL we need to add approximately 4.5 mg of cyanobacteria into the cuvette. For measuring the dry weight of the spirulina, we purchased a milligram scale and some small lab spoons that can pick up 4-5 mg of a substance. We used those spoons and the milligram scale to weigh out approximately 4 mg of cyanobacteria and then added it into the cuvette that contains 4.5 mL of water to make a solution consisting of 1 mg/mL of cyanobacteria. After we weighed out the cyanobacteria, we shined our UV laser module at it to see the fluorescing light.

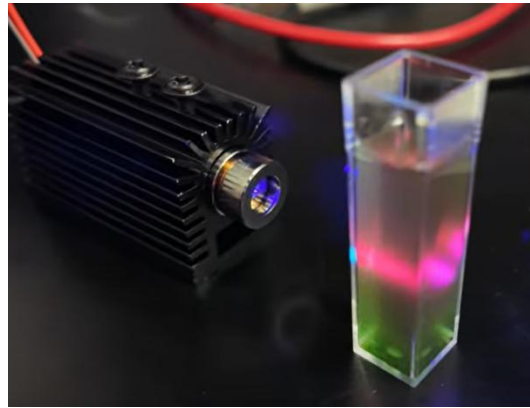


Figure 48. 1 mg/mL sample of cyanobacteria shown to be fluorescing when excited by the 405 nm UV laser

The next step in testing our biological cyanobacteria was to determine if the spectrum of cyanobacteria is differentiable than green algae. This is already known to be true from our preliminary research (see chapter 2 and 3), but we wanted to verify for ourselves. To do this, we temporarily borrowed a spectrometer from the undergraduate lab in CREOL to obtain the spectrums of cyanobacteria and of a bamboo leaf which is mostly chlorophyll just like green algae shown in the figure 40 below.



Figure 49. Spectrum of cyanobacteria (left) and bamboo leaf (mimicking green algae) (right)

As shown in figure 40, the cyanobacteria have a distinct peak at 651.08 nm and the chlorophyll strong bamboo leaf (again, closely mimicking green algae) has a distinct peak at 686.81 nm. This supports our research from chapters 2 and 3 and also demonstrates that we have a strong enough signal to be working with for detection.

9.3 ESP32 Integration and Testing

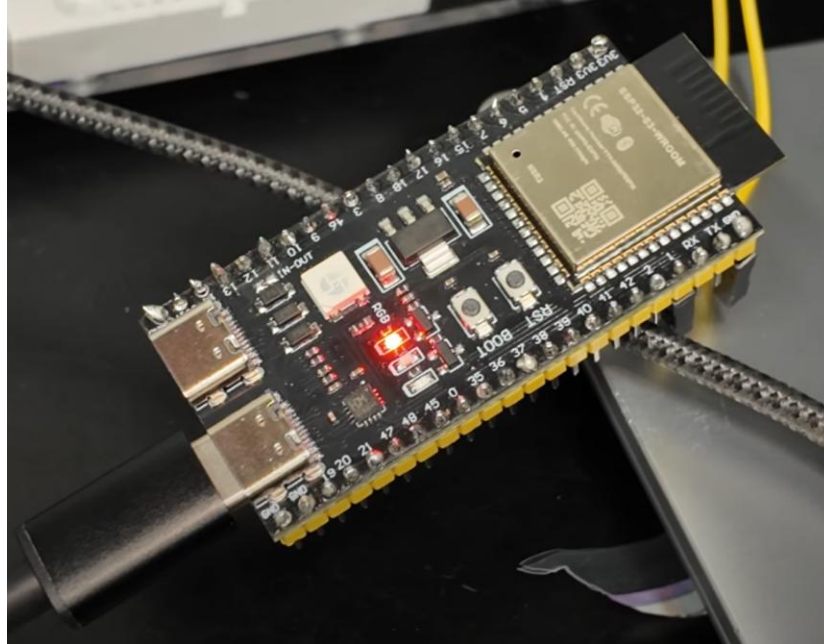


Figure 50. ESP322 Integration testing

The microcontrollers (MCU) we selected for the LUCID project are the ESP32-S3 and the Raspberry Pi 5, as mentioned earlier. During initial testing, we were able to connect and program the ESP32 and Raspberry pi relatively easily after addressing a few early challenges related to USB communication. We discovered that the development board we purchased did not use the more common CP2102 USB-to-UART bridge but instead utilized a CH343 UART bridge. After identifying and installing the correct CH343 USB driver, we were able to consistently upload code to the ESP32-S3 without further issues. Once communication was established, we began using the ESP32-S3 for early hardware testing. Through basic firmware development, we successfully connected the ESP32 to the FDS100 photodiode and connected it to our firebase database. By configuring the ESP32's ADC1 input channels, we were able to read and verify analog voltage outputs from both sensors, confirming that our signal conditioning circuits, and initial software setup were working correctly. These tests demonstrated that the ESP32-S3 provided the necessary flexibility, processing speed, and peripheral support needed for LUCID's real-time data collection and analysis requirements. So far, the ESP32-S3 has proven to be an excellent choice for the project. Its combination of high-speed ADCs, abundant GPIOs, native USB support, and strong developer resources has made it much easier to interface with the critical components in our system. As we continue developing and integrating the full system, the ESP32-S3's performance during prototyping gives us strong confidence that it will meet all the needs of the final LUCID design.

9.4 Raspberry Pi 5 Integration and Testing



Figure 51. Raspberry Pi Camera Integration testing

The Raspberry Pi 5 was one of the core components of the L.U.C.I.D system. The Pi acted as the primary computation unit responsible for image acquisition, spectral analysis, and real-time communication between the system and the website. Its integration involved coordinating between the hardware components such as the camera and the software components used in the python script, such as the NumPy and Matplotlib functions used for image processing, wavelength mapping and firebase data transmission. The raspberry pi was selected for its balance of computational power, and adaptability with our system using packages for the camera module.

The image acquisition process was established using the arducam OV5647 camera connected via the Pis camera port. The Picamera2 library and CV2 package were also imported into the camera script to control setting such as the exposure, gain, and frame settings. Controlling these settings on the camera allowed for consistent capture conditions during sample diagnostics and image captures. Additionally, the python script also oversaw the communication between the pi and the Firebase. Through the Firebase-admin package, the Pi was able to connect to the websites Firebase and listen for commands, once connected to Wi-Fi. These commands were sent through the website to the Firebase to initiate the L.U.C.I.D. system and begin image acquisition and processing.

Post-processing logic was developed within the script to isolate the fluorescent peaks between 3 regions initially, to verify the position of camera with the known wavelength locations. The blue and green region were eventually filtered out to improve accuracy between sample readings. The script then isolated the red region of the spectrum (around 640nm-720nm). By observing the peak intensity at these known wavelengths, the system was able to identify the peak intensity of green algae to fluoresce at around 665nm-690nm and the peak intensity for cyanobacteria samples fluoresced at around 620nm-650nm. The intensity of these peaks was then used to develop a multi-tiered calibration model using linear and quadratic regression models to estimate the concentration of cyanobacteria based on intensity. Once the analysis was complete, the final intensity profile, peak wavelength, estimated concentration, and classification were sent to the Firebase Realtime Database as a payload from the python script. The frontend was then able to retrieve the data to update the spectrum chart, detection alerts, and peak intensity section in real time.

9.5 Software Testing

To ensure the reliability of the LUCID system, a structured software test was applied throughout the development process of the system. Initially unit testing was used to verify the functionality and implementation of the lower-level operations such as, routines that generate timing signals (ROG, CLK, SHSW), or functions that manage the communication between the ESP32 and Firebase, as well as Wi-Fi. These tests will confirm that the digital output timings within the specified tolerances by using an oscilloscope to measure the signal waveform. For testing the integration, the full pipeline will be validated. This involved looking at the control signal output, ADC sampling of the ILX511 & FSD100, and verifying the spectrum data uploaded to the real-time database in Firebase. These specific tests will also assist in identifying issues such as timing drift, ADC lag, or data corruption or inaccuracy during transmission.

There was a need to be a focus on verifying the accuracy and stability of the ADC readings. The analog to digital converter is where the core operation will take place for our system. If the ADC is found to be inaccurate, it will ruin any of the results captured through it and invalidate our entire system. To verify the ADC readings, we will need to conduct repeated scans of a known, stable light source that will be taken under controlled conditions. These results were then be compared over time. This helped validate that the ADC is sampling consistently and that the ILX511 was receiving the correct clock signal required to operate properly as determined by its datasheet. Pixel data that was previously acquired will also be subjected to statistical tests. For example, statistical tests such as calculating the mean signal-to-noise ratio calculating the standard deviation and plotting the full pixel data will provide necessary information to verify the accuracy of the data collected.

Unfortunately, the ILX511 was inconsistent in capturing the full spectrum and our esp32 wasn't fast enough to provide an accurate operating clock signal for the ILX511. Due to these issues, our team pivoted to using the Raspberry Pi 5 and CMOS camera. We found this implementation to be much easier to develop and repair when needed.

9.5.1 Unit Tests for Timing Signal

Table 26. Timing Signal Test

Test ID	Test Description	Expected Result	Tool
T1.1	Generate a ϕ ROG pulse period of 5000ns	Pulse duration measured at 5000ns	Oscilloscope
T1.2	Generate a ϕ CLK signal at 2 MHz as required	Consistent square wave with minimal noise	Oscilloscope
T1.3	ϕ CLK continues for full pixel readout cycle	Proper timing to allow 2048 pulses per scan to for all pixels in the array	Oscilloscope
T1.4	Ensure ϕ ROG/SHSW do not overlap incorrectly	Proper sequence observed	Oscilloscope

These tests were used to validate the generation of control signals require by the ILX511 CCD sensor, such and the ϕ ROG and ϕ CLK. The timing accuracy is critical for ensuring reliable sensor operations and correct pixel data output. By using an oscilloscope, we were able to visually confirm that there was instability in our pulses and the measured frequency of the clock signal were too slow to properly integrate the ILX511 with our ESP32.

9.5.2 Unit Tests for Raspberry Pi

During the development of the Raspberry Pi's python script for capturing fluorescence and conducting spectral analysis, unit testing was essential to ensure stable performance and validate the accuracy of the data. The unit test for our camera script was mostly conducted through test-driven practices and modular validations to confirm the functionality of the Pi cam. This required segmenting our camera python script logic into individual components and testing them individually. An example of this was the region of interest, which needed to be calibrated so that the camera would only capture the fluorescence received from the gradient and not outside noise. Each function was verified through individual testing and feeding known sample images and intensity arrays to verify that our functions were processing the images properly.

9.5.3 Unit Tests for ADC Signal Acquisition

Table 27. ADC Signal Test

Test ID	Test Description	Expected Result	Tool
T2.1	Read stable analog voltage using analogRead()	ADC values that will match the known input	Calibrated voltage reference source
T2.2	Sample ADC while ϕ CLK is active	Pixel values that correspond to the expected brightness	Calibrated light sensor
T2.3	Capture the full 2048 pixels into an array	Array filled with no missing data	Serial Monitor / Plot
T2.4	Validate the ADC noise without light (Dark Readout)	Values show low and consistent signal	Histogram Analysis

This setup ensures the ESP32 will accurately sample the analog voltage from the FDS100 output pin during readout. The ADC must produce reliable results under various conditions, including reading a stable reference voltage and capturing the full 2048 pixel array with minimal data loss.

9.5.4 Unit Tests for Data Processing Accuracy

Table 28. Data Processing Tests

Test ID	Test Description	Expected Result	Tool
T3.1	Apply a 3-point moving average filter	A smooth output and graph	Plot comparison
T3.2	Apply a 3-point median filter	Sharp outliers removed	Plot comparison
T3.3	Peak detection algorithm identifying maximum florescence	Peak index corresponding with the brightest region	Print peak index

After data acquisition, the LUCID system will need to go through digital processing to clean and interpret the raw signal. These tests validate key algorithms such as the 3-point moving average filter, which will be used to smooth the data. The 3-point median filter will then remove the outlier spikes.

9.5.5 Unit Tests for Firebase

Table 29. Firebase Connection Test

Test ID	Test Description	Expected Result	Tool
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T4.1	Consistent connection through hotspot	Stable signal	Inspect element
T4.2	Send JSON of full scan (2048 values)	Firebase write success	Firebase Console
T4.3	Limit data transmission rate to <1Hz	No overloads in data transmission and stable data transfer	Serial + DB monitoring
T4.4	Corrupted network test	Auto-reconnection & retry observed	Wi-Fi Toggling Test

These tests confirm the system's ability to securely connection to the Firebase and upload the spectral data reliably. Authentication using user credentials will ensure that only authorized devices and users can access the database. The tests check the ability to transmit full 2048 scans in JSON format and monitor the rate0limiting by ensuring the data is not sent faster than once per second.

9.5.6 End-to-End System Validation

Table 30. System Validation Test

Test ID	Test Description	Expected Result	Tool
T5.1	Scan diffracted spectrum from grating (i.e. obtaining intensity profile)	Broad, high-intensity spectrum shown	Plot on website interface
T5.2	Verify presence of 650nm	Narrow peak near correct pixel	Intensity + Wavelength Verification
T5.3	Very light intensity and measure ADC linearity	Output increases linearly	SNR calculation
T5.4	UI auto-updates when data is uploaded	Chart refreshes within <1sec	Website live test

End-to-End tests ensure the software and hardware subsystems form sensor controls to web visualization are operating correctly and cohesively. A light test will yield a reference spectrum curve, while the 602nm laser will have a narrow peak at the correct pixel index. This will validate the entire acquisitions and process pathway.

9.5.7 Long-Term Reliability Testing

Table 31. Sustainability and Reliability Test

Test ID	Test Description	Expected Result	Tool
T6.1	Run system for a prolonged time	No memory leakage and stable data	Serial + DB export

T6.2	Upload thousands of scans to Firebase	No data loss, DB structure intact	Firebase Browser
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Long-term tests will assess the systems' durability and stability over extended periods of operation. Continuous runs are going to be used to detect memory leaks in the system or system crashes in the ESP32s firmware, while high-volume data uploads to Firebase will verify the integrity and scalability of the backend infrastructure.

9.6 Final System Testing

After integrating all of our optics, electronics, software, and improved on our initial prototype, we tested the L.U.C.I.D. system's final functionality.

9.6.1 FDS100 Photodiode Laser Output Power Stability Test (1st Engineering Req.)

We first tested the FDS100 photodiode for the same reasons as mentioned earlier in the paper. This test also showed satisfaction of our first engineering requirement, which was to send at least 10 mW of optical power to our sample in order to induce visible fluorescence. The figure below showcases these results and satisfaction of our first engineering requirement.

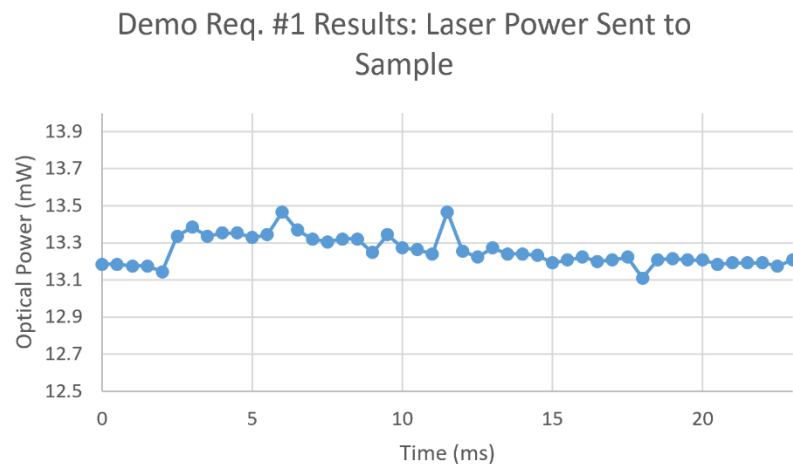


Figure 52. Measurements captured by FDS100 photodiode

As shown in the figure above, we were sending an average optical power of 13.26 mW, which was enough power to induce visible fluorescence and is greater than 10 mW, therefore satisfying our first engineering requirement. The laser was also relatively stable during the test, the data having a standard deviation of 0.08.

9.6.2 Cyanobacteria Concentration Measurement Test

We tested the rest of the L.U.C.I.D. system, which mainly includes the spectrometer and website/data-sending functionality. These results are summarized in the table below:

Table 32. Cyanobacteria concentration measurement test results

Trial #	Expected Concentration (mg/mL)	Measured Concentration (mg/mL)	Peak Wavelength (nm)	Intensity (A.U.)	Time (s)	Sample Identification	% error
1	0.25	0.289	660	149.614	5	Error	15.6
2	0.25	0.278	655	146.598	5	Cyanobacteria	11.2
3	0.25	0.247	650	137.58	30	Cyanobacteria	1.2
4	0.25	0.298	640	152.16	5	Cyanobacteria	19.2
5	0.5	0.418	647	176.4	5	Cyanobacteria	16.4
6	0.5	0.512	650	186.3	5	Cyanobacteria	2.4
7	0.5	0.455	648	181.3	5	Cyanobacteria	9
8	1	1.189	648	201.6	5	Cyanobacteria	18.9
9	1	1.2	650	186.3	5	Cyanobacteria	20
10	1	0.9	648	181.3	5	Cyanobacteria	10

The L.U.C.I.D. system, as shown from the table above, was clearly able to identify and measure cyanobacteria concentrations with less than 20% error in significantly less than 45 seconds. The first trial gave an error that we found to be caused by the peak wavelength being outside the expected range; the root of this error was simply a misalignment of the CMOS sensor. Consequently, this test clearly shows satisfaction with the 2nd and 3rd engineering requirements, and the L.U.C.I.D. system is therefore declared a success.

9.6.3 Website Final Display

During the testing depicted in the previous two sections, 9.5.1 and 9.5.2, all of the results were uniformly displayed on our easy to use website as shown in the figure below.

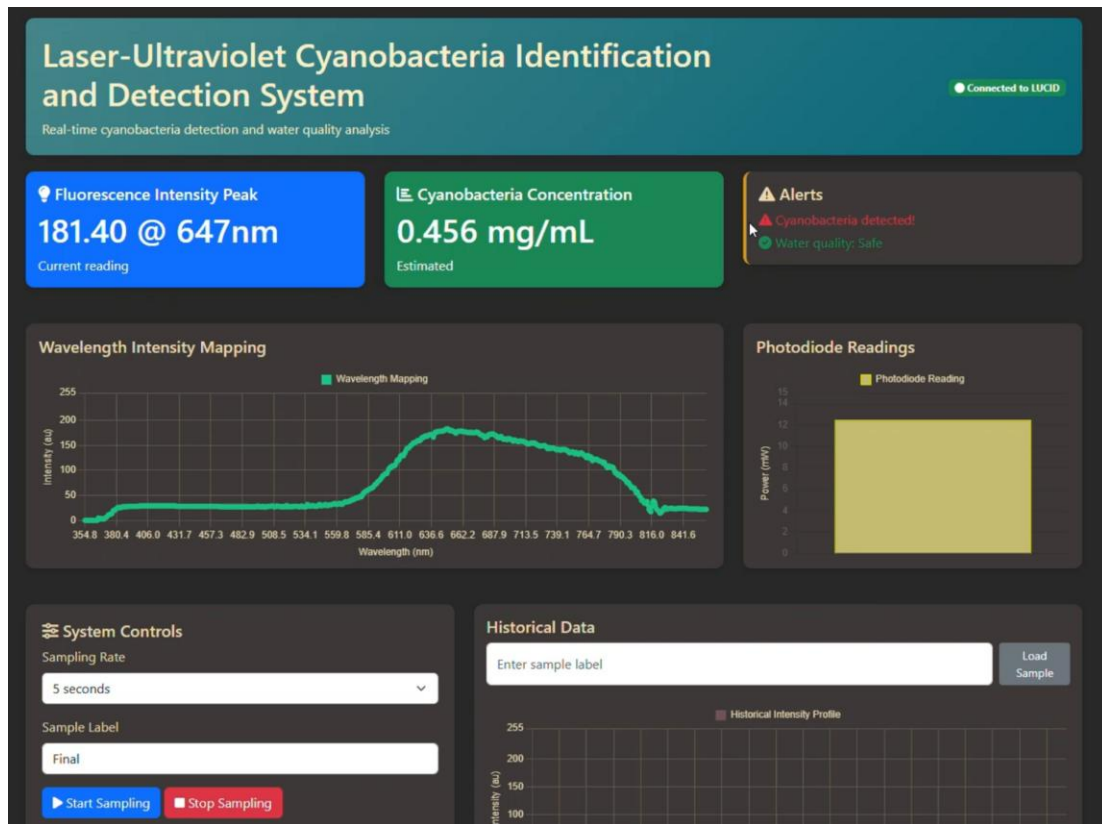


Figure 53. Display of website during a test

As you can see, the website displayed the peak wavelength and the respective intensity (in AU), the concentration of cyanobacteria detected, and a few alerts indicating the safety of the water sample, as well as the laser power reading (labeled photodiode readings in the figure above).

Chapter 10: Administrative Content

The following chapter covers administrative content that consists of the house of quality, a rough timeline of the project, the overall budgeting and finance information of the project, and work distribution.

10.1 House of Quality

The House of Quality in figure five allows our group to determine the factors that we wanted to focus on while doing the project. We did this by dividing and comparing engineering requirements compared to what consumers of the product would want to have. These tradeoffs helped us better understand what kind of product we wanted to make and help us understand the direction we wanted to go in avoiding future confusion when checking the correlations on this table.

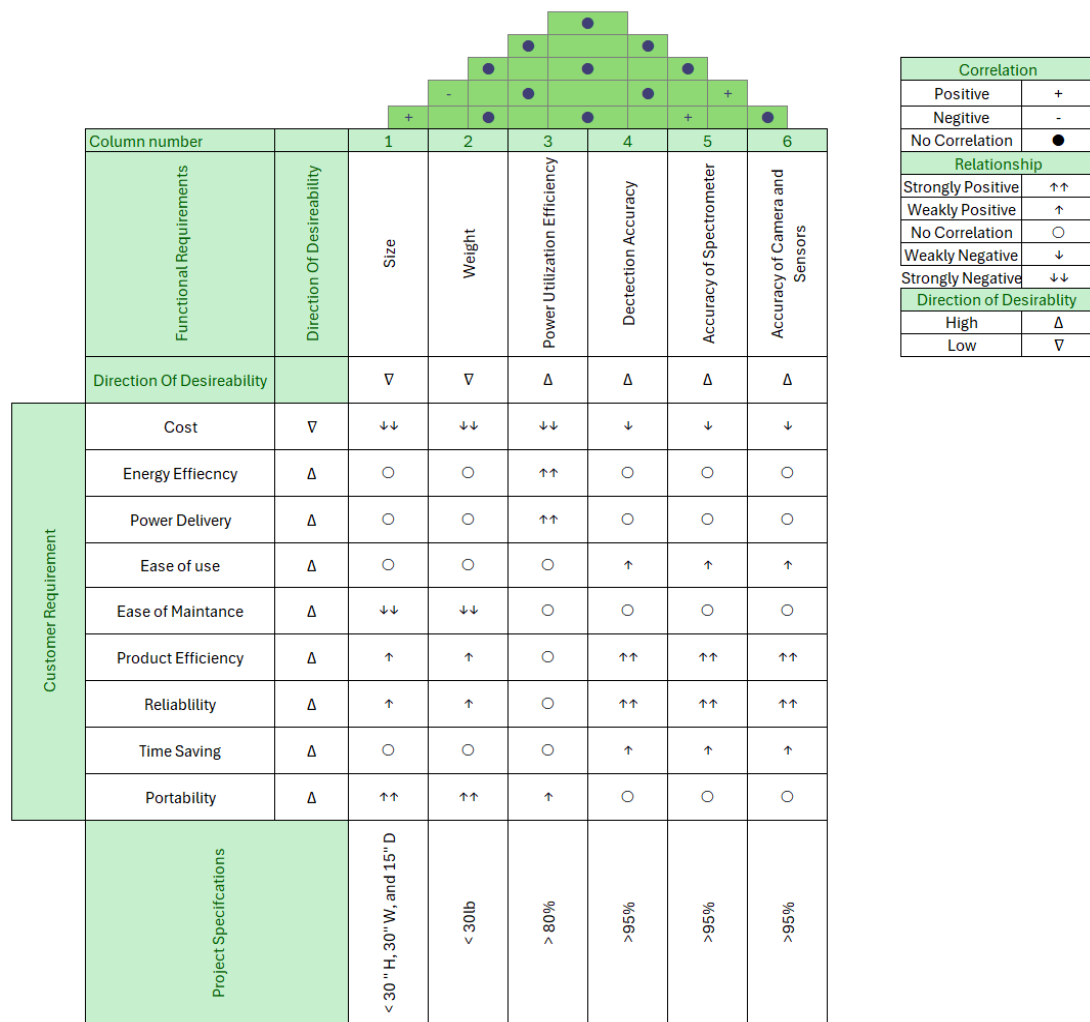


Figure 54. House of Quality Diagram

10.2 Budget & Financing

Much of the cost for this project was attributed to optics since optical components are generally expensive. It is important to note that we continued throughout the project to try to find ways to perform fluorescence spectroscopy at a lower cost by simplifying the optics and electronics without decreasing function, efficiency, or compromising our specifications. For example, the diffraction grating and the aspheric condenser were the most expensive optical components, so finding alternative solutions was a priority. Despite these financial factors, we anticipated our cyanobacteria detection system is to be below our budget of \$1,000. This is significantly cheaper than most fluorescence-based detection systems which are typically on the order of several thousand dollars. Our estimated cost and total cost are explained in detail in table 33 below.

Table 33. Bill of Materials and Budgeting

Component	Est. Cost	Qty.	Unit Cost	Total Cost
Aspheric Condenser Lens	\$100.00	1.00	\$75.26	\$75.26
Diffraction Grating	\$200.00	1.00	\$145.00	\$145.00
CCD Linear Sensor	\$70.00	1.00	\$60.00	\$60.00
Raspberry Pi	-	1.00	\$75.00	\$75.00
Pi-Cam	-	1.00	\$25.00	\$25.00
PCX Lens	\$50.00	1.00	\$40.50	\$40.50
Beamsplitter	\$80.00	1.00	\$54.55	\$54.55
Laser Diode Module	\$50.00	3.00	\$12.58	\$37.74
Photodiode	\$30.00	1.00	\$17.55	\$17.55
Heatsink	\$20.00	2.00	\$6.44	\$12.88
Cuvette	\$20.00	24.00	\$1.04	\$24.98
PCB	\$200.00	2.00	\$70.00	\$140.00
Spirulina Powder	\$15.00	1.00	\$14.00	\$14.00
Chlorella Powder	\$15.00	1.00	\$12.00	\$12.00
PLA Filament (kg)	\$50.00	4.00	\$ 25.00	\$100.00
Misc. Purchases	\$100.00	-	-	~\$60.00
TOTAL	\$1000.00			\$894.46

10.3 Initial Project Milestones

There were big points or milestones in the project that were markers for the overall process of the system and for that reason were important to keep track of, in which case we are using table 34, and 35 below to keep track of such progress.

Table 34. Design Timeline/Schedule

Project Design				
Start Date	Planned Date	End Date	Task	Description
1/6/25	1/14/25	1/14/25	Recruitment	Finding members for our team
1/14/25	1/23/24	1/24/24	Divide and Conquer Paper	First 10 pages of our final document
1/14/25	3/23/25	3/24/25	80-Page Paper	80 pages made for our final draft of the paper
1/14/25	4/21/25	4/22/25	Final Paper	120 pages for the final draft of our project in Senior Design 1

Table 35. Prototyping Timeline/Schedule

Project Prototyping				
Start Date	Planned Date	End Date	Task	Description
5/12/25	5/10/25	4/16/25	BOM/ Part Purchase	Making a BOM based on the parts we decided in senior design 1 and purchasing the parts
4/16/25	5/17/25	5/24/25	Individual Research/ Design	Once the parts have arrived, we will start working individually to make sure each of our parts work to start integrating together.
5/24/25	5/24-31/25	5/31/25	System Research/ Design	Designing the system together with all the parts connected and first look at what the final design will accurately look

5/12/25	5/25/25	5/25/25	PCB Design	Design a PCB to fulfill the requirements of our tested breadboarded project
TBD	TBD	7/1/25	PCB Testing	Testing the PCB out to see the full expected functionality
5/12/25	5/1/25	7/20/25	Prototype Completion	The completion of our Prototype and the end of Senior Design 2

10.4 Distribution of Work

Table 36 below summarizes the tasks assigned to each group member for both SD1 and SD2 to ensure each member is contributing equally and significantly to the project. Tasks were assigned based on expertise and degree. Some primary and secondary roles overlapped. All team members participated in testing during every stage of the project.

Table 36. Work Distribution

Team Member	Primary Role	Secondary Role
Max Baryshnikov	• Team Lead • Spectrometer optical design • Optoelectronic Implementation • Component mounting CAD	• System testing & calibration • 3D printing components & housing
Nick Drennen	• Fluorescence chamber optical design • Optoelectronic Implementation • Structural Housing CAD	• System testing & calibration • 3D printing components & housing
Luis Rodriguez-Rivera	• Software design and testing • Data storage • Website design and integration • MCU integration	• System testing & calibration • Spectrometer noise reduction (software side)
Sean Waddell	• PCB Design • Power supply and management • MCU selection and integration	• System testing & calibration • Hardware wiring

Chapter 11 Conclusion

In conclusion, L.U.C.I.D. focused on using the unique spectral properties of cyanobacteria to differentiate it from green algae and therefore aid in detecting the early stages of eutrophication. Eutrophication devastates aquatic ecosystem which is closely linked to land-based ecosystem health including, but not limited to, human health.

By using a 20 mW 405 nm UV laser and emitting it through a 50:50 beamsplitter, we were able to excite at least 0.25 mg/mL of cyanobacteria which fluoresces a reddish/orange color at a wavelength of 650 nm. Green algae, which contains mostly chlorophyll, fluoresces at a wavelength of 680 nm therefore giving a 30 nm difference between cyanobacteria and green algae. This spectral difference is enough to differentiate the two. Collecting this fluorescing light and sending it to a diffraction grating will split the fluorescent signals constituent wavelengths apart and reflect them towards the Arducam for spectral measurements. We can calibrate the intensity of each spectrum to different concentrations of cyanobacteria to make essentially what is a calibration curve that will be used for real-time measurements. Given the optical signals, our system has carefully selected electronics to provide sufficient enough power to our system without burning or damaging any of our components, especially the more damage-sensitive ones like the optoelectronic components such as the laser and silicon photodiode we are using.

Our raspberry pi MCU is the brain of the operations because it processes the images and received from the samples and transmits them to our website. The ESP32 also is an important MCU that is used to power on external components such as the Laser and the FDS100 photodiode. The ESP32 was also required to take in data from the FDS100 and transmit it to the Firebase Database. All of this data is then sent to our online database and website where results are stored and displayed for the user to look at.

The overarching goal is to protect the environment and prevent the early onsets of eutrophication from harmful algal blooms that may be difficult to detect visually and take too long using other methods and currently existing technologies.

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Appendix B: LLM Prompts

Prompt 1

Prompt:

“Can you suggest some microcontroller boards in price ranges of low, medium, and high that we could use to integrate the ILX511B CCD Linear Sensor?”

Response:

“Sure! The ILX511B is a CCD linear image sensor, so you’ll need a microcontroller board with enough speed to handle its data rate (up to 2 MHz clock), sufficient I/O for interfacing, and ideally ADC capabilities (if you’re reading analog outputs) or a fast external ADC if needed. Here’s a breakdown across low, medium, and high price ranges:

Low-cost range (\$5 - \$20)

1. ESP32-S3
 - Processor: Dual-core Tensilica Xtensa LX7 @ up to 240 MHz
 - Pros: Affordable, Wi-Fi + BLE, decent processing power, multiple ADC channels, low power consumption.
 - Cons: Limited ADC resolution (12-bit) might require external circuitry for CCD timing.
2. Raspberry Pi Pico
 - Processor: Dual-core ARM Cortex-M0+ @ 133 MHz
 - Pros: Ultra-cheap, good for prototyping, programmable IO (PIO) useful for custom CCD timing signals.
 - Cons: No Wi-Fi/Bluetooth on base model, limited RAM compared to higher-end boards.

Mid-range (\$20 - \$50)

1. Teensy 4.1
 - Processor: ARM Cortex-M7 @ 600 MHz
 - Pros: Very fast MCU, powerful enough for real-time CCD data capture, has a high-speed ADC.
 - Cons: Limited onboard peripherals compared to some SBCs, but very solid for embedded.
2. Arduino Portenta H7 Lite
 - Processor: Dual-core ARM Cortex-M7 + M4
 - Pros: Great balance of power and features, supports machine vision workloads, industrial-grade build.
 - Cons: Mid-range price, some features stripped down compared to full Portenta H7.

High-end (\$50 - \$150+)

1. Raspberry Pi 4 Model B (4GB or 8GB)
 - Processor: Quad-core Cortex-A72 @ 1.5 GHz
 - Pros: Runs full Linux, lots of libraries and support for interfacing with CCD sensors, can handle large data buffers, great for heavy image processing.

- Cons: Might need a dedicated ADC or external circuit to interface with ILX511B, as Pi lacks direct ADC.
2. NVIDIA Jetson Nano
- Processor: Quad-core ARM Cortex-A57 + 128-core Maxwell GPU
 - Pros: Ideal for real-time image processing (e.g., running OpenCV or AI on CCD data), ample GPIO, supports high-speed data handling.
 - Cons: Higher power draw requires Linux development experience.

Would you like me to also recommend external ADC options or timing generator circuits to pair with these boards for optimal CCD performance?" (ChatGPT Model 4o, 2025)

Prompt 2

Prompt:

Do you know anything about Zemax?

Response:

Yeah, I can definitely help with Zemax! It's an optical design software used to model, simulate, and analyze optical systems — like lenses, imaging systems, fiber couplers, beam expanders, etc. There are two main flavors:

- **Zemax OpticStudio** – the main GUI-based tool for designing and optimizing optical systems (both sequential and non-sequential modes).
- **Zemax OpticsBuilder** – for integrating Zemax models into CAD environments like SolidWorks.

In **sequential mode**, light travels from one surface to the next in order — great for traditional lens systems.

In **non-sequential mode**, rays can reflect, scatter, split, or refract in any order — useful for illumination systems, light pipes, stray light analysis, etc.

Typical things you can do:

- Model a lens system and optimize its performance
- Simulate spot diagrams, MTF, aberrations
- Use merit functions for automatic optimization
- Export lens drawings and system data
- Simulate stray light paths
- Use built-in glass catalogs and coatings

What are you using it for? I can walk you through specific setups, optimization steps, tolerancing, or help interpret output.

Prompt 3

Prompt:

I'm trying to model a collimated laser source propagating through a beamsplitter. the reflected beam will be 90 degrees to the incident beam, and then focusing through a preselected lens to a point.

Response:

Got it — sounds like a classic free-space laser setup: a collimated laser hits a beamsplitter, reflects 90°, and then goes through a focusing lens to a spot. Here's how you can model that in **Zemax OpticStudio (Sequential Mode)** step by step:

Basic Setup

1. Collimated Laser Source

- In Surface 1: set it to **Object at Infinity** (Object Distance = Infinity) or use a **collimated source** if you know the beam diameter and divergence.
- Set **Wavelength** to your laser (e.g., 405 nm).
- Define the **beam diameter** using an **aperture stop** or entrance pupil diameter.

2. Beamsplitter (90° Reflection)

In sequential mode, it's a bit tricky to do non-axial reflections, so use this trick:

- Insert a **"Fold Mirror"** surface:
 - Surface type: **Standard**
 - Check the box for **"Tilt/Decenter Elements"**
 - Under **"Tilt About X"**, set 90°.
 - This will redirect the beam perpendicularly.
- Alternatively, use a **Mirror** with tilt about X or Y for 90° fold.

✓ You can simulate a beamsplitter coating later with a **coating file** (e.g., BSR or custom coating).

3. Lens for Focusing

- Add your **preselected lens** (e.g., a plano-convex lens):
 - You can either enter each surface manually using its radius and thickness
 - Or import a complete lens file from the Zemax catalog

Set spacing after the lens to allow it to focus:

- Insert an image surface (e.g., Surface 5) far enough beyond the lens
- Use the **"Solve"** on thickness to find the focal plane (pick **para-axial image height = 0** if your beam is centered)

Helpful Tools for This Setup

- **Beam Footprint Diagram**: Shows how your beam spreads or focuses on each surface.
- **Spot Diagram**: Confirms how tightly the lens focuses the beam.
- **Ray Fan/OPD**: Evaluates aberrations if beam quality matters.

Notes

- You may need to turn on **"Use Ray Aiming"** if you find misalignment due to large tilt angles.
- If you're working with physical elements like a cube beamsplitter or need exact reflection/transmission, **Non-Sequential Mode** is more flexible.

Would you like a walkthrough for doing this in **Non-Sequential Mode** instead? That's usually better for angular paths and realistic coatings.